

# Indefinite Integration

## Question1

If  $\frac{x^4}{(x-1)(x-2)} = f(x) + \frac{A}{x-1} + \frac{B}{x-2}$ , then  $f(-2) + A + B =$

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**Options:**

A.

32

B.

28

C.

22

D.

20

**Answer: D**

**Solution:**

$$\frac{x^4}{(x-1)(x-2)} = f(x) + \frac{A}{(x-1)} + \frac{B}{(x-2)}$$

$$\Rightarrow x^4 = f(x)(x-2)(x-1) + A(x-2) + B(x-1)$$

at  $x = 1$

$$1 = -A \Rightarrow A = -1$$

at  $x = 2$

$$16 = B(2-1) \Rightarrow B = 16$$

Put the values of  $A$  and  $B$  and put

$$x = -2$$

$$16 = f(-2)(-3)(-4) - (-4) + 16(-3)$$

$$\Rightarrow 12f(-2) = 60$$

$$\Rightarrow f(-2) = 5$$

$$\text{Hence, } f(-2) + A + B = 5 - 1 + 16 = 20$$

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## Question2

If  $\int \frac{x^4+1}{x^2+1} dx = Ax^3 + Bx^2 + Cx + D \tan^{-1} x + E$ , then  $A + B + C + D =$

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Options:

A.

$$\frac{3}{2}$$

B.

$$\frac{4}{3}$$

C.

$$\frac{1}{3}$$

D.

$$\frac{2}{3}$$

**Answer: B**

**Solution:**

$$\begin{aligned}\frac{x^4+1}{x^2+1} &= \frac{x^4-1+2}{x^2+1} \\ &= \frac{(x^4-1)}{x^2+1} + \frac{2}{x^2+1} \\ &= \frac{(x^2-1)(x^2+1)}{x^2+1} + \frac{2}{x^2+1} \\ &= x^2-1 + \frac{2}{x^2+1}\end{aligned}$$

$$\text{So, } \int \frac{x^4+1}{x^2+1} dx = \int \left( x^2 - 1 + \frac{2}{x^2+1} \right) dx$$

$$= \frac{x^3}{3} - x + 2 \tan^{-1} x + c$$

Comparing with

$$Ax^3 + Bx^2 + Cx + D \tan^{-1} x + E$$

We get,  $A = 1/3, B = 0, C = -1, D = 2$  Therefore,

$$A + B + C + D = \frac{1}{3} + 0 - 1 + 2 = 4/3$$

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## Question3

If  $\int \frac{x^2 - x + 2}{x^2 + x + 2} dx = x - \log(f(x)) + \frac{2}{\sqrt{7}} \tan^{-1}(g(x)) + c$ , then

$$f(-1) + \sqrt{7}g(-1) =$$

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**Options:**

A.

1

B.

0

C.

-1

D.

2

**Answer: A**

**Solution:**

$$\begin{aligned}\frac{x^2 - x + 2}{x^2 + x + 2} &= 1 - \frac{2x}{x^2 + x + 2} \\ &= 1 - \frac{2x + 1 - 1}{x^2 + x + 2} \\ &= 1 - \frac{2x + 1}{x^2 + x + 2} - \frac{1}{x^2 + x + 2} \\ &= 1 - \frac{2x + 1}{x^2 + x + 2} - \frac{1}{\left(x + \frac{1}{2}\right)^2 + \frac{7}{4}} \\ \therefore \int \frac{x^2 - x + 2}{x^2 + x + 2} dx &= \int \left( 1 - \frac{2x + 1}{x^2 + x + 2} - \frac{1}{\left(x + \frac{1}{2}\right)^2 + \frac{7}{4}} \right) dx \\ &= \int 1 \cdot dx - \int \frac{2x + 1}{x^2 + x + 2} dx - \int \frac{1}{\left(x + \frac{1}{2}\right)^2 + \frac{7}{4}} \cdot dx\end{aligned}$$

$$\text{let } x^2 + x + 2 = u \Rightarrow (2x + 1)dx = du$$

$$\begin{aligned}&= x - \int \frac{1}{u} \cdot du - \frac{1}{\sqrt{7/4}} \tan^{-1} \left( \frac{x + 1/2}{\sqrt{7/4}} \right) \\ &= x - \ln(x^2 + x + 2) - \frac{2}{\sqrt{7}} \tan^{-1} \left( \frac{2x + 1}{\sqrt{7}} \right)\end{aligned}$$

Comparing with

$$x - \log(f(x)) + \frac{2}{\sqrt{7}} \tan^{-1}(g(x))$$

$$f(x) = x^2 + x + 2 \text{ and } g(x) = \frac{2x+1}{\sqrt{7}}$$

$$\text{then, } f(-1) = (-1)^2 - 1 + 2 = 2$$

$$\text{and } g(-1) = \frac{2(-1)+1}{\sqrt{7}} = -1/\sqrt{7}$$

$$\text{Hence, } f(-1) + \sqrt{7} \cdot g(-1)$$

$$= 2 + \sqrt{7}(-1/\sqrt{7}) = 1$$

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## Question4

$$\int \sec\left(x - \frac{\pi}{3}\right) \sec\left(x + \frac{\pi}{6}\right) dx =$$

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**Options:**

A.

$$\log \left| \frac{\sec\left(x - \frac{\pi}{3}\right)}{\sec\left(x + \frac{\pi}{6}\right)} \right| + C$$

B.

$$\log \left| \frac{\cos\left(x - \frac{\pi}{3}\right)}{\cos\left(x + \frac{\pi}{6}\right)} \right| + C$$

C.

$$\log \left| \frac{\operatorname{cosec}\left(x - \frac{\pi}{3}\right)}{\operatorname{cosec}\left(x + \frac{\pi}{6}\right)} \right| + C$$

D.

$$\log \left| \frac{\sin\left(x - \frac{\pi}{3}\right)}{\sin\left(x + \frac{\pi}{6}\right)} \right| + C$$

**Answer: B**

**Solution:**

$$\text{In } \sec\left(x - \frac{\pi}{3}\right) \sec\left(x + \frac{\pi}{6}\right)$$

$$\text{Let } A = x - \frac{\pi}{3} \text{ and } B = x + \frac{\pi}{6}$$

$$\text{then } B - A = x/2 \Rightarrow B = A + \pi/2$$

$$\text{So, } \sec B = \sec\left(\frac{\pi}{2} + A\right) = -\operatorname{cosec} A$$

$$\text{Then, } \sec A \cdot \sec B = \sec A(-\operatorname{cosec} A)$$

$$= -\sec A \cdot \operatorname{cosec} A$$

$$\text{Now, } -\sec A \cdot \operatorname{cosec} A = \frac{-1}{\sin A \cdot \cos A}$$

$$= \frac{-2}{\sin 2A} = -2 \operatorname{cosec} 2A$$

$$\begin{aligned} \therefore \int \sec \left( x - \frac{\pi}{3} \right) \sec \left( x + \frac{\pi}{6} \right) dx \\ = -2 \int \operatorname{cosec} \left( 2x - \frac{2\pi}{3} \right) dx \end{aligned}$$

$$\text{Let } 2x - \frac{2\pi}{3} = u$$

$$\begin{aligned} \Rightarrow 2 \cdot dx = du = - \int \operatorname{cosec} u \cdot du \\ = -\ln |\operatorname{cosec}(u) - \cot(u)| + c \end{aligned}$$

$$\text{and } \operatorname{cosec}(u) - \cot(u) = \tan(u/2)$$

$$= -\ln \left| \tan \left( \frac{u}{2} \right) \right| + C = -\ln \left| \tan \left( x - \frac{\pi}{3} \right) \right| + c$$

$$= \ln \left| \cot \left( x - \frac{\pi}{3} \right) \right| + C = \ln \left| \frac{\cos \left( x - \frac{\pi}{3} \right)}{\sin \left( x - \pi/3 \right)} \right| + c$$

$$\begin{aligned} \therefore \sin \left( x - \frac{\pi}{3} \right) &= \sin \left( \frac{\pi}{2} + \left( x - \frac{\pi}{6} \right) \right) \\ &= \cos \left( x - \frac{\pi}{6} \right) \\ &= \ln \left| \frac{\cos(x - \pi/3)}{\cos(x - \pi/6)} \right| + c \end{aligned}$$

## Question5

If  $\int \frac{a \cos x + 3 \sin x}{5 \cos x + 2 \sin x} dx = \frac{26}{29} x - \frac{k}{29} \log |5 \cos x + 2 \sin x| + \dots$  then  $|a + k| =$

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Options:

A.

3

B.

11

C.

12

D.

2

**Answer: B**

**Solution:**

$$\int \frac{a \cos x + 3 \sin x}{5 \cos x + 2 \sin x} \cdot dx$$

$$= \frac{26}{29}x - \frac{k}{29} \log |5 \cos x + 2 \sin x| + C$$

$$\because \frac{d}{dx}(5 \cos x + 2 \sin x) = -5 \sin x + 2 \cos x$$

$$\text{let } a \cos x + 3 \sin x = A(-5 \sin x + 2 \cos x) + B(5 \cos x + 2 \sin x)$$

By comparing

$$\begin{aligned} -5A + 2B &= 3 & \dots (i) \\ 2A + 5B &= a & \dots (ii) \end{aligned}$$

$$[\text{Eq. (ii)}] \times 2 - [\text{Eq. (i)}] \times 5$$

$$29A = 2a - 15 \Rightarrow A = \frac{2a-15}{29}$$

$$\text{So, } I = A \log |5 \cos x + 2 \sin x| + Bx + C$$

Comparing with the original form

$$\text{We get } A = -\frac{k}{29}, B = \frac{26}{29}$$

$$\text{So, } A = \frac{2a-15}{29} = \frac{-k}{29}$$

$$\Rightarrow k = 15 - 2a \Rightarrow |a + k| = |15 - a|$$

$$\text{From Eq. (ii), we get } 2\left(\frac{2a-15}{29}\right) + 3\left(\frac{26}{29}\right) = a$$

$$\Rightarrow 4a - 30 + 130 = 29a \Rightarrow a = 4$$

$$\text{Hence, } |a + k| = |15 - a| = 11$$

## Question 6

$$\text{If } \int \frac{dx}{1 - \sin^4 x} = A \tan x + B \tan^{-1}(\sqrt{2} \tan x) + C, \text{ then } A^2 - B^2 =$$

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Options:

A.

$$\frac{1}{2}$$

B.

$$\frac{3}{4}$$

C.

$$\frac{1}{4}$$

D.

$$\frac{1}{8}$$

**Answer: D**

**Solution:**

$$\begin{aligned}\because 1 - \sin^4 x &= (1 - \sin^2 x)(1 + \sin^2 x) \\ &= \cos^2 x (1 + \sin^2 x) \\ \therefore \int \frac{1}{1 - \sin^4 x} dx &= \int \frac{1}{\cos^2 x \cdot (1 + \sin^2 x)} \cdot dx \\ &= \int \frac{\sec^2 x}{1 + \sin^2 x} dx\end{aligned}$$

Let,  $\tan x = t$

$$\Rightarrow \frac{1}{1 + \sin^2 x} = \frac{1 + t^2}{1 + 2t^2} \quad \left\{ \because \sin x = \frac{1}{\sqrt{1 + t^2}} \right\}$$

and  $\sec^2 x \cdot dx = dt$

$$\begin{aligned}&= \int \frac{1 + t^2}{1 + 2t^2} \cdot dt \\ \therefore \frac{1 + t^2}{1 + 2t^2} &= \frac{1}{2} \left( 1 + \frac{1}{1 + 2t^2} \right) \\ \therefore I &= \frac{1}{2} \int \left( 1 + \frac{1}{1 + 2t^2} \right) dt \\ &= \frac{1}{2} \left( t + \frac{1}{\sqrt{2}} \tan^{-1}(\sqrt{2}t) \right) + C \\ &= \frac{1}{2} \tan x + \frac{1}{2\sqrt{2}} \tan^{-1}(\sqrt{2} \tan x) + C\end{aligned}$$

$$A \tan x + B \tan^{-1}(\sqrt{2} \tan x) + C$$

$$\text{We get, } A = \frac{1}{2}, B = \frac{1}{2\sqrt{2}}$$

$$\therefore A^2 - B^2 = \frac{1}{4} - \frac{1}{8} = \frac{1}{8}$$

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## Question7

If  $\frac{2x^4 - 3x^2 + 4}{(x^2 + 1)(x^2 + 2)} = a + \frac{px + q}{x^2 + 1} + \frac{mx + n}{x^2 + 2}$ , then  $\frac{n}{q} =$

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**Options:**

A.

$$p + m - a$$

B.

$$\frac{p+m}{a}$$

C.

$$\frac{a}{p+m}$$

D.

$$p + m + a$$

**Answer: A**

**Solution:**

Given,

$$\begin{aligned} \frac{2x^4 - 3x^2 + 4}{(x^2 + 1)(x^2 + 2)} &= \frac{2x^4 - 3x^2 + 4}{x^4 + 3x^2 + 2} \\ &= \frac{2x^4 + 6x^2 + 4}{x^4 + 3x^2 + 2} - \frac{9x^2}{x^4 + 3x^2 + 2} \\ &= 2 - \frac{9x^2}{x^4 + 3x^2 + 2} \\ \therefore \frac{9x^2}{x^4 + 3x^2 + 2} &= \frac{A}{x^2 + 1} + \frac{B}{x^2 + 2} \\ \Rightarrow 9x^2 &= A(x^2 + 2) + B(x^2 + 1) \\ \Rightarrow 9 &= A + B \text{ and } 2A + B = 0 \\ \Rightarrow B &= -2A \\ \therefore A &= -9, B = 18 \\ \therefore \frac{2x^4 - 3x^2 + 4}{(x^2 + 1)(x^2 + 2)} &= 2 + \frac{9}{x^2 + 1} - \frac{18}{x^2 + 2} \\ a = 2, p = 0, q = 9, m = 0, n &= -18 \\ \therefore n/q &= -2 \\ p + m - a &= 0 + 0 - 2 = -2 \end{aligned}$$

## Question8

$$\int (\log 2x)^3 dx =$$

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**Options:**

A.

$$x \left[ (\log 2x)^3 - 3(\log 2x)^2 + 6(\log 2x) - 6 \right] + C$$

B.

$$\frac{x}{4} \left[ 4(\log 2x)^3 - 6(\log 2x)^2 + 6(\log 2x) - 3 \right] + C$$

C.

$$\frac{x}{2} \left[ (\log 2x)^3 - 3(\log 2x)^2 + 3(\log 2x) - 6 \right] + C$$

D.

$$x \left[ (\log 2x)^3 - 6(\log 2x)^2 + 18(\log 2x) - 54 \right] + C$$

**Answer: A**

**Solution:**

$$\text{Let } I = \int 1 \cdot (\log 2x)^3 dx$$

Using by part

$$\begin{aligned} &= x(\log 2x)^3 - \int \left( 3(\log 2x)^2 \cdot \frac{1}{x} \cdot x \right) dx \\ &= x(\log 2x)^3 - 3 \int (\log 2x)^2 dx \\ &= (x \log 2x)^3 - 3 \cdot x(\log 2x)^2 + 3 \int \left( x \cdot 2 \cdot \log 2x \cdot \frac{1}{x} \right) dx \\ &= x [(\log 2x)^3 - 3(\log 2x)^2] + 6 \int (\log 2x) dx \\ &= x [(\log 2x)^3 - 3(\log 2x)^2] + 6x \log 2x - 6 \int dx \\ &= x [(\log 2x)^3 - 3(\log 2x)^2 + 6 \log 2x - 6] + C \end{aligned}$$

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## Question9

$$\int \frac{x+1}{(x-2)\sqrt{1-x}} dx =$$

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**Options:**

A.

$$\log(x+1) - \log(x-2)\sqrt{1-x} + C$$

B.

$$\log(x-2)\sqrt{1-x} + C$$

C.

$$6 \tan^{-1} \sqrt{1-x} - 2\sqrt{1-x} + C$$

D.

$$4 \tan^{-1} \sqrt{1-x} - 2\sqrt{1-x} + C$$

**Answer: C**

**Solution:**

$$\text{Let } I = \int \frac{x+1}{(x-2\sqrt{1-x})} dx$$

$$\sqrt{1-x} = t$$

$$\Rightarrow 1-x = t^2$$

$$\Rightarrow x = 1-t^2$$

$$\text{and } \frac{-1}{2\sqrt{1-x}} dx = dt$$

$$\Rightarrow \frac{dx}{\sqrt{1-x}} = -2dt$$

$$\therefore I = \int \frac{1-t^2+1}{(1-t^2-2)}(-2)dt = -2 \int \frac{t^2-2}{t^2+1} dt$$

$$= -2 \int \left( \frac{t^2+1}{t^2+1} - \frac{3}{t^2+1} \right) dx$$

$$= -2(t - 3 \tan^{-1} t) + C$$

$$= -2\sqrt{1-x} + 6 \tan^{-1} \sqrt{1-x} + C$$

$$= 6 \tan^{-1} \sqrt{1-x} - 2\sqrt{1-x} + C$$

## Question10

$$\int \frac{1}{1+x+x^2} dx =$$

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**Options:**

A.

$$\frac{2}{\sqrt{3}} \log \left( \frac{2x+1+\sqrt{3}}{2x-1-\sqrt{3}} \right) + C$$

B.

$$\frac{1}{\sqrt{3}} \log \left( \frac{2x+1-\sqrt{3}}{2x+1+\sqrt{3}} \right) + C$$

C.

$$\frac{2}{\sqrt{3}} \tan^{-1} \left( \frac{2x+1}{\sqrt{3}} \right) + C$$

D.

$$\frac{2}{\sqrt{5}} \tan^{-1} \left( \frac{2x+1}{\sqrt{5}} \right) + C$$

**Answer: C**

**Solution:**

$$\begin{aligned}
 \text{Let } I &= \int \frac{1}{x^2 + x + 1} dx \\
 &= \int \frac{1}{\left(x + \frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} dx \\
 &= \frac{1}{\frac{\sqrt{3}}{2}} \tan^{-1} \left( \frac{x + 1/2}{\sqrt{3}/2} \right) + C \\
 &= \frac{2}{\sqrt{3}} \tan^{-1} \left( \frac{2x + 1}{\sqrt{3}} \right) + C
 \end{aligned}$$


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## Question11

If  $\int \frac{dx}{(x \tan x + 1)^2} = f(x) + C$ , then  $\lim_{x \rightarrow \frac{\pi}{2}} f(x) =$

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**Options:**

A.

$$\frac{\pi}{2}$$

B.

$$\frac{2}{\pi}$$

C.

$$\frac{1}{\pi}$$

D.

$\infty$

**Answer: B**

**Solution:**

$$\begin{aligned}
\text{Let } I &= \int \frac{dx}{(x \tan x + 1)^2} \\
&= \int \frac{\cos^2 x}{(x \sin x + \cos x)^2} dx \\
&= \int \left( \frac{\cos x}{x} \right) \frac{x \cos x}{(x \sin x + \cos x)^2} dx \\
&= \frac{\cos x}{x} \int \frac{x \cos x}{(x \sin x + \cos x)^2} \\
&\quad - \int \frac{(-x \sin x - \cos x)}{x^2} \\
&\quad \left( \int \frac{x \cos x}{(x \sin x + \cos x)^2} dx \right) \\
&= \frac{\cos x}{x} \left( \frac{-1}{x \sin x + \cos x} \right) - \int \frac{x \sin x + \cos x}{x^2} \cdot \frac{1}{(x \sin x + \cos x)} dx \\
&= \frac{-\cos x}{x(x \sin x + \cos x)} dx \\
\therefore f(x) &= \frac{1}{x} + C \\
\therefore \lim_{x \rightarrow \frac{\pi}{2}} f(x) &= \lim_{x \rightarrow \frac{\pi}{2}} \frac{-\cos x}{x(x \sin x + \cos x)} + \frac{1}{x} \\
&= \frac{2}{\pi}
\end{aligned}$$


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## Question12

$$\int \sin^3 x \cos^2 x dx =$$

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**Options:**

A.

$$\frac{\sin^4 x \cos x}{5} - \frac{\sin^2 x \cos x}{15} - \frac{2 \cos x}{15} + C$$

B.

$$- \frac{\sin^4 x \cos x}{5} - \frac{\sin^2 x \cos x}{15} + \frac{2 \cos x}{15} + C$$

C.

$$\frac{\sin^4 x \cos' x}{5} - \frac{\sin^2 x \cos x}{15} + \frac{2x}{15} + C$$

D.

$$\frac{\sin^4 x \cos x}{5} + \frac{\sin^2 x \cos x}{3} - \frac{2x}{15} + C$$

**Answer: A**

**Solution:**

$$I = \int \sin^3 x \cos^2 x dx$$

$$\text{Let } \cos x = t$$

$$\Rightarrow -\sin x dx = dt$$

$$\therefore I = \int (1 - t^2)t^2(-dt)$$

$$= \int (t^4 - t^2)dt = \frac{t^5}{5} - \frac{t^3}{3} + C$$

$$= \frac{\cos^5 x}{5} - \frac{\cos^3 x}{3} + C$$

$$= \frac{\cos x (1 - \sin^2 x)^2}{5} - \frac{\cos x (1 - \sin^2 x)}{3} + C$$

$$= \frac{\sin^4 x \cos x}{5} - \frac{2 \sin^2 x \cos x}{5} + \frac{\cos x}{5} - \frac{\cos x}{3} + \frac{\cos x \sin^2 x}{3} + C$$

$$= \frac{\sin^4 x \cdot \cos x}{5} - \frac{\sin^2 x \cos x}{15} - \frac{2 \cos x}{15} + C$$

## Question13

If

$$\frac{3x^3 - 7x + 1}{(x-2)^5} = \frac{A}{x-2} + \frac{B}{(x-2)^2} + \frac{C}{(x-2)^3} + \frac{D}{(x-2)^4} + \frac{E}{(x-2)^5}, \text{ then } A(B + C + D + E) =$$

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Options:

A.

0

B.

348

C.

64

D.

256

**Answer: A**

**Solution:**

$$\text{Given, } \frac{3x^3 - 7x + 1}{(x-2)^5}$$

$$= \frac{A}{(x-2)} + \frac{B}{(x-2)^2} + \frac{C}{(x-2)^3} + \frac{D}{(x-2)^4} + \frac{E}{(x-2)^5}$$

$$= \frac{A(x-2)^4 + B(x-2)^3 + C(x-2)^2 + D(x-2) + E}{(x-2)^5}$$

$$= Ax^4 + (-8A + B)x^3 + (24A - 6B + C)x^2 + (-32A + 12B - 4C + D)x + 16A - 8B + 4C - 2D + E$$

Comparing the coefficient, we get

$$\begin{aligned}A &= 0 \\-8A + B &= 3 \\\Rightarrow B &= 3 + 8A = 3 \\24A - 6B + C &= 0 \\\Rightarrow C &= 6B - 24A = 18 - 0 = 18 \\-32A + 12B - 4C + D &= -7 \\\Rightarrow D &= -7 + 32A - 12B + 4C = -7 + 0 - 36 + 72 = 29 \\16A - 8B + 4C - 2D + E &= 1 \\\Rightarrow E &= 1 - 16A + 8B - 4C + 2D \\&= 1 - 0 + 24 - 72 + 58 = 11 \\\therefore A(B + C + D + E) &= 0(3 + 18 + 29 + 11) = 0\end{aligned}$$

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## Question 14

$$\int (\sqrt{\tan x} + \sqrt{\cot x}) dx =$$

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Options:

A.

$$2 \tan^{-1} \left( \frac{\tan x - 1}{\sqrt{\tan x}} \right) + C$$

B.

$$\tan^{-1} \left( \frac{\tan x - 2}{2\sqrt{\tan x}} \right) + C$$

C.

$$\sqrt{2} \tan^{-1} \left( \frac{\tan x - 1}{\sqrt{2} \tan x} \right) + C$$

D.

$$\sqrt{2} \tan^{-1} \left( \frac{\tan x + 1}{\sqrt{2} \tan x} \right) + C$$

**Answer: C**

**Solution:**

$$\begin{aligned}&(\sqrt{\tan x} + \sqrt{\cot x}) dx \\&= \int \left( \sqrt{\tan x} + \frac{1}{\sqrt{\tan x}} \right) dx \\&= \int \frac{\tan x + 1}{\sqrt{\tan x}} dx\end{aligned}$$

Let  $u = \sqrt{\tan x}$ , then  $u^2 = \tan x$  and

$$\begin{aligned}
2udu &= \sec^2 x dx \\
&= (1 + \tan^2 x) dx \\
&= (1 + u^4) dx \\
&= \int \frac{u^2 + 1}{u} \cdot \frac{2u}{1 + u^4} du \\
&= \int \frac{2(u^2 + 1)}{1 + u^4} du = \int \frac{2(1 + \frac{1}{u^2})}{\frac{1}{u^2} + u^2} du \\
&= \int \frac{2(1 + \frac{1}{u^2})}{(u - \frac{1}{u})^2 + 2} du
\end{aligned}$$

$$\begin{aligned}
\text{Again, Let } v &= u - \frac{1}{u} \\
\Rightarrow dv &= \left(1 + \frac{1}{u^2}\right) du \\
&= \int \frac{2}{2\left(\frac{v^2}{2} + 1\right)} dv = \int \frac{1}{\left(\frac{v^2}{2} + 1\right)} dv \\
&= \int \frac{1}{\left(\frac{v}{\sqrt{2}}\right)^2 + 1} dv \\
&= \tan^{-1}\left(\frac{v}{\sqrt{2}}\right) \cdot \sqrt{2} + C
\end{aligned}$$

where  $c = \text{constant}$

$$\begin{aligned}
&= \sqrt{2} \tan^{-1}\left(\frac{u - \frac{1}{u}}{\sqrt{2}}\right) + C \\
&= \sqrt{2} \tan^{-1}\left(\frac{\sqrt{\tan x} - \frac{1}{\sqrt{\tan x}}}{\sqrt{2}}\right) + C \\
&= \sqrt{2} \tan^{-1}\left(\frac{\tan x - 1}{\sqrt{2 \tan x}}\right) + C
\end{aligned}$$

## Question 15

$$\int \frac{\sqrt{x-2}}{2x+4} dx =$$

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**Options:**

A.

$$\sqrt{x-2} - \frac{1}{2} \tan^{-1}\left(\frac{\sqrt{x-2}}{2}\right) + C$$

B.

$$\sqrt{x-2} - 2 \tan^{-1}\left(\frac{\sqrt{x-2}}{2}\right) + C$$

C.

$$\sqrt{x-2} + 2 \tan^{-1} \left( \frac{\sqrt{x-2}}{2} \right) + C$$

D.

$$\sqrt{x-2} + \frac{1}{2} \tan^{-1} \left( \frac{\sqrt{x-2}}{2} \right) + C$$

**Answer: B**

**Solution:**

$$\text{Let } I = \int \frac{\sqrt{x-2}}{2x+4} dx$$

$$\text{and } u = \sqrt{x-2}$$

$$\Rightarrow u^2 = x - 2 \text{ and } x = u^2 + 2$$

Differentiate  $x$  w.r.t  $u$ , we get

$$\frac{dx}{du} = 2u \Rightarrow dx = 2u du$$

$$\begin{aligned} I &= \int \frac{\sqrt{x-2}}{2x+4} dx = \int \frac{2u \cdot u}{2(u^2+2)+4} du \\ &= \int \frac{2u^2}{2u^2+8} du = \int \frac{u^2}{u^2+4} du \\ &= \int \frac{u^2+4-4}{u^2+4} du = \int \left( 1 - \frac{4}{u^2+4} \right) du \\ &= u - 4 \cdot \frac{1}{2} \cdot \tan^{-1} \left( \frac{u}{2} \right) + C \end{aligned}$$

where,  $C = \text{constant}$

$$= u - 2 \tan^{-1} \left( \frac{u}{2} \right) + C$$

$$= \sqrt{x-2} - 2 \tan^{-1} \left( \frac{\sqrt{x-2}}{2} \right) + C$$

## Question16

If  $\int x^{49} \left[ \tan^{-1} x^{50} + \frac{x^{50}}{1+x^{100}} \right] dx = \frac{x^n}{k} f(x) + c$ , then

$$f(x) - f \left( \sqrt[k]{x^n} \right) =$$

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**Options:**

A.

$$k + n$$

B.

$$k - n$$

C.

$$1/k$$

D.

$$1/n$$

**Answer: B**

**Solution:**

Given,

$$\begin{aligned} I &= \int x^{49} \left[ \tan^{-1}(x^{50}) + \frac{x^{50}}{1+x^{100}} \right] dx \\ &= \frac{x^n}{k} f(x) + c \end{aligned}$$

Let  $u = x^{50}$ , then  $du = 50x^{49}dx$

$$\begin{aligned} \text{so, } I &= \int x^{49} \left[ \tan^{-1} x^{50} + \frac{x^{50}}{1+x^{100}} \right] dx \\ &= \int x^{49} \left[ \tan^{-1}(u) + \frac{u}{1+u^2} \right] \cdot \frac{du}{50x^{49}} \\ &= \frac{1}{50} \int \left( \tan^{-1}(u) + \frac{u}{1+u^2} \right) du \\ &= \frac{1}{50} \left[ \int \tan^{-1}(u) du + \int \frac{u}{1+u^2} du \right] \\ &= \frac{1}{50} \left[ u \tan^{-1}(u) - \int \frac{u}{1+u^2} du + \int \frac{u}{1+u^2} du \right] \\ &\quad [\text{using product rule}] \\ &= \frac{1}{50} [u \tan^{-1}(u)] + c \\ &= \frac{1}{50} \cdot x^{50} \cdot \tan^{-1}(x^{50}) + c \end{aligned}$$

Comparing to original equation, we get

$$n = 50, k = 50, f(x) = \tan^{-1}(x^{50})$$

$$\text{Now, } f(x) - f\left(\sqrt[k]{x^n}\right)$$

$$\begin{aligned} &= f(x) - f\left(x^{n/k}\right) \\ &= \tan^{-1}(x^{50}) - f\left(x^{50/50}\right) \\ &= \tan^{-1}(x^{50}) - f(x) \\ &= \tan^{-1}(x^{50}) - \tan^{-1}(x^{50}) = 0 \end{aligned}$$

$$\therefore f(x) - f\left(\sqrt[k]{x^n}\right) = 0 \quad \dots (i)$$

$$\text{Now, } k - n = 50 - 50 = 0 \quad \dots (ii)$$

From Eqs. (i) and (ii), we get

$$f(x) - f\left(\sqrt[k]{x^n}\right) = k - n$$

---

## Question17

$$\int \frac{x}{\sqrt{x^2-2x+5}} dx =$$

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Options:

A.

$$\sqrt{x^2 - 2x + 5} + \sinh^{-1} \left( \frac{x-1}{2} \right) + C$$

B.

$$\frac{1}{2}\sqrt{x^2 - 2x + 5} + \sin^{-1} \left( \frac{x-1}{2} \right) + C$$

C.

$$2\sqrt{x^2 - 2x + 5} + \cosh^{-1} \left( \frac{x-1}{2} \right) + C$$

D.

$$\sqrt{x^2 - 2x + 5} - \cos^{-1} \left( \frac{x-1}{2} \right) + C$$

**Answer: A**

**Solution:**

$$\begin{aligned} \text{Let } I &= \int \frac{x}{\sqrt{x^2 - 2x + 5}} dx \\ &= \int \frac{x - 1 + 1}{\sqrt{x^2 - 2x + 5}} dx \\ &= \int \frac{x - 1}{\sqrt{x^2 - 2x + 5}} + \int \frac{1}{\sqrt{x^2 - 2x + 5}} dx \end{aligned}$$

$$\text{Let } u = x^2 - 2x + 5, \text{ then } du = (2x - 2)dx$$

$$\begin{aligned} &= 2(x - 1)dx \\ &= \frac{1}{2} \int \frac{du}{\sqrt{u}} + \int \frac{1}{\sqrt{(x-1)^2 + 4}} dx \\ &= \frac{1}{2} \left( 2u^{\frac{1}{2}} \right) + \sinh^{-1} \frac{(x-1)}{2} + C \\ &\left[ \because \int \frac{1}{\sqrt{x^2 + a^2}} = \sinh^{-1} \frac{x}{a} \right] \end{aligned}$$

Where,  $C$  = constant

$$\begin{aligned} &= \sqrt{u} + \sinh^{-1} \left( \frac{(x-1)}{2} \right) + C \\ &= \sqrt{x^2 - 2x + 5} + \sinh^{-1} \left( \frac{x-1}{2} \right) + C \end{aligned}$$

---

## Question18

For  $0 < x < 1$ ,  $\int [\tan^{-1}(1 - x + x^2) + \tan^{-1}(1 - x)] dx =$

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Options:

A.

$$x \cot^{-1} x + \log \sqrt{1 + x^2} + C$$

B.

$$x \tan^{-1} x - \log(1 + x^2) + C$$

C.

$$x \cot^{-1} x + \frac{3}{4} \log(1 + x^2) + C$$

D.

$$x \tan^{-1} x - \frac{3}{4} \log \sqrt{1 + x^2} + C$$

**Answer: A**

**Solution:**

For  $0 < x < 1$ ,

$$\begin{aligned} I &= \int [\tan^{-1}(1 - x + x^2) + \tan^{-1}(1 - x)] dx \\ &= \int \tan^{-1} \left[ \frac{(1 - x + x^2) + (1 - x)}{1 - (1 - x + x^2)(1 - x)} \right] dx \\ &= \int \tan^{-1} \left( \frac{2 - 2x + x^2}{x(2 - 2x + x^2)} \right) dx \\ &= \int \tan^{-1} \left( \frac{1}{x} \right) dx \\ &= \int \cot^{-1}(x) dx = \int \left( \frac{\pi}{2} - \tan^{-1} x \right) dx \\ &= \int \frac{\pi}{2} dx - \int \tan^{-1} x dx \\ &= \frac{\pi}{2} x - \left[ x \tan^{-1} x - \frac{1}{2} \ln(1 + x^2) + C \right] \end{aligned}$$

[Using product rule]

$$= \frac{\pi}{2} x - x \tan^{-1} x + \frac{1}{2} \ln(1 + x^2) + C$$

where  $C$  = constant of integration

$$\begin{aligned}
&= x \left( \frac{\pi}{2} - \tan^{-1} x \right) + \frac{1}{2} \ln(1 + x^2) + C \\
&= x \cot^{-1} x + \frac{1}{2} \ln(1 + x^2) + C \\
&= x \cot^{-1} x + \log \sqrt{1 + x^2} + C
\end{aligned}$$


---

## Question19

If  $\frac{3x+1}{(x-1)(x^2+2)} = \frac{A}{x-1} + \frac{Bx+C}{x^2+2}$ , then  $5(A - B) =$

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Options:

A.

$$A + C$$

B.

$$8C$$

C.

$$C + 8$$

D.

$$\frac{C}{8}$$

**Answer: B**

**Solution:**

$$\begin{aligned}
\because \frac{3x+1}{(x-1)(x^2+2)} &= \frac{A}{x-1} + \frac{Bx+C}{x^2+2} \\
\Rightarrow 3x+1 &= A(x^2+2) + (Bx+C)(x-1) \\
\Rightarrow 3x+1 &= (A+B)x^2 + (C-B)x + (2A-C)
\end{aligned}$$

Comparing both sides, we get

$$\begin{aligned}
&A + B = 0 \\
\Rightarrow A &= -B \quad \dots (i)
\end{aligned}$$

$$\text{and } C - B = 3 \quad \dots (ii)$$

$$\text{and } 2A - C = 1 \quad \dots (iii)$$

put the value of  $A$  from Eq. (i) to Eq. (iii)

$$\begin{aligned}
\Rightarrow -2B - C &= 1 \\
\Rightarrow C &= -2B - 1 \quad \dots (iv)
\end{aligned}$$

By Eqs. (ii) and (iv),

We get

$$(-2B - 1) - B = 3 \Rightarrow B = -4/3$$

$$\text{and } A = -(-4/3) = 4/3$$

$$\begin{aligned}\text{Also, } C &= -2B - 1 = -2(-4/3) - 1 \\ &= \frac{8}{3} - 1 = 5/3\end{aligned}$$

$$\begin{aligned}\text{thus, } 5(A - B) &= 5\left(\frac{4}{3} - (-4/3)\right) \\ &= 5\left(\frac{8}{3}\right) = \frac{40}{3}\end{aligned}$$

$$\text{and } 8C = 8 \times \frac{5}{3} = \frac{40}{3}$$

$$\text{Therefore, } 5(A - B) = 8C$$

---

## Question20

$$\int \frac{\sec^2 x}{\sin^7 x} dx - \int \frac{7}{\sin^7 x} dx =$$

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Options:

A.

$$\frac{1}{\sin^6 x \cos x} + C$$

B.

$$\frac{\tan x}{\sin^8 x} + C$$

C.

$$\sin^8 x \cos x + C$$

D.

$$\sec x \tan^7 x + C$$

**Answer: A**

**Solution:**

$$\text{Let } I = \int \frac{\sec^2 x - 7}{\sin^7 x} dx$$

$$\Rightarrow I = \int \frac{1 - 7 \cos^2 x}{\sin^7 x \cos^2 x} dx$$

$$\Rightarrow I = \int \frac{\sin^5 x (1 - 7 \cos^2 x) dx}{\sin^{12} x \cos^2 x}$$

Put  $\sin^6 x \cos x = t$

$$\begin{aligned}(6 \sin^5 x \cos^2 x - \sin^7 x) dx &= dt \\ \Rightarrow \sin^5 x (6 \cos^2 x - \sin^2 x) dx &= dt \\ \Rightarrow \sin^5 x (6 \cos^2 x - 1 + \cos^2 x) dx &= dt \\ \Rightarrow \sin^5 x (1 - 7 \cos^2 x) dx &= -dt \\ \therefore I &= \int \frac{-dt}{t^2} \\ \Rightarrow I &= \frac{1}{t} + C \\ \Rightarrow I &= \frac{1}{\sin^6 x \cos x} + C\end{aligned}$$

---

## Question21

If  $\int (x^6 + x^4 + x^2) \sqrt{2x^4 + 3x^2 + 6} dx = f(x) + c$ , then  $f(3) =$

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**Options:**

A.

$$\frac{3}{2}(95)^{3/2}$$

B.

$$\frac{3}{2}(195)^{3/2}$$

C.

$$\frac{3}{2}(265)^{3/2}$$

D.

$$\frac{3}{2}(175)^{3/2}$$

**Answer: B**

**Solution:**

$$\begin{aligned}
& \int (x^6 + x^4 + x^2) \sqrt{2x^4 + 3x^2 + 6} dx \\
&= \int x (x^5 + x^3 + x) \sqrt{2x^4 + 3x^2 + 6} dx \\
&= \int (x^5 + x^3 + x) \sqrt{2x^6 + 3x^4 + 6x^2} dx \\
&\text{Put } 2x^6 + 3x^4 + 6x^2 = t \\
&\Rightarrow 12 (x^5 + x^3 + x) dx = dt \\
&= \frac{1}{12} \int \sqrt{t} \cdot dt = \frac{1}{12} \frac{t^{\frac{3}{2}}}{\left(\frac{3}{2}\right)} + C \\
&= \frac{t^{\frac{3}{2}}}{18} + C = \frac{1}{18} (2x^6 + 3x^4 + 6x^2)^{\frac{3}{2}} + C \\
&\text{So, } f(x) = \frac{1}{18} (2x^6 + 3x^4 + 6x^2)^{3/2} \\
&\Rightarrow f(3) = \frac{1}{18} (9(195))^{\frac{3}{2}} = \frac{3}{2} (195)^{\frac{3}{2}}
\end{aligned}$$


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## Question22

$$\int \frac{dx}{(x+1)\sqrt{x^2+1}} =$$

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**Options:**

A.

$$\frac{1}{\sqrt{2}} \sinh^{-1} \left( \frac{1+x}{1-x} \right) + C$$

B.

$$\frac{1}{\sqrt{2}} \sinh^{-1} \left( \frac{1-x}{1+x} \right) + C$$

C.

$$-\frac{1}{\sqrt{2}} \sinh^{-1} \left( \frac{1-x}{1+x} \right) + C$$

D.

$$-\frac{1}{\sqrt{2}} \sinh^{-1} \left( \frac{1+x}{1-x} \right) + C$$

**Answer: C**

**Solution:**

$$\text{Let } I = \int \frac{dx}{(x+1)\sqrt{x^2+1}}$$

$$\text{Put } x+1 = \frac{1}{t} \Rightarrow dx = -\frac{1}{t^2} dt$$

$$x = \frac{1}{t} - 1$$

$$\therefore I = \int \frac{-\frac{1}{t^2} dt}{\frac{1}{t} \sqrt{\left(\frac{1}{t} - 1\right)^2 + 1}}$$

$$I = \int \frac{-dt}{\sqrt{2t^2 - 2t + 1}}$$

$$I = \frac{-1}{\sqrt{2}} \int \frac{dt}{\sqrt{\left(t - \frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2}}$$

$$I = -\frac{1}{2} \sinh^{-1} 2 \left(t - \frac{1}{2}\right) + C$$

$$I = -\frac{1}{\sqrt{2}} \sinh^{-1} \left(\frac{1-x}{1+x}\right) + C$$

## Question23

$$\text{If } \int \frac{dx}{2 \cos x + 3 \sin x + 4} = \frac{2}{\sqrt{3}} f(x) + C, \text{ then } f\left(\frac{2\pi}{3}\right) =$$

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Options:

A.

$$\frac{\pi}{12}$$

B.

$$\frac{\pi}{8}$$

C.

$$\frac{5\pi}{12}$$

D.

$$\frac{5\pi}{8}$$

**Answer: C**

**Solution:**

$$\int \frac{dx}{2 \cos x + 3 \sin x + 4}$$

Let  $t = \tan \frac{x}{2} \Rightarrow \sin x = \frac{2t}{1+t^2}$

$$\cos x = \frac{1-t^2}{1+t^2}$$

and  $dx = \frac{2dt}{1+t^2}$

$$= \int \frac{2dt}{(1+t^2) \left[ \frac{2(1-t^2)}{(1+t^2)} + \frac{6t}{(1+t^2)} + 4 \right]}$$

$$= \int \frac{2dt}{2-2t^2+6t+4+4t^2}$$

$$= \int \frac{dt}{t^2+3t+3} = \int \frac{dt}{\left(t+\frac{3}{2}\right)^2 + \frac{3}{4}}$$

put  $u = t + \frac{3}{2} \Rightarrow du = dt$

$$= \int \frac{1}{u^2 + \left(\frac{\sqrt{3}}{2}\right)^2} du$$

$$= \frac{1}{\left(\frac{\sqrt{3}}{2}\right)} \tan^{-1} \left( \frac{u}{\left(\frac{\sqrt{3}}{2}\right)} \right) + C$$

$$= \frac{2}{\sqrt{3}} \tan^{-1} \left( \frac{2u}{\sqrt{3}} \right) + C$$

$$= \frac{2}{\sqrt{3}} \tan^{-1} \left( \frac{2t+3}{\sqrt{3}} \right) + C$$

$$= \frac{2}{\sqrt{3}} \tan^{-1} \left( \frac{2 \tan \frac{x}{2} + 3}{\sqrt{3}} \right) + C$$

Thus,  $f(x) = \tan^{-1} \left( \frac{2 \tan \left(\frac{x}{2}\right) + 3}{\sqrt{3}} \right)$

$$\therefore f\left(\frac{2\pi}{3}\right) = \tan^{-1} \left( \frac{2 \tan \left(\frac{\pi}{3}\right) + 3}{\sqrt{3}} \right)$$

$$= \tan^{-1} \left( \frac{2\sqrt{3} + 3}{\sqrt{3}} \right)$$

$$= \tan^{-1}(2 + \sqrt{3}) = \frac{5\pi}{12}$$

## Question24

If  $\int \frac{1}{((x+4)^3(x+1)^5)^{1/4}} dx = A \cdot \left(\frac{x+4}{x+1}\right)^n + C$

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**Options:**

A.

n A=3

B.

$$n + \frac{1}{A} = -\frac{1}{2}$$

C.

$$A + n = 1$$

D.

$$A = n$$

**Answer: B**

**Solution:**

$$\therefore \int \frac{1}{((x+4)^3(x+1)^5)^{\frac{1}{4}}} dx$$

$$= A \cdot \left( \frac{x+4}{x+1} \right)^n + C$$

$$= \int \frac{1}{(x+1)^2 \left( \frac{x+4}{x+1} \right)^{3/2}} dx$$

$$\text{Let } t = \frac{x+4}{x+1}$$

$$\Rightarrow dt = \frac{-3}{(x+1)^2} dx = - \int \frac{1}{3} t^{-3/4} dt$$

$$= -\frac{1}{3} \left[ \frac{\frac{-3}{4} + 1}{\frac{-3}{4} + 1} \right] + C$$

$$= -\frac{1}{3} [4t^{1/4}] + C$$

$$= -\frac{4}{3} \left( \frac{x+4}{x+1} \right)^{1/4} + C$$

$$\text{Comparing with } A = \frac{(x+4)^n}{x+1} + C$$

$$A = -\frac{4}{3}, n = \frac{1}{4}$$

$$\text{Therefore, } n + \frac{1}{A} = \frac{1}{4} + \frac{1}{(-\frac{4}{3})}$$

$$= \frac{1}{4} - \frac{3}{4}$$

$$= \frac{-2}{4} = \frac{-1}{2}$$

---

## Question25

$$\int \frac{x+1}{x^3-1} dx =$$

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**Options:**

A.

$$\frac{1}{3} \log \left( \frac{x+1}{x^2+x+1} \right) + C$$

B.

$$\frac{1}{3} \log \left( \frac{(x-1)^2}{x^2+x+1} \right) + C$$

C.

$$\frac{1}{3} \log \left( \frac{x-1}{x^2+x+1} \right) + C$$

D.

$$\frac{1}{3} \log \left( \frac{(x+1)^2}{x^2-x+1} \right) + C$$

**Answer: B**

**Solution:**

$$\begin{aligned} \text{Let } I &= \int \frac{x+1}{x^3-1} dx \\ &= \int \frac{2}{3(x-1)} dx + \int \frac{-2x-1}{3(x^2+x+1)} dx \\ &= \frac{2}{3} \ln |x-1| - \frac{1}{3} \ln |x^2+x+1| + C \end{aligned}$$

where  $C$  is constant of integration.

$$= \frac{1}{3} \ln \left| \frac{(x-1)^2}{x^2+x+1} \right| + C$$

---

## Question 26

$$\int \frac{x^4 - 16x^2 + 2x + 8}{x^3 - 4x^2 + 2} dx =$$

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**Options:**

A.

$$\frac{x^2 + 8x + C}{2}$$

B.

$$x^2 + 8x + C$$

C.

$$x^3 - 4x + C$$

D.

$$\frac{x^2 - 8x + C}{2}$$

**Answer: A**

**Solution:**

$$\begin{aligned} \text{Let } I &= \int \frac{x^4 - 16x^2 + 2x + 8}{x^3 - 4x^2 + 2} dx \\ &= \int \frac{(x + 4)(x^3 - 4x^2 + 4)}{(x^3 - 4x^2 + 2)} dx \\ &= \int (x + 4) dx = \frac{x^2}{2} + 4x + C_1 \\ &= \frac{x^2 + 8x + 2C_1}{2} \\ &= \frac{x^2 + 8x + C}{2} \end{aligned}$$

where  $C_1$  is constant of integration  $C = 2C_1$ .

-----

## Question 27

$$\int \frac{\sec^2 x}{(\sec x + \tan x)^{\frac{5}{2}}} dx =$$

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**Options:**

A.

$$-\frac{(\sec x + \tan x)^{\frac{5}{2}}}{5} - \frac{(\sec x + \tan x)^{\frac{7}{2}}}{7} + C$$

B.

$$-\frac{(\sec x - \tan x)^{\frac{5}{2}}}{5} - \frac{(\sec x - \tan x)^{\frac{7}{2}}}{7} + C$$

C.

$$-\frac{(\sec x + \tan x)^{\frac{3}{2}}}{3} - \frac{(\sec x + \tan x)^{\frac{7}{2}}}{7} + C$$

D.

$$-\frac{(\sec x - \tan x)^{\frac{3}{2}}}{3} - \frac{(\sec x - \tan x)^{\frac{7}{2}}}{7} + C$$

**Answer: D**

**Solution:**

$$\text{Let } I = \int \frac{\sec^2 x}{(\sec x + \tan x)^{\frac{5}{2}}} dx$$

$$\text{Put } \sec x + \tan x = t$$

$$\therefore \sec x - \tan x = \frac{1}{t}$$

$$\therefore \sec x = \frac{1}{2} \left( t + \frac{1}{t} \right) \text{ and } \tan x = \frac{1}{2} \left( t - \frac{1}{t} \right)$$

Differentiate w.r.t.  $x$ , we get

$$\Rightarrow (\sec x \tan x + \sec^2 x) dx = dt$$

$$\Rightarrow dx = \frac{dt}{\sec x \tan x + \sec^2 x}$$

$$I = \int \frac{\sec^2 x}{t^{\frac{5}{2}}} \times \frac{dt}{\sec x \tan x + \sec^2 x}$$

$$= \int \frac{\sec x}{t^{\frac{5}{2}}} \times \frac{dt}{(\sec x + \tan x)}$$

$$= \int \left( \frac{1}{2} \left( t + \frac{1}{t} \right) \right) \cdot \frac{1}{t^{\frac{5}{2}}} \times \frac{1}{t} dt$$

$$= \frac{1}{2} \int \left( t + \frac{1}{t} \right) t^{-\frac{7}{2}} dt$$

$$= \frac{1}{2} \left[ \int t^{-\frac{5}{2}} dt + \int t^{-\frac{9}{2}} dt \right]$$

$$= \frac{1}{2} \left[ \frac{t^{-\frac{3}{2}}}{-\frac{3}{2}} + \frac{t^{-\frac{7}{2}}}{-\frac{7}{2}} \right] + C$$

$$= \frac{1}{2} \left[ \frac{-2}{3} \times \frac{1}{t^{\frac{3}{2}}} - \frac{2}{7} \frac{1}{t^{\frac{7}{2}}} \right] + C$$

$$= \frac{-1}{3} (\sec x - \tan x)^{\frac{3}{2}} - \frac{1}{7} (\sec x - \tan x)^{\frac{7}{2}} + C$$

## Question 28

$$\int \frac{1}{\cos x} \left[ \frac{1}{\sin x} - \frac{1}{\sin x + 3 \cos x} \right] dx =$$

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Options:

A.

$$\frac{1}{3} \log \left| \frac{\sin x}{\sin x + 3 \cos x} \right| + C$$

B.

$$\log \left| \frac{\cos x}{\sin x + 3 \cos x} \right| + c$$

C.

$$\frac{1}{3} \log \left| \frac{\cos x}{\sin x + 3 \cos x} \right| + C$$

D.

$$\log \left| \frac{\sin x}{\sin x + 3 \cos x} \right| + c$$

**Answer: D**

**Solution:**

$$\begin{aligned} \text{Let } I &= \int \frac{1}{\cos x} \left[ \frac{1}{\sin x} - \frac{1}{\sin x + 3 \cos x} \right] dx \\ &= \int \frac{2}{\sin 2x} dx - \int \frac{\sec^2 x}{\tan x + 3} dx \\ &= 2 \int \operatorname{cosec} 2x dx - \int \frac{dz}{z + 3}, \text{ where} \\ z &= \tan x \\ &= \frac{2 \log_e |\operatorname{cosec} 2x - \cot 2x|}{2} - \ln |z + 3| + C \\ &= \ln |\operatorname{cosec} 2x - \cot 2x| - \ln |\tan x + 3| + C \\ &= \ln \left| \frac{\frac{1}{\sin 2x} - \frac{\cos 2x}{\sin 2x}}{\frac{\sin x}{\cos x} + 3} \right| + C \\ &= \ln \left| \frac{\frac{1 - \cos 2x}{\sin 2x}}{\frac{\sin x + 3 \cos x}{\cos x}} \right| + C \\ &= \ln \left| \frac{2 \sin^2 x}{2 \sin x \cdot \cos x} \times \frac{\cos x}{\sin x + 3 \cos x} \right| + C \\ &= \ln \left| \frac{\sin x}{\sin x + 3 \cos x} \right| + C \end{aligned}$$


---

## Question29

$$\int \cos^{-1} \left( \frac{1-x^2}{1+x^2} \right) dx =$$

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**Options:**

A.

$$2 \left[ x \tan^{-1} x - \log \sqrt{1+x^2} \right] + C$$

B.

$$2x \tan^{-1} x + \log \sqrt{1+x^2} + C$$

C.

$$x \tan^{-1} x + \log \sqrt{1-x^2} + C$$

D.

$$2 \left[ \tan^{-1} x - \log \sqrt{1+x^2} \right] + C$$

**Answer: A**

**Solution:**

$$\begin{aligned} \text{Let } I &= \int \cos^{-1} \left( \frac{1-x^2}{1+x^2} \right) dx \\ &= \int 2 \tan^{-1} x dx = 2 \int (\tan^{-1} x) \cdot 1 dx \\ &= 2 \left[ x \tan^{-1} x - \int \frac{x}{1+x^2} dx \right] \\ &= 2 \left[ x \tan^{-1} x - \frac{1}{2} \int \frac{2x}{1+x^2} dx \right] \\ &= 2 \left[ x \tan^{-1} x - \ln(1+x^2) \right] + C \end{aligned}$$

### Question30

**If  $\frac{ax+5}{(x^2+b)(x+3)} = \frac{x+21}{12(x^2+b)} + \frac{c}{12(x+3)}$ , then  $b^2 =$**

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### Options:

A.

$$a^3 - c$$

B.

$$a^2 + c$$

C.

$a - c$

D.

$$a + c$$

**Answer: A**

**Solution:**

$$\begin{aligned} \frac{ax+5}{(x^2+b)(x+3)} &= \frac{x+21}{12(x^2+b)} + \frac{c}{12(x+3)} \\ \Rightarrow 12(ax+5) &= (x+21)(x+3) + c(x^2+b) \\ \Rightarrow 12ax+60 &= x^2+3x+21x+63+cx^2+bc \\ \Rightarrow 12ax+60 &= (1+c)x^2+24x+63+bc \end{aligned}$$

Comparing coefficient

$$\Rightarrow 1 + c = 0 \quad 12a = 24 \quad 63 + bc = 60$$

$$c = -1 \quad a = 2 \quad 63 - b = 60$$

$$\Rightarrow b = 3$$

$$b^2 = 9 = a^3 - c$$

$$\therefore b^2 = a^3 - c$$

---

## Question31

$$\int \left( \sum_{r=0}^{\infty} \frac{x^r 2^r}{r!} \right) dx =$$

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Options:

A.

$$e^x + C$$

B.

$$\frac{-2}{1-2x} + C$$

C.

$$2e^{2x} + C$$

D.

$$\frac{e^{2x}}{2} + C$$

**Answer: D**

**Solution:**

$$\begin{aligned} & \int \left( \sum_{r=0}^{\infty} \frac{x^r 2^r}{r!} \right) dx \\ &= \int \left( 1 + \frac{(2x)^1}{1!} + \frac{(2x)^2}{2!} + \frac{(2x)^3}{3!} + \dots \right) dx \\ &= \int e^{2x} dx = \frac{e^{2x}}{2} + C \end{aligned}$$

---

## Question32

$$\int \frac{dx}{12 \cos x + 5 \sin x} =$$

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Options:

A.

$$\frac{1}{13} \log \left| \tan \left( \frac{\pi}{4} + \frac{x}{2} - \frac{1}{2} \tan^{-1} \frac{5}{12} \right) \right| + C$$

B.

$$\frac{5}{12} \log \left| \tan \left( \frac{\pi}{4} + \frac{x}{2} - \frac{1}{2} \tan^{-1} \frac{5}{12} \right) \right| + C$$

C.

$$\frac{1}{13} \log \left| \tan \left( \frac{\pi}{4} + \frac{x}{2} + \frac{1}{2} \tan^{-1} \frac{5}{12} \right) \right| + C$$

D.

$$\frac{5}{12} \log \left| \tan \left( \frac{\pi}{4} + \frac{x}{2} + \frac{1}{2} \tan^{-1} \frac{5}{12} \right) \right| + C$$

**Answer: A**

**Solution:**

$$\begin{aligned} & \int \frac{1}{12 \cos x + 5 \sin x} dx \\ R &= \sqrt{12^2 + 5^2} = \sqrt{169} = 13 \\ \tan \alpha &= \frac{5}{12} \\ &= \int \frac{1}{13 \left( \frac{12}{13} \cos x + \frac{5}{13} \sin x \right)} dx \\ &= \frac{1}{13} \int \frac{1}{\cos(x - \alpha)} dx \\ &= \frac{1}{13} \log \left| \tan \left( \frac{\pi}{4} + \frac{x}{2} - \frac{1}{2} \tan^{-1} \frac{5}{12} \right) \right| + C \end{aligned}$$

---

## Question33

If  $\int \frac{\cos^3 x}{\sin^2 x + \sin^4 x} dx = c - \operatorname{cosec} x - f(x)$ , then  $f\left(\frac{\pi}{2}\right) =$

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Options:

A.

B.

0

C.

$\pi/2$

D.

$\pi$

**Answer: C**

**Solution:**

$$\begin{aligned} & \int \frac{\cos^3 x}{\sin^2 x + \sin^4 x} dx \\ &= \int \frac{\cos x (1 - \sin^2 x)}{\sin^2 x (1 + \sin^2 x)} dx \end{aligned}$$

Put  $u = \sin x$

$$\begin{aligned} du &= \cos x dx = \int \frac{1 - u^2}{u^2 (1 + u^2)} du \\ &= \int \left( \frac{1}{u^2} - \frac{2}{1 + u^2} \right) du \\ &= -\frac{1}{u} - 2 \tan^{-1}(u) + C \\ &= -\frac{1}{\sin x} - 2 \tan^{-1}(\sin x) + C \\ \Rightarrow f(x) &= 2 \tan^{-1}(\sin x) \\ \Rightarrow f\left(\frac{\pi}{2}\right) &= 2 \tan^{-1}(1) = 2 \times \frac{\pi}{4} = \frac{\pi}{2} \end{aligned}$$

---

## Question34

$$\int \frac{13 \cos 2x - 9 \sin 2x}{3 \cos 2x - 4 \sin 2x} dx =$$

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**Options:**

A.

$$3x - \frac{1}{2} \log |3 \cos 2x - 4 \sin 2x| + C$$

B.

$$\frac{x}{2} - 3 \log |3 \cos 2x - 4 \sin 2x| + C$$

C.

$$3x + \frac{1}{2} \log |3 \cos 2x - 4 \sin 2x| + C$$

D.

$$x + \frac{3}{2} \log |3 \cos 2x - 4 \sin 2x| + C$$

**Answer: A**

**Solution:**

$$\begin{aligned} & \int \frac{13 \cos 2x - 9 \sin 2x}{3 \cos 2x - 4 \sin 2x} dx \\ &= \int \left( 3 - \frac{1(-6 \sin 2x - 8 \cos 2x)}{2(3 \cos 2x - 4 \sin 2x)} \right) dx \\ &= 3x - \frac{1}{2} \log |3 \cos 2x - 4 \sin 2x| + C \end{aligned}$$


---

## Question35

$$\int \sqrt{x^2 + x + 1} dx$$

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**Options:**

A.

$$\frac{(2x+1)}{4} \sqrt{x^2 + x + 1} + \frac{3}{8} \sinh^{-1} \left( \frac{2x+1}{\sqrt{3}} \right) + C$$

B.

$$\frac{x+1}{4} \sqrt{x^2 + x + 1} + \frac{3}{8} \sinh^{-1} \left( \frac{2x+1}{\sqrt{3}} \right) + C$$

C.

$$\frac{x+1}{4} \sqrt{x^2 + x + 1} - \frac{3}{8} \sinh^{-1} \left( \frac{2x+1}{\sqrt{3}} \right) + C$$

D.

$$\frac{(2x+1)}{4} \sqrt{x^2 + x + 1} - \frac{3}{8} \sinh^{-1} \left( \frac{2x+1}{\sqrt{3}} \right) + C$$

**Answer: A**

**Solution:**

$$\begin{aligned}
& \int \sqrt{x^2 + x + 1} dx \\
&= \int \sqrt{x^2 + x + \frac{1}{4} - \frac{1}{4} + 1} dx \\
&= \int \sqrt{\left(x + \frac{1}{2}\right)^2 + \frac{3}{4}} dx \\
&= \int \sqrt{\left(x + \frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} dx \\
&= \frac{x + \frac{1}{2}}{2} \sqrt{x^2 + x + 1} + \frac{3}{8} \ln \left| x + \frac{1}{2} + \sqrt{x^2 + x + 1} \right| + C \\
&= \frac{2x + 1}{4} \sqrt{x^2 + x + 1} + \frac{3}{8} \sin^{-1} \left( \frac{2x + 1}{\sqrt{3}} \right) + C
\end{aligned}$$


---

## Question 36

If  $\frac{x+1}{(x-1)^2(x^2+1)} = \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{Cx+D}{x^2+1}$ , then  $\sqrt{3A^2 + 4D^2 + 5C^2 + B^2} =$

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Options:

A.

$$\frac{3}{2}$$

B.

$$\frac{1}{2}$$

C.

$$1$$

D.

$$2$$

**Answer: D**

**Solution:**

$$\begin{aligned}
\frac{x+1}{(x-1)^2(x^2+1)} &= \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{Cx+D}{x^2+1} \\
x+1 &= A(x-1)(x^2+1) + B(x^2+1) + (Cx+D)(x-1)^2 \quad \dots (i)
\end{aligned}$$

Put  $x = 1$

$$2 = 2B \Rightarrow B = 1$$

In Eq. (i) put  $x = 0$

$$1 = -A + B + D$$

$$\Rightarrow A = D$$

... (ii)  
... (ii)

In Eq. (i) put  $x = -1$

$$0 = -4A + 2 - 4C + 4D$$

$$4C = 2 \Rightarrow C = \frac{1}{2}$$

... (iii)

In Eq. (i) put  $x = 2$

$$3 = 5A + 5 + 1 + D \Rightarrow 6A = -3$$

$$A = -\frac{1}{2} = D$$

$$\sqrt{3A^2 + 4D^2 + 5C^2 + B^2}$$

$$= \sqrt{\frac{3}{4} + \frac{4}{4} + \frac{5}{4} + 1}$$

$$= \sqrt{\frac{16}{4}} = 2$$

## Question 37

$$\int \frac{1}{9 \cos^2 x - 24 \sin x \cos x + 16 \sin^2 x} dx =$$

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Options:

A.

$$\frac{\cos x}{4(3 \cos x - 4 \sin x)} + C$$

B.

$$\frac{\sin x}{4(3 \cos x - 4 \sin x)} + C$$

C.

$$\frac{\cos x}{3 \cos x - 4 \sin x} + C$$

D.

$$\frac{\sin x}{3 \cos x - 4 \sin x} + C$$

**Answer: A**

**Solution:**

$$\begin{aligned}
 I &= \int \frac{1}{9 \cos^2 x - 24 \sin x \cos x + 16 \sin^2 x} dx = \\
 \Rightarrow I &= \int \frac{1}{(3 \cos x - 4 \sin x)^2} dx \\
 \Rightarrow I &= \int \frac{\sec^2 x}{(3 - 4 \tan x)^2} dx \\
 3 - 4 \tan x &= t \\
 -4 \sec^2 x dx &= dt \\
 \Rightarrow I &= \frac{1}{-4} \int t^{-2} dt \\
 &= \frac{1}{4} \cdot \frac{1}{t} + C = \frac{\cos x}{4(3 \cos x - 4 \sin x)} + C
 \end{aligned}$$


---

## Question 38

If  $\int \frac{1}{\cot \frac{x}{2} \cot \frac{x}{3} \cot \frac{x}{6}} dx = A \log \left| \cos \frac{x}{2} \right| + B \log \left| \cos \frac{x}{3} \right| + C \log \left| \cos \frac{x}{6} \right| + k$ , then  $A + B + C =$

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Options:

A.

7

B.

-7

C.

11

D.

1

**Answer: A**

**Solution:**

$$\begin{aligned}
& \int \frac{1}{\cot \frac{x}{2} \cot \frac{x}{3} \cot \frac{x}{6}} dx \\
&= \int \tan \frac{x}{2} \tan \frac{x}{3} \tan \frac{x}{6} dx \\
&\because \frac{x}{6} = \frac{x}{2} - \frac{x}{3} \\
&\Rightarrow \tan \frac{x}{6} = \frac{\tan \frac{x}{2} - \tan \frac{x}{3}}{1 + \tan \frac{x}{2} \tan \frac{x}{3}} \\
&\int (\tan \frac{x}{2} - \tan \frac{x}{3} - \tan \frac{x}{6}) dx \\
&= -2 \log \left| \cos \frac{x}{2} \right| + 3 \log \left| \cos \frac{x}{3} \right| \\
&\quad + 6 \log \left| \sec \frac{x}{6} \right| + k \\
&\therefore A = -2, B = 3, C = 6 \\
&\Rightarrow A + B + C = 7.
\end{aligned}$$


---

## Question39

$$\int \frac{\sin x + \cos x}{\sin x - \cos x} dx =$$

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Options:

A.

$$-x + \log |\cos x - \sin x| + C$$

B.

$$x - \log |\cos x - \sin x| + C$$

C.

$$-\log |\cos x - \sin x| + C$$

D.

$$\log |\cos x - \sin x| + C$$

**Answer: C**

**Solution:**

$$I = \int \frac{\sin x + \cos x}{\sin x - \cos x} dx \text{ put } \sin x - \cos x = t$$

$$\Rightarrow -(\cos x + \sin x) dx = dt$$

$$I = - \int \frac{1}{t} dt$$

$$= -\log |\sin x - \cos x| + C$$

$$= -\log |\cos x - \sin x| + C$$


---

## Question40

$$\int \frac{x^4-1}{x^2\sqrt{x^4+x^2+1}} dx =$$

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Options:

A.

$$\frac{2\sqrt{x^4+x^2+1}}{x} + C$$

B.

$$\frac{\sqrt{x^4+x^2+1}}{x} + C$$

C.

$$\frac{\sqrt{x^4+x^2+1}}{2x} + C$$

D.

$$\frac{4\sqrt{x^4+x^2+1}}{x} + C$$

**Answer: B**

**Solution:**

$$\begin{aligned} I &= \int \frac{x^4-1}{x^2\sqrt{x^4+x^2+1}} dx \\ \Rightarrow I &= \int \frac{x^2-\frac{1}{x^2}}{\sqrt{x^4+x^2+1}} dx \\ &= \int \frac{x-\frac{1}{x^3}}{\sqrt{x^2+\frac{1}{x^2}+1}} dx \end{aligned}$$

$$\text{Put } x^2 + \frac{1}{x^2} + 1 = t^2$$

$$\Rightarrow 2x - \frac{2}{x^3} dx = 2t dt$$

$$I = \int dt = t + C = \frac{\sqrt{x^4+x^2+1}}{x} + C$$

---

## Question41

$$\int \frac{(3x-2) \tan\left(\sqrt{9x^2-12x+1}\right)}{\sqrt{9x^2-12x+1}} dx =$$

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Options:

A.

$$\frac{1}{3} \sec^2 \sqrt{9x^2 - 12x + 1} + C$$

B.

$$\frac{1}{3} \sec^2 x + C$$

C.

$$\frac{1}{2} \log \left| \sec \sqrt{9x^2 - 12x + 1} \right| + C$$

D.

$$\frac{1}{3} \log \left| \sec \sqrt{9x^2 - 12x + 1} \right| + C$$

**Answer: D**

**Solution:**

$$I = \int \frac{(3x-2) \tan(\sqrt{9x^2-12x+1})}{\sqrt{9x^2-12x+1}} dx$$

$$\text{Put } 9x^2 - 12x + 1 = t^2$$

$$\Rightarrow (18x - 12)dx = 2tdt$$

$$\Rightarrow (3x - 2)dx = \frac{1}{3}tdt$$

$$\Rightarrow I = \frac{1}{3} \int \frac{t \tan t}{t} dt = \frac{1}{3} \int \tan t dt$$

$$= \frac{1}{3} \log |\sec t| + C$$

$$= \frac{1}{3} \log \left| \sec \sqrt{9x^2 - 12x + 1} \right| + C$$

---

## Question42

If  $\int e^{\sin x} (1 + \sec x \tan x) dx = e^{\sin x} f(x) + c$ , then in  $0 \leq x \leq 2\pi$ , then number of solutions of  $f(x) = 1$  is

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Options:

A.

0

B.

4

C.

3

D.

2

**Answer: D**

**Solution:**

Given  $\int e^{\sin x} (1 + \sec x \tan x) dx$

$$= e^{\sin x} f(x) + c$$
$$\int e^{\sin x} \left( \frac{1}{\cos x} + \frac{\sin x}{\cos^3 x} \right) \cos x dx$$

Put  $\sin x = t$

$$\cos x dx = dt$$

$$\cos x = \sqrt{1 - t^2}$$

$$\int e^t \left[ \frac{1}{\sqrt{1-t^2}} + \frac{t}{(1-t^2)^{3/2}} \right] dt$$

[Using result  $\int e^x [f(x) + f'(x)] dx$

$$= e^x f(x) + C]$$
$$\int e^t \left[ \frac{1}{\sqrt{1-t^2}} + \left( \frac{1}{\sqrt{1-t^2}} \right)^1 \right] dt = e^t \cdot \frac{1}{\sqrt{1-t^2}}$$
$$= e^{\sin x} \cdot \frac{1}{\cos x} = e^{\sin x} \cdot \sec x + C$$

Comparing with  $e^{\sin x} f(x) + C$

$$\therefore f(x) = \sec x = 1$$
$$x = 0, 2\pi$$

$\therefore$  Number of solution is 2 .

---

## Question43

If  $\int \frac{dx}{(x-1)^{\frac{3}{2}}(x-3)^{\frac{1}{2}}} = \sqrt{f(x)} + C$ , then  $f(-1) - f(0) =$

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**Options:**

A.

-3

B.

-4

C.

-2

D.

-1

**Answer: D**

**Solution:**

$$\int \frac{dx}{(x-1)^{3/2}(x-3)^{1/2}}$$

Let  $I = \int \frac{dx}{(x-1)^2 \left(\frac{x-3}{x-1}\right)^{\frac{1}{2}}}$

Put  $\frac{x-3}{x-1} = t^2$

$$\frac{x-1-x+3}{(x-1)^2} dx = 2t dt$$

$$\frac{1}{(x-1)^2} dx = t dt$$

$$I = \int \frac{t dt}{t} = \int dt = t = \sqrt{\frac{x-3}{x-1}} + C$$

$$\therefore f(x) = \frac{x-3}{x-1}$$

$$f(-1) - f(0) = \frac{-4}{-2} - 3 = 2 - 3 = -1$$

---

## Question44

$$\int \frac{x}{(1-x^2)\sqrt{2-x^2}} dx =$$

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**Options:**

A.

$$\log \left| \frac{\sqrt{2-x^2}+1}{\sqrt{2-x^2}-1} \right| + C$$

B.

$$\frac{1}{2} \log \left| \frac{\sqrt{2-x^2}}{1-x^2} \right| + C$$

C.

$$\frac{1}{2} \log \left| \frac{1+\sqrt{2-x^2}}{1-\sqrt{2-x^2}} \right| + C$$

D.

$$\log \left| \frac{1-x^2}{\sqrt{2-x^2}} \right| + C$$

**Answer: C**

**Solution:**

$$\text{Let } I = \int \frac{x}{(1-x^2)\sqrt{2-x^2}} dx$$

$$\text{Put } 2-x^2 = t^2$$

$$\therefore -x dx = t dt \text{ and } x^2 = 2-t^2$$

$$\Rightarrow I = \int \frac{-t dt}{(t^2-1)t}$$

$$\begin{aligned} \Rightarrow I &= \int \frac{dt}{1-t^2} = \frac{1}{2} \ln \left( \frac{1+t}{1-t} \right) \\ &= \frac{1}{2} \ln \left( \frac{1+\sqrt{2-x^2}}{1-\sqrt{2-x^2}} \right) + C \end{aligned}$$

## Question45

$$\int \left( \frac{1+x+\sqrt{x+x^2}}{\sqrt{x}+\sqrt{1+x}} \right) dx =$$

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**Options:**

A.

$$\frac{1}{2} \sqrt{1+x} + C$$

B.

$$\frac{2}{3} (1+x)^{\frac{3}{2}} + C$$

C.

$$\sqrt{1+x} + C$$

D.

$$2(1+x)^{\frac{3}{2}} + C$$

**Answer: B**

**Solution:**

$$\begin{aligned} I &= \int \frac{1+x+\sqrt{x+x^2}}{\sqrt{x}+\sqrt{1+x}} dx \\ &= \int \frac{(\sqrt{1+x})^2 + \sqrt{x(1+x)}}{\sqrt{x}+\sqrt{1+x}} dx \\ &= \int \frac{\sqrt{1+x}(\sqrt{1+x} + \sqrt{x})}{\sqrt{x}+\sqrt{1+x}} dx \\ I &= \int \sqrt{1+x} dx = \frac{2}{3}(1+x)^{3/2} + C \end{aligned}$$

---

## Question46

If  $\int x^2 \cos^2 x dx = \frac{1}{6} f(x) + g(x) \sin 2x + h(x) \cos 2x + c$ , then  $f(1) + g(2) + h\left(\frac{1}{2}\right) =$

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**Options:**

A.

0

B.

2

C.

1

D.

-1

**Answer: B**

**Solution:**

Given,  $\int x^2 \cos^2 x dx = \frac{1}{6} f(x) + g(x) \sin 2x + h(x) \cos 2x + C$

LHS

$$I = \int x^2 \cos^2 x dx = \int x^2 \left( \frac{1 + \cos 2x}{2} \right) dx$$

$$\begin{aligned} I &= \frac{1}{2} \int (x^2 + x^2 \cos 2x) dx \\ &= \frac{1}{2} \int x^2 dx + \frac{1}{2} \int x^2 \cos 2x dx \\ &= \frac{1}{2} \cdot \frac{x^3}{3} + \frac{1}{2} I_1 \end{aligned}$$

$$\text{Where } I_1 = \int x^2 \cos^2 x dx$$

$$\begin{aligned} I_1 &= \frac{x^2}{2} \sin 2x + \frac{1}{2} x \cos 2x - \frac{1}{4} \sin 2x + C \\ \Rightarrow I &= \frac{x^3}{6} + \frac{1}{2} \left\{ \sin 2x \left( \frac{x^2}{2} - \frac{1}{4} \right) + \frac{x}{2} \cos 2x \right\} \\ &= \frac{x^3}{6} + \left( \frac{x^2}{4} - \frac{1}{8} \right) \sin 2x + \frac{x}{4} \cos 2x \\ &= \frac{1}{6} f(x) + g(x) \sin 2x + h(x) \cos 2x + C \\ \therefore f(x) &= x^3, g(x) = \frac{x^2}{4} - \frac{1}{8} \text{ and } h(x) = \frac{x}{4} \\ \Rightarrow f(1) &= 1, g(2) = \frac{7}{8} h\left(\frac{1}{2}\right) = \frac{1}{8} \\ \therefore f(1) + g(2) + h\left(\frac{1}{2}\right) &= 1 + \frac{7}{8} + \frac{1}{8} = 2 \end{aligned}$$

## Question47

$$\int \frac{e^{\sin x} (\sin 2x - 8 \cos x)}{2(\sin x - 3)^2} dx =$$

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**Options:**

A.

$$e^{\sin x} (\sin x - 3) + C$$

B.

$$\frac{e^{\sin x}}{(\sin x - 3)^2} + C$$

C.

$$e^{\sin x} (\sin x - 3)^2 + C$$

D.

$$\frac{e^{\sin x}}{\sin x - 3} + C$$

**Answer: D**

## Solution:

We have,

$$I = \int \frac{e^{\sin x} (\sin 2x - 8 \cos x)}{2(\sin x - 3)^2} dx$$
$$\Rightarrow I = \int \frac{e^{\sin x} (2 \sin x - 8) \cos x}{2(\sin x - 3)^2} dx$$

Put  $\sin x = t \Rightarrow \cos x dx = dt$

$$\Rightarrow I = \int \frac{e^t (t-4)}{(t-3)^2} dt$$
$$I = \int e^t \left( \frac{1}{t-3} - \frac{1}{(t-3)^2} \right) dt$$
$$[\because \int e^x [f(x) + f'(x)] dx = e^x f(x) + C]$$
$$\Rightarrow I = e^t \cdot \frac{1}{t-3} + C$$
$$\Rightarrow I = e^{\sin x} \frac{1}{\sin x - 3} + C$$

---

## Question48

If  $\int \left( 3t^2 \sin \frac{1}{t} - t \cos \frac{1}{t} \right) dt = f(t) \sin \left( \frac{1}{t} \right) + C$  then  $f(2) =$

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Options:

A.

2

B.

-12

C.

8

D.

-16

**Answer: C**

## Solution:

We have,

$$\int \left( 3t^2 \sin \frac{1}{t} - t \cos \frac{1}{t} \right) dt = f(t) \sin \frac{1}{t} + C$$

$$\int \sin \frac{1}{t} \cdot 3t^2 dt - \int t \cos \frac{1}{t} dt = f(t) \sin \frac{1}{t} + C$$

$$\sin \frac{1}{t} \cdot t^3 - \int \cos \left( \frac{1}{t} \right) \left( -\frac{1}{t^2} \right) t^3 dt - \int t \cos \frac{1}{t} dt$$

$$\Rightarrow t^3 \sin \frac{1}{t} + \int t \cos \frac{1}{t} dt - \int t \cos \frac{1}{t} dt = f(t) \sin \frac{1}{t} + C$$

$$\Rightarrow t^3 \sin \frac{1}{t} = f(t) \sin \frac{1}{t} + C$$

$$\therefore f(t) = t^3$$

$$f(2) = 2^3 = 8$$

---

## Question49

$$\int (\log x)^3 x^4 dx =$$

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**Options:**

A.  $x^5 \left[ \frac{1}{5} (\log x)^3 - \frac{3}{25} (\log x)^2 + \frac{6}{125} \log x - \frac{6}{625} \right] + C$

B.  $x^5 \left[ \frac{1}{5} (\log x)^3 - \frac{2}{25} (\log x)^2 + \frac{6}{125} \log x - \frac{12}{125} \right] + C$

C.

$$x^5 \left[ \frac{1}{5} (\log x)^3 - \frac{4}{25} (\log x)^2 - \frac{9}{125} \log x - \frac{8}{125} \right] + C$$

D.

$$x^5 \left[ \frac{1}{5} (\log x)^3 + \frac{3}{25} (\log x)^2 - \frac{6}{125} \log x - \frac{6}{125} \right] + C$$

**Answer: A**

**Solution:**

$$\text{Let } I = \int (\log x)^3 \cdot x^4 dx$$

$$I = \int t^3 e^{4t} \cdot e^t dt \quad \text{Put } \log x = t$$

$$x = e^t$$

$$dx = e^t dt$$

$$I = \int t^3 e^{5t} dt$$

$$\Rightarrow I = e^{5t} \left[ \frac{t^3}{5} - \frac{3t^2}{25} + \frac{6t}{125} - \frac{6}{625} \right] + C$$

$$\Rightarrow I = x^5$$

$$\left[ \frac{1}{5} (\log x)^3 - \frac{3}{25} (\log x)^2 + \frac{6}{125} (\log x) - \frac{6}{625} \right] + C$$

## Question50

$$\int \frac{\sin 2x}{\sin^2 x + 3 \cos x - 3} dx$$

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**Options:**

A.

$$2 \log \left| \frac{\cos x - 2}{\cos x - 1} \right| + C$$

B.

$$\log \left( \frac{(\cos x - 2)^2}{(\cos x - 1)^4} \right) + C$$

C.

$$\log \left( \frac{(\cos x - 2)^2}{|\cos x - 1|} \right) + C$$

D.

$$\log \left( \frac{(\cos x - 2)^4}{(\cos x - 1)^2} \right) + C$$

**Answer: D**

**Solution:**

$$\begin{aligned}
\text{Let } I &= \int \frac{\sin 2x}{\sin^2 x + 3 \cos x - 3} dx \\
&= - \int \frac{\sin 2x}{1 - \cos^2 x + 3 \cos x - 3} dx \\
&= \int \frac{2 \sin x \cdot \cos x}{-\cos^2 x + 3 \cos x - 2} dx \\
&= \int \frac{2t(-dt)}{-t^2 + 3t - 2} \quad \text{Put } \cos x = t \\
\sin x dx &= -dt \\
&= \int \frac{2t dt}{t^2 - 3t + 2} = \int \frac{2t}{(t-2)(t-1)} dt \\
&= 2 \int \left( \frac{2}{t-2} - \frac{1}{t-1} \right) dt \\
&= 2[2 \ln(t-2) - \ln(t-1)] \\
&= 2 \ln \left( \frac{(t-2)^2}{t-1} \right) = \ln \frac{(t-2)^4}{(t-1)^2} + C \\
&= \ln \left( \frac{(\cos x - 2)^4}{(\cos x - 1)^2} \right) + C
\end{aligned}$$


---

## Question 51

If  $\int \frac{dx}{\sin^3 x + \cos^3 x} = A \log \left| \frac{\sqrt{2}+t}{\sqrt{2}-t} \right| + B \tan^{-1}(t) + C$ , then  $\left( \frac{B}{A}, t \right) =$

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Options:

A.

$$(3\sqrt{2}, \sin x - \cos x)$$

B.

$$(2\sqrt{2}, \sin x - \cos x)$$

C.

$$\left( \frac{\sqrt{2}}{3}, \sin x - \cos x \right)$$

D.

$$\left( \frac{3}{\sqrt{2}}, \sin x + \cos x \right)$$

**Answer: B**

**Solution:**

$$\begin{aligned}
\text{Let } I &= \int \frac{1}{\sin^3 x + \cos^3 x} dx \\
&= \int \frac{1}{(\sin x + \cos x)(\sin^2 x + \cos^2 x - \sin x \cos x)} dx \\
&= \int \frac{\sin x + \cos x}{(\sin x + \cos x)^2 (1 - \sin x \cos x)} dx \\
&= \int \frac{\sin x + \cos x}{(1 + 2 \sin x \cos x)(1 - \sin x \cos x)} dx \\
&= \int \frac{dt}{(1 + 1 - t^2) \left(1 - \frac{1-t^2}{2}\right)}
\end{aligned}$$

$$\begin{aligned}
&\text{Put } \sin x - \cos x = t \\
&(\cos x + \sin x) dx = dt
\end{aligned}$$

$$\begin{aligned}
1 - 2 \sin x \cos x &= t^2 \\
2 \sin x \cos x &= 1 - t^2 \\
&= \int \frac{2dt}{(2 - t^2)(1 + t^2)} = -2 \int \frac{dt}{(t^2 + 1)(t^2 - 2)} \\
&= -\frac{2}{3} \int \left( \frac{1}{t^2 - 2} - \frac{1}{t^2 + 1} \right) dt \\
&= -\frac{2}{3} \left[ \frac{1}{2\sqrt{2}} \ln \left( \frac{t - \sqrt{2}}{t + \sqrt{2}} \right) - \tan^{-1} t \right] \\
&= -\frac{1}{3\sqrt{2}} \ln \left( \frac{\sin x - \cos x - \sqrt{2}}{\sin x - \cos x + \sqrt{2}} \right) + \frac{2}{3} \tan^{-1}(\sin x - \cos x) + C
\end{aligned}$$

$$= \frac{1}{3\sqrt{2}} \ln \left| \frac{\sin x - \cos x + \sqrt{2}}{\sin x - \cos x - \sqrt{2}} \right| + \frac{2}{3} \tan^{-1}(\sin x - \cos x) + C$$

$$\therefore A = \frac{1}{3\sqrt{2}}, B = \frac{2}{3} \text{ and } t = \sin x - \cos x$$

$$\therefore \left( \frac{B}{A}, t \right) = (2\sqrt{2}, \sin x - \cos x)$$

## Question 52

$$\frac{4x^2+5}{(x-2)^4} = \frac{A}{(x-2)} + \frac{B}{(x-2)^2} + \frac{C}{(x-2)^3} + \frac{D}{(x-2)^4}, \text{ then } \sqrt{\frac{A}{C} + \frac{B}{C} + \frac{D}{C}} \text{ is equal to}$$

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Options:

A.  $\frac{\sqrt{29}}{4}$

B.  $\frac{\sqrt{23}}{4}$

C.  $\frac{5}{4}$

D.  $\frac{4}{5}$

**Answer: C**

## Solution:

We have,

$$\begin{aligned}\frac{4x^2 + 5}{(x-2)^4} &= \frac{A}{x-2} + \frac{B}{(x-2)^2} \\ &+ \frac{C}{(x-2)^3} + \frac{D}{(x-2)^4} \\ \Rightarrow 4x^2 + 5 &= A(x-2)^3 + B(x-2)^2 \\ &+ C(x-2) + D \\ \Rightarrow 4x^2 + 5 &= A(x^3 - 8 - 3(x^2)(2) \\ &+ 3(x)(4) + B(x^2 + 4 - 4x) \\ &+ C(x-2) + D \\ \Rightarrow 4x^2 + 5 &= Ax^3 + x^2(-6A + B) \\ &+ x(12A - 4B + C) \\ &+ (-8A + 4B - 2C + D) \\ \Rightarrow A &= -6A + B = 4 \\ \Rightarrow B &= 4 \\ 12A - 4B + C &= 0 \\ \Rightarrow 0 - 16 + C &= 0 \Rightarrow C = 16\end{aligned}$$

$$\text{and, } -8A + 4B - 2C + D = 5$$

$$\Rightarrow 0 + 16 - 32 + D = 5$$

$$\Rightarrow D = 21$$

$$\therefore \sqrt{\frac{A+B+D}{C}} = \sqrt{\frac{0+4+21}{16}} = \sqrt{\frac{25}{16}} = \frac{5}{4}$$

---

## Question53

If  $\int \frac{\sqrt[4]{x}}{\sqrt{x} + \sqrt[4]{x}} dx = \frac{2}{3} \left[ A\sqrt[4]{x^3} + B\sqrt[4]{x^2} + C\sqrt[4]{x} + D \log(1 + \sqrt[4]{x}) \right] + K$ , then  $\frac{2}{3}(A + B + C + D)$  is equal to

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Options:

A.  $2/3$

B.  $-2/3$

C.  $4/3$

D.  $-4/3$

**Answer: B**

## Solution:

$$\text{Let } I = \int \frac{\sqrt[4]{x}}{\sqrt{x} + \sqrt[4]{x}} dx$$

$$\text{Let } x = t^4 \Rightarrow dx = 4t^3 dt$$

$$\begin{aligned}
 I &= \int \frac{t}{(t^2 + t)} \times 4t^3 dt \\
 &= 4 \int \frac{t^4}{t^2 + t} dt = 4 \int \frac{t^3}{t + 1} dt \\
 &= 4 \int \left( (t^2 - t + 1) - \frac{1}{t + 1} \right) dt \\
 &= 4 \left[ \frac{t^3}{3} - \frac{t^2}{2} + t - \log |t + 1| \right] + C \\
 &= \frac{4}{6} \left[ 2\sqrt[4]{x^3} - 3\sqrt[4]{x^2} \right. \\
 &\quad \left. + 6\sqrt[4]{x} - 6 \log |1 + \sqrt[4]{x}| \right] + C
 \end{aligned}$$

Here,  $A = 2, B = -3, C = 6, D = -6$

$$\begin{aligned}
 &\frac{2}{3}(A + B + C + D) \\
 &= \frac{2}{3}(2 - 3 + 6 - 6) = -\frac{2}{3}
 \end{aligned}$$


---

## Question54

$\int (\log x)^m x^n dx$  is equal to

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**Options:**

- A.  $\int t^m e^{nt} dt, \quad t = e^x$
- B.  $\int t^m e^{(n+1)t} dt, \quad t = e^x$
- C.  $\int t^m e^{(n+1)t} dt, \quad x = e^t$
- D.  $\int t^m e^{nt} dt, \quad x = e^t$

**Answer: C**

**Solution:**

$$\text{Let } I = \int (\log x)^m x^n dx$$

$$\text{Let } \log x = t \Rightarrow \frac{1}{x} dx = dt \text{ and } x = e^t$$

$$I = \int t^m \times (e^t)^{n+1} dt = \int t^m \times e^{(n+1)t} dt$$


---

## Question55

$\int \sin^{-1} \left( \sqrt{\frac{x-a}{x}} \right) dx$  is equal to

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**Options:**

A.  $x \cos^{-1} \sqrt{\frac{a}{x}} - \sqrt{ax - a^2} + c$

B.  $x \sec^{-1} \cdot \sqrt{\frac{a}{x}} + \sqrt{x^2 - ax} + c$

C.  $x \sin^{-1} \sqrt{\frac{x}{a}} + \sqrt{x^2 + ax} + c$

D.  $\frac{x}{a} \sin^{-1} \frac{x}{a} + \frac{x^2}{a} \sqrt{1 + a^2} + c$

**Answer: A**

**Solution:**

Let  $I = \int \sin^{-1} \left( \sqrt{\frac{x-a}{x}} \right) dx$

Let  $x = a \sec^2 \theta$

$$\Rightarrow dx = 2a \sec^2 \theta \tan \theta d\theta$$

$$= \int \sin^{-1} \sqrt{\frac{a \tan^2 \theta}{a \sec^2 \theta}} \times 2a \sec^2 \theta \tan \theta d\theta$$

$$= \int \theta \times 2a \sec^2 \theta \tan \theta d\theta$$

Let,  $\tan \theta = t \Rightarrow \sec^2 \theta d\theta = dt$

$$= 2a \int t \times \tan^{-1} t dt$$

$$= 2a \left[ \tan^{-1} t \times \frac{t^2}{2} - \int \frac{1}{1+t^2} \times \frac{t^2}{2} dt \right]$$

$$= 2a \left[ \tan^{-1} t \times \frac{t^2}{2} - \frac{1}{2} \int \left[ 1 - \frac{1}{(1+t^2)} \right] dt \right]$$

$$= 2a \left[ \tan^{-1} t \times \frac{t^2}{2} - \frac{1}{2} t + \frac{1}{2} \tan^{-1} t \right] + C$$

$$= 2a \left[ \frac{\tan^{-1} t}{2} [(t^2 + 1)] - \frac{t}{2} \right] + C$$

$$= 2a \left[ \frac{\theta}{2} (1 + \tan^2 \theta) - \frac{\tan \theta}{2} \right] + C$$

$$= 2a \left[ \sec^{-1} \sqrt{\frac{x}{a}} \times \frac{x}{a} - \frac{1}{2} \sqrt{\frac{x}{a} - 1} \right] + C$$

$$= x \cos^{-1} \sqrt{\frac{a}{x}} - \sqrt{ax - a^2} + C$$

---

## Question 56

If  $\int \frac{\sin x \cos x}{\sqrt{\cos^4 x - \sin^4 x}} dx = -\frac{f(x)}{2} + c$ , then domain of  $f(x)$  is

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**Options:**

A.  $[2n\pi, (2n+1)\pi], n = 0, 1, 2, \dots$

B.  $[(4n-1)\frac{\pi}{2}, (4n+1)\frac{\pi}{2}], n = 0, 1, 2, \dots$

C.  $[(4n-1)\frac{\pi}{4}, (4n+1)\frac{\pi}{4}], n = 0, 1, 2, \dots$

D.  $[(2n\frac{\pi}{4}, (2n+1)\frac{\pi}{4}], n = 0, 1, 2, \dots$

**Answer: C**

**Solution:**

Let  $I = \int \frac{\sin x \cos x}{\sqrt{\cos^4 x - \sin^4 x}} dx$

Let  $\sin^2 x = t \Rightarrow 2 \sin x \cos x dx = dt$

$$= \frac{1}{2} \int \frac{1}{\sqrt{(1-t)^2 - t^2}} dt$$

$$= \frac{1}{2} \int \frac{1}{\sqrt{1-2t}} dt$$

$$= \frac{\sqrt{1-2t}}{-2} + C$$

$$= \frac{\sqrt{1-2\sin^2 x}}{-2} + C$$

Here,  $f(x) = \sqrt{1-2\sin^2 x} = \sqrt{\cos 2x}$

$\therefore$  Domain of  $f(x)$  is

$$[(4n-1)\frac{\pi}{4}, (4n+1)\frac{\pi}{4}], n = 0, 1, 2, \dots$$

[ $\because \cos 2x$  is positive in (I) and IV quadrant and  $\cos(-\theta) = \cos \theta$ ]

## Question57

If  $\frac{13x+43}{2x^2+17x+30} = \frac{A}{2x+5} + \frac{B}{x+6}$ , then  $A + B$  is equal to

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**Options:**

A. 8

B. 18

C. 3

D. 5

**Answer: A**

## Solution:

Given,

$$\begin{aligned}\frac{13x + 43}{2x^2 + 17x + 30} &= \frac{A}{2x + 5} + \frac{B}{x + 6} \\ \Rightarrow \frac{13x + 43}{2x^2 + 17x + 30} &= \frac{A(x + 6) + B(2x + 5)}{(2x + 5)(x + 6)} \\ \Rightarrow 13x + 43 &= x(A + 2B) + 6A + 5B\end{aligned}$$

On compare both sides, we get

$$\begin{array}{rcl}A + 2B & = & 13 \\ \text{and } 6A + 5B & = & 43 \\ \Rightarrow 6A + 12B & = & 78 \\ 6A + 5B & = & 43 \\ \hline - & - & - \\ 7B & = & 35 \\ B = 5 & \Rightarrow & A = 3\end{array}$$

So,  $A + B = 3 + 5 = 8$

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## Question58

$\int e^{4x^2+8x-4}(x+1)\cos(3x^2+6x-4)dx$  is equal to

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Options:

- A.  $\frac{e^{4x^2+8x-4}}{25} [3 \sin(3x^2 + 6x - 4) - 4 \cos(3x^2 + 6x - 4)] + c$
- B.  $\frac{e^{4x^2+8x-4}}{50} [4 \cos(3x^2 + 6x - 4) + 3 \sin(3x^2 + 6x - 4)] + c$
- C.  $\frac{e^{4x^2+8x-4}}{25} [3 \cos(3x^2 + 6x - 4) + 4 \sin(3x^2 + 6x - 4)] + c$
- D.  $\frac{e^{4x^2+8x-4}}{50} [4 \sin(3x^2 + 6x - 4) + 3 \cos(3x^2 + 6x - 4)] + c$

**Answer: B**

## Solution:

$$\begin{aligned}\therefore I &= \int e^{4x^2+8x-4}(x+1)\cos(3x^2+6x-4)dx \\ &= \int e^{4(x^2+2x)-4}(x+1)(\cos(3(x^2+2x)-4))dx \\ \text{Put } x^2 + 2x &= t \Rightarrow 2(x+1)dx = dt \\ \therefore I &= \frac{1}{2} \int e^{4t-4}\cos(3t-4)dt\end{aligned}$$

We know that,

$$\begin{aligned}& \int e^{ax+b} \cos(cx+d) dx \\&= \frac{e^{ax+b}}{a^2+b^2} (-a \cos(cx+d) + c \sin(cx+d)) + c \\&\therefore I = \frac{1}{2} \frac{e^{4t-4}}{3^2+(4)^2} [4 \cos(3t-4) + 3 \sin(3t-4)] + c \\&= \frac{e^{4t-4}}{50} [4 \cos(3t-4) + 3 \sin(3t-4)] + c \\&= \frac{e^{4x^2+8x-4}}{50} [4 \cos(3x^2+6x-4) + 3 \sin(3x^2+6x-4)] + c\end{aligned}$$

---

## Question59

$\int [(\log 2x)^2 + 2 \log 2x] dx$  is equal to

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**Options:**

- A.  $(\log 2x)^2 + c$
- B.  $2x \log 2x + c$
- C.  $x(\log 2x)^2 + c$
- D.  $2x(\log x)^2 + c$

**Answer: C**

**Solution:**

$$I = \int [(\log 2x)^2 + 2 \log 2x] dx$$

$$\text{Let } \log 2x = t \Rightarrow x = e^t/2$$

$$\begin{aligned}\Rightarrow \frac{1}{2x} \cdot 2dx &= dt \Rightarrow dx = xdt \Rightarrow dx = \frac{e^t}{2} dt \\ \therefore \int \frac{1}{2} (t^2 + 2t) e^t dt \\&= \frac{1}{2} \left[ (t^2 + 2t) e^t - \int (2t + 2) e^t dt \right] \\&= \frac{1}{2} \left[ (t^2 + 2t) e^t - \left\{ (2t + 2) e^t - \int 2e^t dt \right\} \right] \\&= \frac{1}{2} [(t^2 + 2t) e^t - (2t + 2) e^t + 2e^t] \\&= \frac{e^t}{2} [t^2 + 2t - 2t - 2 + 2] \\&= \frac{e^t}{2} t^2 + c \quad \dots (i)\end{aligned}$$

On putting value of  $t$  and  $e^t/2$  in Eq. (i)

$$= x(\log 2x)^2 + c$$

---

## Question60

If  $\int \log (6 \sin ^2 x+17 \sin x+12) \cos x d x=f(x)+c$ , then  $f\left(\frac{\pi}{2}\right)$  is equal to

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Options:

A.  $\frac{1}{6}[\log 5^5+\log 7^7-12]$

B.  $\frac{1}{6}[7 \log 5+5 \log 7+29]$

C.  $\frac{1}{6}[14 \log 5+15 \log 7+12]$

D.  $\frac{1}{6}[15 \log 5+14 \log 7-29]$

**Answer: D**

**Solution:**

Let

$$I=\int \log \left(6 \sin ^2 x+17 \sin x+12\right) \cos x d x$$

Let  $\sin x=t$

$$\cos x \cdot d x=d t$$

$$=\int \cos x \log \left(6 \sin ^2 x+17 \sin x+12\right) d x$$

$$I=\int 1 \cdot \log \left(6 t^2+17 t+12\right) d t$$

II I

$$=\log \left(6 t^2+17 t+12\right) \cdot t-\int \frac{12 t+17}{6 t^2+17 t+12} \cdot t d t$$

$$=t \log \left(6 t^2+17 t+12\right)-\int \frac{12 t^2+17 t}{6 t^2+17 t+12} d t$$

$$=t \log \left(6 t^2+17 t+12\right)$$

$$-\int\left(-\frac{4}{3 t+4}-\frac{3}{2 t+3}+2\right) d t$$

$$=t \log \left(6 t^2+17 t+12\right)+\frac{4}{3} \log (3 t+4)+\frac{3}{2} \log (2 t+3)-2 t$$

Now, put  $t=\sin x$

$$=\sin x \log \left(6 \sin ^2 x+17 \sin x+12\right)+\frac{4}{3} \log$$

$$(3 \sin x+4)+\frac{3}{2} \log (2 \sin x+5)-2 \sin x+C$$

$$\text { Now, } f\left(\frac{\pi}{2}\right)=1 \cdot \log (6+17+12)$$

$$+\frac{4}{3} \log (3+4)+\frac{3}{2} \log (2+3)-2$$

$$=\frac{1}{6}[15 \log 5+14 \log 7-29]$$

-----

## Question61

$\int \frac{1}{(1+x^2)\sqrt{x^2+2}} dx$  is equal to

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**Options:**

A.  $-\tan^{-1} \frac{\sqrt{x^2+2}}{|x|} + c$

B.  $-\tan^{-1} \sqrt{x^2+2} + c$

C.  $-\tan^{-1} \sqrt{\frac{x^2+1}{x^2+2}} + c$

D.  $-\tan^{-1} \sqrt{\frac{x^2+2}{x^2+1}} + c$

**Answer: A**

**Solution:**

$$I = \int \frac{1}{(1+x^2)\sqrt{x^2+2}} dx.$$

$$\text{Let } x = \sqrt{2} \tan u$$

$$\begin{aligned} dx &= \sqrt{2} \sec^2 u \cdot du \\ &= \int \frac{\sqrt{2} \sec^2 u}{(1+2 \tan^2 u) \sqrt{2 \tan^2 u + 2}} du \\ &= \int \frac{\sqrt{2} \sec^2 u}{(1+2 \tan^2 u) \sqrt{2} \cdot \sec u} du \\ &= \int \frac{\sec u}{1+2 \tan^2 u} du \\ &= \int \frac{\sec u}{1+2 \tan^2 u} \times \frac{\cos^2 u}{\cos^2 u} \cdot du \\ &= \int \frac{\cos u}{\cos^2 u + 2 \sin^2 u} du \\ &= \int \frac{\cos u}{1 + \sin^2 u} du \end{aligned}$$

$$\text{Let } \sin u = t$$

$$\begin{aligned} \cos u \cdot du &= dt = \int \frac{1}{1+t^2} dt \\ &= \tan^{-1}(t) = \tan^{-1}(\sin u) \\ &= \tan^{-1} \left( \sin \left( \tan^{-1} \left( \frac{x}{\sqrt{2}} \right) \right) \right) + C \\ I &= -\tan^{-1} \left( \frac{\sqrt{x^2+2}}{|x|} \right) + C \end{aligned}$$

---

## Question62

$\int \sin^4 x \cos^4 x dx$  is equal to

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**Options:**

A.

$$\frac{1}{128}(-2 \sin^3 x \cos x - 3 \sin x \cos x + 3) + c$$

B.

$$\frac{1}{256}(-2 \sin^3 2x \cos 2x - 3 \sin 2x \cos 2x + 6x) + c$$

C.

$$\frac{1}{128}(2 \sin^3 x \cos x - 3 \sin x \cos x + 3x) + c$$

D.

$$\frac{1}{256}(3 \sin^3 x \cos x - 2 \sin x \cos x + 2) + c$$

**Answer: B**

**Solution:**

$$\begin{aligned} \therefore \int \sin^4 x \cos^4 x dx &= \int (\sin^2 x \cos^2 x)^2 dx \\ &= \int \left( \frac{1}{4} \sin^2 2x \right)^2 dx = \frac{1}{16} \int \sin^4 2x dx \\ &= \frac{1}{16} \int \left( \frac{3}{8} - \frac{\cos 4x}{2} + \frac{\cos 8x}{8} \right) dx \\ &= \frac{1}{128} \int (3 - 4 \cos 4x + \cos 8x) dx \\ &= \frac{1}{128} \left( 3x - \sin 4x + \frac{\sin 8x}{8} \right) + c \end{aligned}$$

From option (b),

$$\begin{aligned} &\frac{1}{256} \left( -\sin^2 2x \sin 4x - \frac{3}{2} \sin 4x + 6x \right) + c \\ &= \frac{1}{256} \left( \frac{(\cos 4x - 1)}{2} \sin 4x - \frac{3}{2} \sin 4x + 6x \right) + c \\ &= \frac{1}{256} \left( \frac{1}{4} \sin 8x - 2 \sin 4x + 6x \right) + c \\ &= \frac{1}{128} \left( 3x - \sin 4x + \frac{\sin 8x}{8} \right) + c \end{aligned}$$

$$\text{Hence, } \int \sin^4 x \cos^4 x dx = \frac{1}{256}$$

$$(-2 \sin^3 2x \cos 2x - 3 \sin 2x \cos 2x + 6x) + c$$

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## Question63

$$\int \frac{x^2-1}{x^3\sqrt{2x^4-2x^2+1}} dx$$

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Options:

A.  $\frac{1+2x^2+2x^4}{2x^2} + c$

B.  $\frac{(1+2x^2+2x^4)^{\frac{1}{2}}}{2x^2} + c$

C.  $\frac{1-2x^2+2x^4}{2x^2} + c$

D.  $\frac{(1-2x^2+2x^4)^{\frac{1}{2}}}{2x^2} + c$

**Answer: D**

**Solution:**

$$I = \int \frac{x^2-1}{x^3\sqrt{2x^4-2x^2+1}} dx$$

#### Step 1: Substitution

We observe the expression inside the square root, so let:

$$u = 2x^4 - 2x^2 + 1$$

Differentiate with respect to  $x$ :

$$\frac{du}{dx} = 8x^3 - 4x$$

Thus,  $du = (8x^3 - 4x) dx$ .

#### Step 2: Simplify the integral

We need to adjust the numerator  $x^2 - 1$  to match the differential. By substituting  $u$  and simplifying, we obtain:

$$I = \int \frac{1}{x^3\sqrt{u}} dx$$

#### Step 3: Final answer

After performing the substitution and solving the integral, the result is:

$$I = \frac{(1 - 2x^2 + 2x^4)^{\frac{1}{2}}}{2x^3} + C$$

Thus, the correct answer is:

**Option D:**

$$\frac{(1 - 2x^2 + 2x^4)^{\frac{1}{2}}}{2x^3} + C$$

---

## Question64

$$\int \frac{x^3 \tan^{-1} x^4}{1+x^8} dx =$$

**AP EAPCET 2024 - 22th May Morning Shift**

**Options:**

A.  $\frac{(\tan^{-1}(x^4))^2}{8} + c$

B.  $\frac{((\tan^{-1}(x^4))^3)}{3} + c$

C.  $\frac{(\tan^{-1}(x^4))^2}{4} + c$

D.  $\frac{(\tan^{-1}(x^4))^2}{2} + c$

**Answer: A**

**Solution:**

$$I = \int \frac{x^3 \tan^{-1} x^4}{1+x^8} dx$$

$$\text{Let } x^4 = t$$

$$4x^3 dx = dt$$

$$x^3 dx = \frac{1}{4} dt$$

$$I = \int \frac{\frac{1}{4} \tan^{-1} t}{1+t^2} dt = \frac{1}{4} \int \frac{\tan^{-1} t}{1+t^2} dt$$

$$\text{Let } \tan^{-1} t = z$$

$$\frac{1}{1+t^2} dt = dz$$

$$= \frac{1}{4} \int z dz$$

$$= \frac{1}{4} \cdot \frac{z^2}{2} + C = \frac{z^2}{8} + c$$

$$= \frac{(\tan^{-1} t)^2}{8} + c$$

$$I = \frac{(\tan^{-1} (x^4))^2}{8} + C$$

## Question 65

$$\int \frac{2}{1+x+x^2} dx =$$

**AP EAPCET 2024 - 22th May Morning Shift**

**Options:**

A.  $\frac{4}{\sqrt{3}} \tan^{-1} \left( \frac{2x-1}{\sqrt{3}} \right) + c$

B.  $\frac{4}{\sqrt{3}} \tan^{-1} \left( \frac{2x+1}{\sqrt{3}} \right) + c$

C.  $\frac{2}{\sqrt{3}} \tan^{-1} \left( \frac{2x-1}{\sqrt{3}} \right) + c$

D.  $\frac{2}{\sqrt{3}} \tan^{-1} \left( \frac{2x+1}{\sqrt{3}} \right) + c$

**Answer: B**

**Solution:**

$$\begin{aligned}
&\text{We have, } \int \frac{2}{x^2 + x + 1} dx \\
&= 2 \int \frac{1}{x^2 + x + 1} dx \\
&= 2 \int \frac{1}{\left(x + \frac{1}{2}\right)^2 + \frac{3}{4}} dx \\
&\quad u = \frac{2x + 1}{\sqrt{3}} \rightarrow du = \frac{2}{\sqrt{3}} dx \\
&\quad = \int \frac{\sqrt{3}}{\left(\frac{3}{4}u^2 + \frac{3}{4}\right)} du \\
\text{On substitute } &= \frac{4}{\sqrt{3}} \int \frac{1}{u^2 + 1} du \\
&= \frac{4}{\sqrt{3}} \tan^{-1} \left( \frac{2x + 1}{\sqrt{3}} \right) + C
\end{aligned}$$


---

## Question66

$$\int \frac{1}{x^2(\sqrt{1+x^2})} dx =$$

### AP EAPCET 2024 - 22th May Morning Shift

**Options:**

A.  $\frac{-\sqrt{x^2+1}}{x} + c$

B.  $\frac{\sqrt{x^2+1}}{x} + c$

C.  $\frac{-\sqrt{x^2-1}}{x} + c$

D.  $\frac{\sqrt{x^2-1}}{x} + c$

**Answer: A**

**Solution:**

$$I = \int \frac{1}{x^2 \sqrt{1+x^2}} dx$$

$$\text{let } x = \tan \theta$$

$$dx = \sec^2 \theta d\theta$$

$$\begin{aligned}
I &= \int \frac{\sec^2 \theta d\theta}{\tan^2 \theta \sqrt{1 + \tan^2 \theta}} \\
&= \int \frac{\sec^2 \theta}{\tan^2 \theta \cdot \sec \theta} d\theta = \int \frac{\frac{1}{\cos \theta}}{\frac{\sin^2 \theta}{\cos^2 \theta}} d\theta \\
&= \int \frac{\cos \theta}{\sin \theta} \cdot \frac{1}{\sin \theta} d\theta
\end{aligned}$$

$$\begin{aligned}
 &= \int \operatorname{cosec} \theta \cdot \cot \theta d\theta \\
 I &= -\operatorname{cosec} \theta + C \\
 &= \frac{-\sqrt{1+\tan^2 \theta}}{\tan \theta} + C \\
 I &= -\frac{\sqrt{1+x^2}}{x} + C
 \end{aligned}$$


---

## Question67

$$\int \frac{\sin 7x}{\sin 2x \sin 5x} dx =$$

### AP EAPCET 2024 - 22th May Morning Shift

Options:

- A.  $\log(\sin 5x \sin 2x) + c$
- B.  $\log(\sin 5x) + \log(\sin 2x) + c$
- C.  $\frac{1}{5} \log(\sin 5x) + \frac{1}{2} \log(\sin 2x) + c$
- D.  $\frac{1}{5} \log(\sin x) + \frac{1}{2} \log(\sin x) + c$

**Answer: C**

**Solution:**

$$\begin{aligned}
 &\text{We have,} \\
 I &= \int \frac{\sin 7x}{\sin 2x \cdot \sin 5x} dx \\
 &= \int \left[ \frac{\sin 2x \cos 5x}{\sin 2x \sin 5x} + \frac{\sin 5x \cos 2x}{\sin 2x \sin 5x} \right] dx \\
 &= \int (\cot 5x + \cot 2x) dx \\
 &= \frac{1}{5} \log(\sin 5x) + \frac{1}{2} \log(\sin 2x) + c \\
 &\left[ \because \int \cot x dx = \frac{1}{a} \log \sin ax \right]
 \end{aligned}$$


---

## Question68

If  $\frac{x+2}{(x^2+3)(x^4+x^2)(x^2+2)} = \frac{Ax+B}{x^2+3} + \frac{Cx+D}{x^2+2} + \frac{Ex^3+Fx^2+Gx+H}{x^4+x^2}$ , then  
 $(E+F)(C+D)(A) =$

### AP EAPCET 2024 - 21th May Evening Shift

**Options:**

A.  $-\frac{1}{4}$

B.  $-\frac{3}{4}$

C.  $\frac{3}{4}$

D.  $\frac{1}{4}$

**Answer: D****Solution:**Given,  $\frac{x+2}{(x^2+3)(x^4+x^2)(x^2+2)}$ 

$$\begin{aligned}
&= \frac{Ax+B}{x^2+3} + \frac{Cx+D}{x^2+2} + \frac{Ex^3+Fx^2+Gx+H}{x^4+x^2} \\
x+2 &= (x^2+2)(x^4+x^2)(Ax+B) + (x^2+3)(x^4+x^2)(Cx+D) \\
&\quad + (Ex^3+Fx^2+Gx+H)(x^2+2)(x^2+3) \\
&= (A+C+E)x^7 + (B+D+F)x^6 \\
&\quad + (3A+4C+5E+G)x^5 \\
&\quad + (3B+4D+5F+H)x^4 + (2A \\
&\quad + 3C+6E+5G)x^3 \\
&\quad + (2B+3D+6F+5H)x^2 + (6G)x + 6H
\end{aligned}$$

On comparing coefficients both sides, we get

$6G = 1 \Rightarrow G = \frac{1}{6} \quad \dots (i)$

$6H = 2 \Rightarrow H = \frac{1}{3} \quad \dots (ii)$

$A + C + E = 0 \quad \dots (iii)$

$3A + 4C + 5E + G = 0 \quad \dots (iv)$

$2A + 3C + 6E + 5G = 0 \quad \dots (v)$

$B + D + F = 0 \quad \dots (vi)$

$3B + 4D + 5F + H = 0 \quad \dots (vii)$

$2B + 3D + 6F + 5H = 0 \quad \dots (viii)$

On putting the value of  $G$  in Eq. (iv) and (v) and multiplying 2 by Eq. (iv) and 3 by Eq. (v), we get

$$6A + 8C + 10E = -\frac{2}{6}$$

$$6A + 9C + 18E = \frac{-15}{6}$$

$$\begin{array}{r}
- \quad - \quad - \\
\hline
-C - 8E = +\frac{13}{6}
\end{array}$$

$$\Rightarrow \quad -C + 8E = -\frac{13}{6} \quad \dots (ix)$$

On multiply by 3 Eq. (iii) and subtracting from Eq. (iv), we get

$$3A + 4C + 5E = -\frac{1}{6}$$

$$3A + 3C + 3E = 0$$

$$\begin{array}{r} - \quad - \quad - \quad - \\ \hline \end{array}$$

$$C + 2E = -\frac{1}{6} \quad \dots(x)$$

Now, from Eqs. (ix) and (x), we get

$$C + 8E = -\frac{13}{6}$$

$$C + 2E = -\frac{1}{6}$$

$$\begin{array}{r} - \quad - \quad - \\ \hline 6E = -\frac{12}{6} \end{array}$$

$\Rightarrow$

$$E = -\frac{1}{3} \text{ and } C = \frac{1}{2}$$

Again, from Eqs. (vii) and (viii), we get

$$(3B + 4D + 5F = -\frac{1}{3}) \times 2 \left[ \because H = \frac{1}{3} \right]$$

$$(2B + 3D + 6F = -\frac{5}{3}) \times 3$$

$$-D - 8F = \frac{13}{3}$$

$$\Rightarrow D + 8F = -\frac{13}{3} \quad \dots(xi)$$

On multiply by 3 in Eq. (vi), we get

$$3B + 4D + 5F = -\frac{1}{3}$$

$$3B + 3D + 3F = 0$$

$$\begin{array}{r} - \quad - \quad - \\ \hline D + 2F = \frac{1}{3} \end{array} \quad \dots(e)$$

From Eqs. (xi) and (xii), we get

$$F = -\frac{2}{3} \text{ and } D = 1$$

On putting the values of  $C$  and  $E$  in Eq. (iii), we get

$$A + C + E = 0$$

$$A + \frac{1}{2} - \frac{1}{3} = 0 \Rightarrow A = -\frac{1}{6}$$

On putting the values of  $F$  and  $D$  in Eq. (vi), we get

$$B = -\frac{1}{3}$$

Now,

$$\begin{aligned} (E + F)(C + D)(A) &= \left(-\frac{1}{3} - \frac{2}{3}\right) \left(\frac{1}{2} + 1\right) \left(-\frac{1}{6}\right) \\ &= (-1) \left(\frac{3}{2}\right) \left(-\frac{1}{6}\right) = \frac{1}{4} \end{aligned}$$

## Question69

$$\int \frac{\sin^6 x}{\cos^8 x} dx =$$

**AP EAPCET 2024 - 21th May Evening Shift**

**Options:**

A.  $\tan 7x + c$

B.  $\frac{\tan^7 x}{7} + c$

C.  $\frac{\tan 7x}{7} + c$

D.  $\sec^7 x$

**Answer: B**

## Solution:

We have,

$$\begin{aligned} I &= \int \frac{\sin^6 x}{\cos^8 x} dx \\ \Rightarrow I &= \int \frac{\sin^6 x}{\cos^6 x} \times \frac{1}{\cos^2 x} dx \\ \Rightarrow I &= \int \tan^6 x \cdot \sec^2 x dx \quad \dots (i) \end{aligned}$$

Let  $\tan x = u$

$$\begin{aligned} \Rightarrow \frac{du}{dx} &= \sec^2 x \\ du &= \sec^2 x dx \end{aligned}$$

On putting in Eq. (i), we get

$$\begin{aligned} \Rightarrow I &= \int u^6 du \\ \Rightarrow I &= \frac{u^7}{7} + C \end{aligned}$$

Put the value of  $u$

$$I = \frac{\tan^7 x}{7} + C$$

---

## Question70

$$\int \frac{x^5}{x^2+1} dx =$$

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Options:

- A.  $\frac{x^4}{4} + \frac{x^3}{3} - \tan^{-1} x + c$
- B.  $\frac{x^4}{4} - \frac{x^2}{2} + \frac{1}{2} \log(x^2 + 1) + c$
- C.  $\frac{x^4}{4} + \frac{x^3}{3} + \tan^{-1} x + c$
- D.  $\frac{x^4}{4} + \frac{x^2}{2} - \frac{1}{2} \log(x^2 + 1) + c$

**Answer: B**

## Solution:

We have,

$$I = \int \frac{x^5}{x^2+1} dx \Rightarrow I = \int \frac{x^4 \cdot x}{x^2+1} dx$$

$$\text{Let } x^2 + 1 = u \Rightarrow x^2 = u - 1$$

$$\Rightarrow 2x = \frac{du}{dx} \Rightarrow \frac{du}{2} = x dx \text{ then, } I \text{ become}$$

$$\Rightarrow I = \int \frac{(u-1)^2}{u} \cdot \frac{du}{2} \Rightarrow I = \int \frac{u^2 + 1 - 2u}{u} \frac{du}{2}$$

$$\Rightarrow I = \frac{1}{2} \int \left[ u + \frac{1}{u} - 2 \right] du$$

On integrating, we get

$$I = \frac{1}{2} \left[ \frac{u^2}{2} + \log u - 2u \right] + c'$$

On putting the value of  $u$ , we get

$$I = \frac{1}{2} \left[ \frac{(x^2 + 1)^2}{2} + \log(x^2 + 1) - 2(x^2 + 1) \right] + c'$$

$$\Rightarrow I = \frac{x^4}{4} - \frac{x^2}{2} + \frac{1}{2} \log(x^2 + 1) + c$$

## Question 71

$$\int \left( \sum_{r=0}^{\infty} \frac{x^r 3^r}{r!} \right) dx =$$

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Options:

A.  $e^x + c$

B.  $\frac{e^{3x}}{3} + c$

C.  $3e^{3x} + c$

D.  $3e^x + c$

**Answer: B**

**Solution:**

$$\text{We have, } I = \int \left( \sum_{r=0}^{\infty} \frac{x^r 3^r}{r!} \right) dx$$

We know that

$$a^m \cdot b^m = (ab)^m \Rightarrow I = \int \left( \sum_{r=0}^{\infty} \frac{(x3)^r}{r!} \right) dx$$

$$\Rightarrow I = \int \left[ \frac{(3x)^0}{0!} + \frac{(3x)^1}{1!} + \frac{(3x)^2}{2!} + \dots \infty \right] dx$$

$$\Rightarrow I = \int [1 + 3x + (3x)^2 + \dots \infty] dx$$

As, we know that

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \dots \infty$$

$$\text{Then, } I = \int e^{3x} dx \Rightarrow I = \frac{e^{3x}}{3} + C$$


---

## Question72

$$\int \frac{x^4+1}{x^6+1} dx =$$

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**Options:**

A.  $\tan^{-1} x - \tan^{-1} x^3 + c$

B.  $\tan^{-1} x - \frac{1}{3} \tan^{-1} x^3 + c$

C.  $\tan^{-1} x + \tan^{-1} x^3 + c$

D.  $\tan^{-1} x + \frac{1}{3} \tan^{-1} x^3 + c$

**Answer: D**

**Solution:**

We have,

$$\begin{aligned} I &= \int \frac{x^4+1}{x^6+1} dx = \int \left( \frac{1+x^4-x^2}{x^6+1} + \frac{1}{3} \times \frac{3x^2}{1+x^6} \right) dx \\ &= \int \frac{1}{1+x^2} dx + \frac{1}{3} \int \frac{3x^2}{1+(x^3)^2} dx \\ &= \tan^{-1} x + \frac{1}{3} \tan^{-1} (x^3) + C \quad [\because 1+x^6 = (1+x^2)(1+x^4-x^2)] \end{aligned}$$


---

## Question73

$$\int e^x (x+1)^2 dx =$$

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**Options:**

A.  $xe^x + c$

B.  $e^x x^2 + c$

C.  $e^x (x^2 + 1) + c$

D.  $e^x (x+1) + c$

**Answer: C**

## Solution:

We have,

$$\Rightarrow I = \int e^x (x+1)^2 dx \Rightarrow I = \int e^x (x^2 + 1 + 2x) dx$$

$$\Rightarrow I = \int e^2 x^2 dx + \int e^x dx + 2 \int x e^2 dx$$

$$\Rightarrow I = x^2 \int e^x dx - \left( \int \left[ \frac{d}{dx} x^2 \right] \int e^x dx + \int e^x dx + 2 \left[ x \int e^x dx - \int \left( \frac{d}{dx} x \int e^x dx \right) dx \right] \right)$$

$$\Rightarrow I = x^2 e^x - 2 \int x e^x dx + e^x + 2x e^x - 2 \int e^x dx$$

$$\Rightarrow I = x^2 e^x - 2x e^x + 2e^x + e^x + 2x e^x - 2e^x + C$$

$$\Rightarrow I = e^x (x^2 + 1) + C$$

---

## Question 74

If  $\frac{1}{(3x+1)(x-2)} = \frac{A}{3x+1} + \frac{B}{x-2}$  and  $\frac{x+1}{(3x+1)(x-2)} = \frac{C}{3x+1} + \frac{D}{x-2}$ , then

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Options:

A.

$$A + 3B = 0, A : C = 1 : 3, B : D = 2 : 3$$

B.

$$A + 3B = 0, A : C = 3 : 1, B : D = 3 : 2$$

C.

$$A - 3B = 0, A : C = 3 : 2, B : D = 1 : 3$$

D.

$$A + 3B = 0, A : C = 3 : 2, B : D = 1 : 3$$

**Answer: D**

## Solution:

$$\begin{aligned} \frac{1}{(3x+1)(x-2)} &= \frac{A}{(3x+1)} + \frac{B}{(x-2)} \\ &= \frac{(x-2)A + B(3x+1)}{(3x+1)(x-2)} \\ &= \frac{Ax - 2A + 3Bx + B}{(3x+1)(x-2)} \\ \frac{1}{(3x+1)(x-2)} &= \frac{x(A+3B) - 2A + B}{(3x+1)(x-2)} \\ A + 3B &= 0 \\ -2A + B &= 1 \end{aligned}$$

$$\begin{aligned}
 \text{and } \frac{(x+1)}{(3x+1)(x-2)} &= \frac{C}{(3x+1)} + \frac{D}{(x-2)} = \frac{C(x-2) + D(3x+1)}{(3x+1)(x-2)} \\
 \frac{(x+1)}{(3x+1)(x-2)} &= \frac{Cx - 2C + 3Dx + D}{(3x+1)(x-2)} \\
 \frac{(x+1)}{(3x+1)(x-2)} &= \frac{x(C+3D) - 2C + D}{(3x+1)(x-2)} \\
 C + 3D &= 1 \\
 -2C + D &= 1
 \end{aligned}$$

From Eq. (i)  $\times 28$  and (ii) Eq., we get

$$2A + 6B = 0$$

$$-2A + B = 1$$

$$\hline 7B = 1$$

$$B = \frac{1}{7}$$

On putting in Eq. (i), we get

$$A = -3 \times \frac{1}{7} = -\frac{3}{7}$$

From Eqs. (iii) and (iv)  $\times 3$ , we get

$$C + 3D = 1$$

$$-6C + 3D = 3$$

$$+ \quad - \quad -$$

$$7C = -2$$

$$C = -\frac{2}{7} \quad D = \frac{3}{7}$$

$$A : C = -\frac{3}{7} ; -\frac{2}{7} \Rightarrow 3 : 2$$

$$\text{and } B : D = 1 : 3$$

## Question 75

If  $x \in \left[2n\pi - \frac{\pi}{4}, 2n\pi + \frac{3\pi}{4}\right]$  and  $n \in Z$ , then  $\int \sqrt{1 - \sin 2x} dx =$

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**Options:**

A.  $-\cos x + \sin x + c$

B.  $\cos x + \sin x + c$

C.  $-\cos x - \sin x + c$

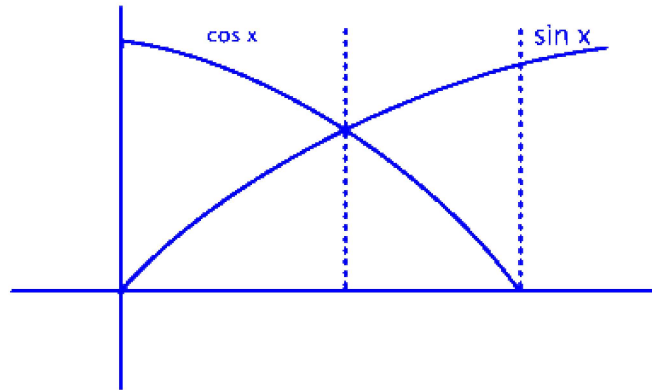
D.  $\cos x - \sin x + c$

**Answer: B**

**Solution:**

Simplify using trigonometric identities.

$$\begin{aligned}
 & \int \sqrt{1 - \sin 2x} dx \quad [\because \sin 2x = 2 \sin x \cos x] \\
 &= \int \sqrt{\sin^2 x + \cos^2 x - 2 \sin x \cos x} dx \\
 & \quad [\because 1 = \sin^2 x + \cos^2 x] \\
 &= \int \sqrt{(\cos x - \sin x)^2} dx \\
 &= \int (\cos x - \sin x) dx \\
 &= \sin x - (-\cos x) + c = \sin x + \cos x + c \\
 &= \cos x + \sin x + c
 \end{aligned}$$



Therefore, the integral of  $\sqrt{1 - \sin 2x}$  w.r.t.  $x$ , is  $\sin x + \cos x + c$ , where  $c$  is the constant of integration.

## Question 76

$$\int e^x \left( \frac{x+2}{x+4} \right)^2 dx =$$

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**Options:**

A.  $-\frac{xe^x}{(x+4)^2} + c$

B.  $-\frac{xe^x}{(x+4)} + c$

C.  $\frac{xe^x}{(x+4)} + c$

D.  $\frac{2xe^x}{(x+4)} + c.$

**Answer: C**

**Solution:**

$$\begin{aligned}
 & \int e^x \cdot \left( \frac{x+2}{x+4} \right)^2 dx \\
 &= \int e^x \frac{x^2 + 4x + 4}{(x+4)^2} dx \\
 &= \int e^x \cdot \left[ \frac{x}{x+4} + \frac{4}{(x+4)^2} \right] dx
 \end{aligned}$$

We know that,

$$\begin{aligned}\int e^x(f(x)+f'(x))dx &= e^x \cdot f(x) + c \\&= \int e^x \left[ \frac{x}{x+4} + \frac{4}{(x+4)^2} \right] dx \\&= e^x \cdot \frac{x}{(x+4)} + c\end{aligned}$$

Or

(c) Alternate solution

$$\begin{aligned}&\int e^x \left( \frac{x+2}{x+4} \right)^2 dx \\&\text{Let } u = x+4 \\&\quad du = dx; x = u-4 \\&\int e^{u-4} \left( \frac{u-4+2}{u} \right)^2 du \\&\int e^{u-4} \left( \frac{u-2}{u} \right)^2 dx \Rightarrow \int e^{u-4} \left( 1 - \frac{2}{u} \right)^2 du \\&\int e^{u-4} \left( 1 + \frac{4}{u^2} - \frac{4}{u} \right) dx \\&\int \left[ e^{u-4} + 4 \frac{e^{u-4}}{u^2} - \frac{4e^{u-4}}{u} \right] du \\&e^{-4} \left[ e^u - 4 \int \left( \frac{e^u}{u} - \frac{e^u}{u^2} \right) du \right]\end{aligned}$$

But Eq. (i)

$$e^{-4} \left[ e^u - 4 \frac{e^u}{u} \right] + c$$

On putting the value of  $u = x+4$

$$e^{-4} \left[ e^{x+4} - \frac{4e^{x+4}}{x+4} \right] + c$$

$$\begin{aligned}&e^x - \frac{4e^x}{(x+4)} + c \\&= e^x \left( \frac{x+4-4}{x+4} \right) + c = \frac{xe^x}{x+4} + c\end{aligned}$$

---

## Question77

If  $\int \frac{1}{1-\cos x} dx = \tan \left( \frac{x}{\alpha} + \beta \right) + c$ , then one of the values of  $\frac{\pi\alpha}{4} - \beta$  is

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**Options:**

A.  $-\frac{\pi}{2}$

B.  $\pi$

C. 0

D.  $\frac{\pi}{4}$

**Answer: B**

**Solution:**

$$\text{If } \int \frac{1}{1 - \cos x} dx = \tan\left(\frac{x}{\alpha} + \beta\right) + c$$

$$\frac{\pi\alpha}{4} - \beta = ?$$

$$\int \frac{1}{1 - \cos x} dx = \tan\left(\frac{x}{\alpha} + \beta\right) + c$$

$$\int \frac{1}{1 - 1 + 2 \sin^2\left(\frac{x}{2}\right)} dx = \tan\left(\frac{x}{\alpha} + \beta\right) + c$$

$$\frac{1}{2} \int \operatorname{cosec}^2\left(\frac{x}{2}\right) dx = \tan\left(\frac{x}{\alpha} + \beta\right) + c \left( \because \sin^2 \frac{x}{2} = \frac{1}{\operatorname{cosec}^2 \frac{x}{2}} \right)$$

$$\frac{1}{2} x - 2 \cot\left(\frac{x}{2}\right) = \tan\left(\frac{x}{\alpha} + \beta\right) + c$$

$$- \cot\left(\frac{x}{2}\right) = \tan\left(\frac{x}{\alpha} + \beta\right) + c$$

$$- \cot\left(\frac{x}{2}\right) = \tan\left(\frac{x}{\alpha} + \beta\right) + c \quad \left( \because \cot y = \frac{1}{\tan y} \right)$$

$$\Rightarrow - \cot\left(\frac{x}{2}\right) = \tan\left(\frac{\pi}{2} - \frac{x}{2}\right)$$

$$\Rightarrow \tan\left(\frac{x}{2} - \frac{\pi}{2}\right) + C = \tan\left(\frac{x}{\alpha} + \beta\right) + c$$

On comparison, we get

$$-\frac{\pi}{2} + \frac{x}{2} = \frac{x}{\alpha} + \beta$$

On solving these and we get

$$\alpha = 2, \beta = -\frac{\pi}{2}$$

Now, the value of  $\frac{\pi\alpha}{4} - \beta = \pi$

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## Question 78

If  $n \geq 2$  is a natural number and  $0 < \theta < \frac{\pi}{2}$ , then  $\int \frac{(\cos^n \theta - \cos \theta)^{1/n}}{\cos^{n+1} \theta} \sin \theta d\theta =$

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**Options:**

A.  $\frac{n}{n-1} (\cos^{(1-n)} \theta - 1)^2 + c$

B.  $\frac{n}{(n+1)(1-n)} (\cos^{(1-n)} \theta - 1)^{1+\frac{1}{n}} + c$

C.  $\frac{1}{n-1} (\cos^{(n-1)} \theta - 1)^2 + c$

D.  $\frac{n}{1-n^2} (1 - \cos^{(1-n)} \theta)^{(n+1)/n} + c$

**Answer: D**

**Solution:**

$$I = \int \frac{(\cos^n \theta - \cos \theta)^{\frac{1}{n}}}{\cos^{n+1} \theta} \sin \theta d\theta$$

$$\begin{aligned} \text{Let } &= \int \left( \frac{\cos^n \theta - \cos \theta}{\cos^n \theta} \right)^{\frac{1}{n}} \frac{\sin \theta}{\cos^n \theta} d\theta \\ &= \int \left( 1 - \cos^{(1-n)} \theta \right)^{\frac{1}{n}} \cos^{(-n)} \theta \sin \theta d\theta \end{aligned}$$

$$\text{Let } t = 1 - \cos^{(1-n)} \theta$$

$$\Rightarrow dt = -(1-n) \cos^{(-n)} \theta (-\sin \theta) d\theta$$

$$\Rightarrow \frac{dt}{1-n} = \cos^{(-n)} \theta \sin \theta d\theta$$

$$\begin{aligned} \therefore I &= \frac{1}{1-n} \int (t)^{\frac{1}{n}} dt = \frac{1}{1-n} \frac{t^{\frac{1+n}{n}}}{\frac{1+n}{n}} + C \\ &= \frac{n}{1-n^2} \left( 1 - \cos^{(1-n)} \theta \right)^{\frac{(n+1)}{n}} + C \end{aligned}$$

---

## Question79

$$\text{If } \frac{x^2+3}{x^4+2x^2+9} = \frac{Ax+B}{x^2+ax+b} + \frac{Cx+D}{x^2+cx+b}, \text{ then } aA + bB + cC + D =$$

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**Options:**

A. 1

B. 0

C. -1

D. 2

**Answer: D**

**Solution:**

Let

$$\begin{aligned} \frac{x^2+3}{x^4+2x^2+9} &= \frac{x^2+3}{(x^2-2x+3)(x^2+2x+3)} \\ &= \frac{Ax+B}{x^2-2x+3} + \frac{Cx+D}{x^2+2x+3} \\ &= \frac{(x^2+2x+3)(Ax+B) + (x^2-2x+3)(Cx+D)}{(x^2-2x+3)(x^2+2x+3)} \end{aligned}$$

On comparing the numerator, we get

$$\begin{aligned} \Rightarrow x^2+3 &= (x^2-2x+3)(Cx+D) + (x^2+2x+3)(Ax+B) \\ &= x^3(A+C) + x^2(2A+B-2C+D) + x(3A+2B+3C-2D) + 3B+3D \end{aligned}$$

Comparing the like terms, we get

$$A + C = 0$$

$$2A + B - 2C + D = 1$$

$$3A + 2B + 3C - 2D = 0$$

$$3B + 3D = 3$$

On solving these equations, we get

$$A = 0, B = \frac{1}{2}, C = 0 \text{ and } D = \frac{1}{2}$$

$$\begin{aligned} \therefore \frac{x^2 + 3}{(x^2 - 2x + 3)(x^2 + 2x + 3)} \\ = \frac{1}{2(x^2 - 2x + 3)} + \frac{1}{2(x^2 + 2x + 3)} \end{aligned}$$

$$\therefore a = -2, b = 3 \text{ and } c = 2$$

$$A = 0, B = \frac{1}{2},$$

$$C = 0, D = \frac{1}{2}$$

Thus,  $aA + bB + cC + D$

$$= (-2)(0) + \frac{1}{2}(3) + 0 + \frac{1}{2} = 2$$

---

## Question 80

$$\int \frac{dx}{x(x^4+1)} =$$

### AP EAPCET 2024 - 20th May Evening Shift

**Options:**

A.  $\log\left(\frac{x}{x^4+1}\right) + c$

B.  $\frac{3}{4}\log(x^4 + 1) + c$

C.  $\frac{1}{3}\log\left(\frac{x^3}{x^4+1}\right) + c$

D.  $\frac{1}{4}\log\left(\frac{x^4}{x^4+1}\right) + c$

**Answer: D**

**Solution:**

$$\int \frac{dx}{x(x^4+1)}$$

Let  $x^4 = t$

then,  $4x^3 dx = dt$

$$\begin{aligned}
 \int \frac{dx}{x(x^4+1)} &= \int \frac{dt}{4x^3 \cdot x(t+1)} = \int \frac{dt}{4t(t+1)} \\
 &= \frac{1}{4} \int \left[ \frac{-1}{(t+1)} + \frac{1}{t} \right] dt \\
 &= \frac{1}{4} [-\log|t+1| + \log|t|] + C \\
 &= \frac{1}{4} \log \left[ \frac{t}{t+1} \right] + C \\
 &= \frac{1}{4} \log \left| \frac{x^4}{x^4+1} \right| + C, \quad (\because x^4 = t)
 \end{aligned}$$


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## Question 81

$$\int \frac{dx}{\sqrt{\sin^3 x \cos(x-a)}} =$$

### AP EAPCET 2024 - 20th May Evening Shift

Options:

- A.  $\frac{1}{\cos \alpha} \sqrt{\cot x + \tan \alpha} + c$
- B.  $\frac{1}{\sqrt{\cos \alpha}} \sqrt{\cot x - \tan \alpha} + c$
- C.  $\frac{-1}{\sqrt{\sin \alpha}} \sqrt{\cot x + \tan \alpha} + c$
- D.  $\frac{-2}{\sqrt{\cos \alpha}} \sqrt{\cot x + \tan \alpha} + c$

**Answer: D**

**Solution:**

$$\begin{aligned}
 \text{Let } I &= \int \frac{dx}{\sqrt{\sin^3 x \cos(x-\alpha)}} \\
 \therefore I &= \int \frac{dx}{\sqrt{\sin^4 x \frac{(\cos x \cos \alpha + \sin x \sin \alpha)}{\sin x}}} \\
 &= \int \frac{\operatorname{cosec}^2 x dx}{\sqrt{\cot x \cos^2 + \sin \alpha}}
 \end{aligned}$$

Let  $\cot x = z$

$$\begin{aligned}
 &\Rightarrow -\operatorname{cosec}^2 x dx = dz \\
 &= - \int \frac{dz}{\sqrt{\cos \alpha z + \sin \alpha}} \\
 &= - \int \frac{dz}{\sqrt{\cos \alpha (z + \tan \alpha)}} \\
 &= - \frac{1}{\sqrt{\cos \alpha}} \int \frac{dz}{\sqrt{z + \tan \alpha}}
 \end{aligned}$$

Let  $z + \tan \alpha = t^2$

$$\begin{aligned}
 \Rightarrow dz &= 2tdt = \frac{-2}{\sqrt{\cos \alpha}} \int \frac{tdt}{\sqrt{t^2}} \\
 &= \frac{-2}{\sqrt{\cos \alpha}} \int dt = \frac{-2}{\sqrt{\cos \alpha}} \cdot t + c \\
 &= \frac{-2}{\sqrt{\cos \alpha}} \cdot \sqrt{z + \tan \alpha} + c \\
 &= \frac{-2\sqrt{\cot x + \tan \alpha}}{\sqrt{\cos \alpha}} + c
 \end{aligned}$$


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## Question82

$$\int \frac{e^{2x}}{\sqrt[4]{e^x+1}} dx =$$

### AP EAPCET 2024 - 20th May Evening Shift

Options:

- A.  $\frac{4}{7}(e^x + 1)^{\frac{4}{3}}(3e^x - 1) + c$
- B.  $\frac{2}{21}(e^x + 1)^{\frac{3}{4}}(3e^x - 7) + c$
- C.  $\frac{4}{21}(e^x + 1)^{3/4}(3e^x - 4) + c$
- D.  $\frac{8}{21}(e^x + 1)^{3/4}(3e^x - 1) + c$

**Answer: C**

**Solution:**

$$\text{Let } I = \int \frac{e^{2x}}{\sqrt[4]{e^x+1}} dx$$

$$\begin{aligned}
 \text{Let } (e^x + 1)^{\frac{1}{4}} &= t \\
 \Rightarrow e^x &= t^4 - 1 \\
 \Rightarrow e^{2x} &= (t^4 - 1)^2 \\
 \Rightarrow 2e^{2x} \cdot dx &= 2(t^4 - 1) \cdot 4t^3 \cdot dt \\
 I &= \int \frac{(t^4 - 1) \cdot 4t^3 dt}{t} \\
 &= 4 \int t^2(t^4 - 1) dt \\
 &= 4 \left[ \frac{t^7}{7} - \frac{t^3}{3} \right] + C \\
 &= 4t^3 \left[ \frac{t^4}{7} - \frac{1}{3} \right] + C \\
 &= \frac{4}{21} t^3 [3t^4 - 1] + C \\
 &= \frac{4}{21} (e^x + 1)^{\frac{3}{4}} (3e^x - 4) + C
 \end{aligned}$$


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## Question83

$$\int \frac{2-\sin x}{2 \cos x+3} dx =$$

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**Options:**

A.  $\frac{2}{\sqrt{5}} \tan^{-1} \left( \frac{1}{\sqrt{3}} \tan \frac{x}{2} \right) - \log \sqrt{2 \cos x + 3} + c$

B.  $\frac{4}{\sqrt{5}} \tan^{-1} \left( \frac{1}{\sqrt{5}} \tan \frac{x}{2} \right) + \log \sqrt{2 \cos x + 3} + c$

C.  $\frac{3}{\sqrt{5}} \tan^{-1} \left( \frac{1}{\sqrt{5}} \tan \frac{x}{2} \right) + \log \sqrt{2 \cos x - 3} + c$

D.  $\frac{1}{\sqrt{5}} \tan^{-1} \left( \frac{1}{\sqrt{5}} \tan \frac{x}{3} \right) - \log \sqrt{2 \cos x - 3} + c$

**Answer: B**

**Solution:**

$$\begin{aligned} & \int \frac{2-\sin x}{2 \cos x+3} dx \\ &= \int \frac{2}{2 \cos x+3} dx - \int \frac{\sin x}{2 \cos x+3} \\ &= 2 \int \frac{\sec^2\left(\frac{x}{2}\right)}{\tan^2 \frac{x}{2}+5} dx + \frac{1}{2} \int \frac{1}{u} du \\ & \left[ \begin{array}{l} \text{let } 2 \cos x + 3 = u \\ \Rightarrow -2 \sin x dx = du \end{array} \right] \\ &= 2 \int \frac{2\sqrt{5}}{5t^2+5} dt + \frac{1}{2} \log |u| \\ & \left[ \text{Let } \frac{1}{\sqrt{5}} \tan \left( \frac{x}{2} \right) = t \Rightarrow \frac{1}{2\sqrt{5}} \sec^2 \frac{x}{2} dx = dt \right] \\ &= \frac{4\sqrt{5}}{5} \int \frac{dt}{t^2+1} + \frac{1}{2} \log |2 \cos x + 3| \\ &= \frac{4}{\sqrt{5}} \tan^{-1} \frac{t}{1} + \frac{\log |2 \cos x + 3|}{2} + C \\ &= \frac{4}{\sqrt{5}} \tan^{-1} \left( \frac{\tan \frac{x}{2}}{\sqrt{5}} \right) + \frac{\log(2 \cos x + 3)}{2} + C \end{aligned}$$

---

## Question84

$$\int \sin^{-1} \sqrt{\frac{x}{a+x}} dx =$$

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**Options:**

A.  $(a+x) \tan^{-1} \sqrt{\frac{x}{a}} - \sqrt{ax} + c$

B.  $\frac{1}{a+x} \tan^{-1} \left( \frac{x}{a} \right) - \sqrt{ax} + c$

C.  $(a+x) \tan^{-1} \left( \frac{a}{x} \right) + \sqrt{ax} + c$

D.  $\sqrt{a+x} \tan^{-1} \left( \frac{x}{a} \right) + ax + c$

**Answer: A**

**Solution:**

$$I = \int \sin^{-1} \sqrt{\frac{x}{a+x}} dx$$

Let  $x = a \tan^2 t$

$$\Rightarrow dx = 2a \tan t \sec^2 t dt$$

$$I = \int \sin^{-1} \sqrt{\frac{a \tan^2 t}{a + a \tan^2 t}} (2a \tan t \sec^2 t) dt$$

$$= 2a \int \sin^{-1} \sqrt{\frac{a \tan^2 t}{a \sec^2 t}} \tan t \sec^2 t dt$$

$$= 2a \int t (\tan t \sec^2 t) dt$$

$$= 2a [t \int \tan t \sec^2 t dt - \int \tan t \sec^2 t dt]$$

$$= 2a \left[ \frac{t \tan^2 t}{2} - \int \frac{\tan^2 t}{2} dt \right]$$

$$= at \tan^2 t - a \tan t + at + C$$

$$= a \left[ \frac{x}{a} \tan^{-1} \left( \sqrt{\frac{x}{a}} \right) - \sqrt{\frac{x}{a}} + \tan^{-1} \sqrt{\frac{x}{a}} \right] + C = (a+x) \tan^{-1} \sqrt{\frac{x}{a}} - \sqrt{ax} + C$$

## Question 85

If  $\frac{A}{x-a} + \frac{Bx+C}{x^2+b^2} = \frac{1}{(x-a)(x^2+b^2)}$ , then C =

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**Options:**

A.  $\frac{-1}{a^2+b^2}$

B.  $\frac{1}{a^2+b^2}$

C.  $\frac{-a}{a^2+b^2}$

D.  $\frac{a}{a^2+b^2}$

**Answer: C**

**Solution:**

We have,  $\frac{A}{x-a} + \frac{Bx+C}{x^2+b^2} = \frac{1}{(x-a)(x^2+b^2)}$

$$\begin{aligned} \Rightarrow \frac{A(x^2 + b^2) + (Bx + C)(x - a)}{(x - a)(x^2 + b^2)} &= \frac{1}{(x - a)(x^2 + b^2)} \\ \Rightarrow A(x^2 + b^2) + (Bx + C)(x - a) &= 1 \\ \Rightarrow Ax^2 + Ab^2 + Bx^2 - Bxa + Cx - Ca &= 1 \\ \Rightarrow (A + B)x^2 + (C - Ba)x + (Ab^2 - Ca) &= 1 \end{aligned}$$

On comparing constants and variables, we get

$$\begin{aligned} A + B &= 0 \Rightarrow B = -A \\ C - Ba &= 0 \Rightarrow C - (-A)a = 0 \Rightarrow C = -Aa \\ Ab^2 - Ca &= 1 \\ \Rightarrow Ab^2 - (-Aa)a &= 1 \Rightarrow Ab^2 + Aa^2 = 1 \\ \Rightarrow A(b^2 + a^2) &= 1 \Rightarrow A = \frac{1}{a^2 + b^2} \end{aligned}$$

Therefore,  $C = -Aa = \frac{-a}{a^2 + b^2}$

---

## Question 86

$\int \frac{2x^2 - 3}{(x^2 - 4)(x^2 + 1)} dx = A \tan^{-1} x + B \log(x - 2) + C \log(x + 2)$ , then  
 $6A + 7B - 5C =$

### AP EAPCET 2024 - 20th May Morning Shift

**Options:**

- A. 9
- B. 10
- C. 6
- D. 8

**Answer: A**

**Solution:**

$$\begin{aligned} \text{Let } I &= \int \frac{2x^2 - 3}{(x^2 - 4)(x^2 + 1)} dx \\ &= \int \frac{(x^2 - 4)}{(x^2 - 4)(x^2 + 1)} + \frac{(x^2 + 1)}{(x^2 - 4)(x^2 + 1)} dx \\ &= \int \left( \frac{1}{x^2 + 1} + \frac{1}{x^2 - 4} \right) dx \\ &= \tan^{-1} x + \int \frac{1}{(x - 2)(x + 2)} dx \\ &= \tan^{-1} x + \frac{1}{4} \int \left( \frac{1}{x - 2} - \frac{1}{x + 2} \right) dx \\ &= \tan^{-1} x + \frac{1}{4} \log(x - 2) - \frac{1}{4} \log(x + 2) + C \end{aligned}$$

Here,  $A = 1$ ,  $B = \frac{1}{4}$  and  $C = -\frac{1}{4}$

$$\begin{aligned}
 6A \rightarrow 7B - 5C &= 6 + \frac{7}{4} - 5\left(-\frac{1}{4}\right) \\
 &= 6 + \frac{7}{4} + \frac{5}{4} = 9
 \end{aligned}$$


---

## Question87

$$\int \frac{3x^9 + 7x^8}{(x^2 + 2x + 5x^8)^2} dx =$$

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**Options:**

A.  $\frac{x^7}{5x^7 + x + 2} + c$

B.  $\frac{x^7}{2(5x^7 + x + 2)} + c$

C.

$\frac{1}{2(5x^7 + x + 2)} + c$

D.  $\frac{-x^7}{2(5x^7 + x + 2)} + c$

**Answer: B**

**Solution:**

$$\begin{aligned}
 \text{We have } I &= \int \frac{3x^9 + 7x^8}{(x^2 + 2x + 5x^8)^2} dx \\
 &= \int \frac{3x^{-7} + 7x^{-8}}{(x^{-6} + 2x^{-7} + 5)^2} dx \quad [\because \text{divide by } x^{16}]
 \end{aligned}$$

$$\begin{aligned}
 \text{Let } x^{-6} + 2x^{-7} + 5 &= t \\
 (-6x^{-7} - 14x^{-8})dx &= dt
 \end{aligned}$$

$$\begin{aligned}
 (3x^{-7} + 7x^{-8})dx &= -\frac{1}{2}dt \\
 I &= -\frac{1}{2} \int \frac{1}{t^2} dt = \frac{1}{2t} + C \\
 &= \frac{1}{2(x^{-6} + 2x^{-7} + 5)} + C \\
 &= \frac{x^7}{2(x + 2 + 5x^7)} + C
 \end{aligned}$$


---

## Question88

$$\int \frac{\cos x + x \sin x}{x(x + \cos x)} dx =$$

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**Options:**

A.  $\log |x^2 + x \cos x| + c$

B.  $\log \left| \frac{x}{x + \cos x} \right| + c$

C.  $\log \left| \frac{\cos x}{x + \cos x} \right| + c$

D.  $\log \left| \frac{1}{x + \cos x} \right| - \log x + c$

**Answer: B**

**Solution:**

$$\begin{aligned}\text{We have } I &= \int \frac{\cos x + x \sin x}{x(x + \cos x)} dx \\ &= \int \frac{(x + \cos x) - x(1 - \sin x)}{x(x + \cos x)} dx \\ &= \int \left( \frac{1}{x} - \frac{1 - \sin x}{x + \cos x} \right) dx \\ &= \log |x| - \log(x + \cos x) + C \\ &= \log \left| \frac{x}{x + \cos x} \right| + C\end{aligned}$$

---

## Question 89

If  $\int \sqrt{\frac{2}{1 + \sin x}} dx = 2 \log |A(x) - B(x)| + C$  and  $0 \leq x \leq \frac{\pi}{2}$ , then  $B\left(\frac{\pi}{4}\right) =$

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**Options:**

A.  $\frac{1}{\sqrt{2+3\sqrt{3}}}$

B.  $\frac{1}{\sqrt{3+2\sqrt{2}}}$

C.  $\frac{-1}{\sqrt{3+2\sqrt{2}}}$

D.  $\frac{2}{\sqrt{2+\sqrt{2}}}$

**Answer: B**

**Solution:**

$$\begin{aligned}
\text{We have } I &= \int \sqrt{\frac{2}{1 + \sin x}} dx \\
&= \int \sqrt{\frac{\frac{2}{1 + 2 \tan x/2}}{\frac{1 + \tan^2 x/2}{1 + \tan^2 x/2}}} dx \\
&= \int \sqrt{\frac{2 \sec^2 x/2}{1 + \tan^2 x/2 + 2 \tan x/2}} dx \\
&= \sqrt{2} \int \frac{\sec x/2}{1 + \tan x/2} dx \\
&= \sqrt{2} \int \frac{1}{\cos \frac{x}{2} + \sin \frac{x}{2}} dx \\
&= \int \frac{1}{\frac{1}{\sqrt{2}} \cos \frac{x}{2} + \frac{1}{\sqrt{2}} \sin \frac{x}{2}} dx \\
&= \int \frac{1}{\sin \left( \frac{\pi}{4} + \frac{x}{2} \right)} dx = \int \operatorname{cosec} \left( \frac{\pi}{4} + \frac{x}{2} \right) dx \\
&= 2 \log \left| \operatorname{cosec} \left( \frac{\pi}{4} + \frac{x}{2} \right) - \cot \left( \frac{\pi}{4} + \frac{x}{2} \right) \right| + C \\
B(x) &= \cot \left( \frac{\pi}{4} + \frac{x}{2} \right) \\
B \left( \frac{\pi}{4} \right) &= \cot \left( \frac{\pi}{4} + \frac{\pi}{8} \right) = \cot \frac{3\pi}{8} = \frac{1}{\sqrt{3 + 2\sqrt{2}}}
\end{aligned}$$


---

## Question90

If  $\int \frac{3}{2 \cos^3 x \sqrt{2 \sin 2x}} dx = \frac{3}{2} (\tan x)^B + \frac{3}{10} (\tan x)^A + C$ , then  
 $A =$

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Options:

- A.  $\frac{1}{2}$
- B. 1
- C. 5
- D.  $\frac{5}{2}$

**Answer: D**

**Solution:**

We begin with the integral:

$$I = \int \frac{3}{2 \cos^3 x \sqrt{2 \sin 2x}} dx$$

Let's simplify this expression:

$$I = \frac{3}{2} \int \frac{\sec^3 x}{\sqrt{4 \sin x \cos x}} dx$$

This can be further rewritten as:

$$I = \frac{3}{4} \int \frac{\sec^2 x \times \sec^2 x dx}{\sqrt{\tan x}}$$

Now, make the substitution  $\tan x = t^2$ . Then, the differential  $\sec^2 x dx$  becomes:

$$\sec^2 x dx = 2t dt$$

Substituting back, the integral becomes:

$$I = \frac{3}{4} \times 2 \int \frac{1}{t} \times (1 + t^4) t dt$$

Simplifying further, we have:

$$I = \frac{3}{2} \int (1 + t^4) dt$$

Evaluating this integral:

$$I = \frac{3}{2} t + \frac{3}{2} \times \frac{t^5}{5} + C$$

Substituting back for  $t = \sqrt{\tan x}$ :

$$I = \frac{3}{2} (\tan x)^{1/2} + \frac{3}{10} (\tan x)^{5/2} + C$$

From this result, we identify that  $B = \frac{1}{2}$  and  $A = \frac{5}{2}$ .

## Question91

If  $\frac{1}{x^4+1} = \frac{Ax+B}{x^2+\sqrt{2}x+1} + \frac{Cx+D}{x^2-\sqrt{2}x+1}$ , then  $BD - AC =$

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**Options:**

A.  $\frac{3}{8}$

B.  $\frac{1}{8}$

C. 1

D. 0

**Answer: A**

**Solution:**

To solve for  $BD - AC$  given the equation  $\frac{1}{x^4+1} = \frac{Ax+B}{x^2+\sqrt{2}x+1} + \frac{Cx+D}{x^2-\sqrt{2}x+1}$ , we follow these steps:

Starting with:

$$\frac{1}{x^4+1} = \frac{(Ax+B)(x^2-\sqrt{2}x+1) + (Cx+D)(x^2+\sqrt{2}x+1)}{(x^2+\sqrt{2}x+1)(x^2-\sqrt{2}x+1)}$$

Simplify the denominator:

$$(x^2 + \sqrt{2}x + 1)(x^2 - \sqrt{2}x + 1) = (x^2 + 1)^2 - (\sqrt{2}x)^2 = x^4 + 1$$

Thus, we expand the numerator:

$$Ax^3 - \sqrt{2}Ax^2 + Ax + Bx^2 - \sqrt{2}Bx + B + Cx^3 + \sqrt{2}Cx^2 + Cx + Dx^2 + \sqrt{2}Dx + D$$

Collecting like terms:

$$x^3 \text{ terms: } A + C = 0$$

$$x^2 \text{ terms: } -\sqrt{2}A + B + \sqrt{2}C + D = 0$$

$$x \text{ terms: } A - \sqrt{2}B + C + \sqrt{2}D = 0$$

$$\text{Constant terms: } B + D = 1$$

From the equations:

$$A + C = 0$$

$$-\sqrt{2}A + B + \sqrt{2}C + D = 0$$

$$A - \sqrt{2}B + C + \sqrt{2}D = 0$$

$$B + D = 1$$

From equation 4, if  $B = D = \frac{1}{2}$ , substituting into equations, we derive:

From equation 1,  $C = -A$ .

Substitute into equation 2:

$$-\sqrt{2}A + B + \sqrt{2}(-A) + D = 0$$

This becomes:

$$-2\sqrt{2}A + \frac{1}{2} + \frac{1}{2} = 0 \implies -2\sqrt{2}A = -1 \implies A = \frac{1}{2\sqrt{2}}$$

$$\text{Then } C = -\frac{1}{2\sqrt{2}}.$$

Now calculate  $BD - AC$ :

$$BD - AC = \left(\frac{1}{2} \times \frac{1}{2}\right) - \left(\frac{1}{2\sqrt{2}} \times \left(-\frac{1}{2\sqrt{2}}\right)\right) = \frac{1}{4} + \frac{1}{8} = \frac{3}{8}$$

---

## Question92

$$\int \frac{2x^2 \cos x^2 - \sin x^2}{x^2} dx =$$

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**Options:**

A.  $\frac{\sin x^2}{x^2} + c$

B.  $\frac{\cos x^2}{x^2} + c$

C.  $\sin x^2 + c$

D.  $\frac{\sin x^2}{x} + c$

**Answer: D**

## Solution:

We know,

$$\begin{aligned}\int [xf'(x) + f(x)] dx &= xf(x) + C \\ \text{We have, } \int \frac{2x^2 \cos x^2 - \sin x^2}{x^2} dx \\ &= \int \frac{x \times \cos x^2 \times 2x - \sin x^2}{x^2} dx \\ &= \\ &= \int \frac{x(\sin x^2)' - (\sin x^2)}{x^2} dx \\ &= \\ &= \frac{\sin x^2}{x} + C\end{aligned}$$

---

## Question93

If  $\int \frac{\log(1+x^4)}{x^3} dx = f(x) \log\left(\frac{1}{g(x)}\right) + \tan^{-1}(h(x)) + c$ , then  $h(x) \left[ f(x) + f\left(\frac{1}{x}\right) \right] =$

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Options:

- A.  $h(x)g(-x)$
- B.  $\frac{g(x)}{2}$
- C.  $g(x) + g(-x)$
- D.  $g(x)h(x)$

**Answer: B**

## Solution:

To evaluate the given integral, we start with:

$$I = \int \frac{\log(1+x^4)}{x^3} dx$$

Applying integration by parts, let's simplify:

$$= \log(1+x^4) \int x^{-3} dx - \int \left( \frac{d}{dx} (\log(1+x^4)) \cdot \int x^{-3} dx \right) dx$$

The derivative  $\frac{d}{dx} \log(1+x^4) = \frac{4x^3}{1+x^4}$ , so the expression becomes:

$$= \frac{\log(1+x^4)}{-2x^2} + \int \frac{2x}{1+x^4} dx$$

The next part simplifies further:

$$= -\frac{1}{2x^2} \log(1+x^4) + \tan^{-1}(x^2) + C$$

Rewriting the logarithm expression:

$$= \frac{1}{2x^2} \log \left( \frac{1}{1+x^4} \right) + \tan^{-1}(x^2) + C$$

We identify the parts:

$$f(x) = \frac{1}{2x^2}$$

$$g(x) = 1 + x^4$$

$$h(x) = x^2$$

Now, consider:

$$h(x) \left[ f(x) + f\left(\frac{1}{x}\right) \right] = x^2 \left[ \frac{1}{2x^2} + \frac{x^2}{2} \right]$$

Simplifying this expression:

$$= x^2 \left[ \frac{1+x^4}{2x^2} \right] = \frac{g(x)}{2}$$

---

## Question94

**Let  $f(x) = \int \frac{x}{(x^2+1)(x^2+3)} dx$ . If  $f(3) = \frac{1}{4} \log \left( \frac{5}{6} \right)$ , then  $f(0) =$**

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**Options:**

A.  $\frac{1}{4} \log \left( \frac{1}{3} \right)$

B. 0

C.  $\frac{1}{2} \log \left( \frac{1}{3} \right)$

D.  $\log \left( \frac{1}{3} \right)$

**Answer: A**

**Solution:**

$$(f(x) = \int \frac{x}{(x^2+1)(x^2+3)} dx = I \text{ (let) Let } x^2 = t, 2x dx = dt$$

$$\begin{aligned}
 I &= \frac{1}{2} \int \frac{1}{(t+1)(t+3)} dt \\
 &= \frac{1}{4} \int \left( \frac{1}{t+1} - \frac{1}{t+3} \right) dt \\
 &= \frac{1}{4} \log \left| \frac{t+1}{t+3} \right| + \log C \\
 &= \frac{1}{4} \log \left| \frac{x^2+1}{x^2+3} \right| + \log c \\
 &= \frac{1}{4} \left( \log \left( \frac{(x^2+1)c^4}{x^2+3} \right) \right) \\
 f(3) &= \frac{1}{4} \log \left| \frac{10}{12} c^4 \right| = \frac{1}{4} \log \frac{5}{6} \\
 c^4 &= \frac{5}{6} \times \frac{12}{10} = 1 \Rightarrow c = 1 \\
 f(0) &= \frac{1}{4} \log \left( \frac{1}{3} \right)
 \end{aligned}$$


---

## Question95

$$\int \frac{2 \cos 2x}{(1+\sin 2x)(1+\cos 2x)} dx =$$

### AP EAPCET 2024 - 19th May Evening Shift

**Options:**

- A.  $2 \tan x + \log(1 + \tan x) + c$
- B.  $\tan x - 2 \log(1 + \tan x) + c$
- C.  $2 \log(1 + \tan x) + \tan x + c$
- D.  $2 \log(1 + \tan x) - \tan x + c$

**Answer: D**

**Solution:**

$$\begin{aligned}
\text{Let } I &= \int \frac{2 \cos 2x}{(1 + \sin 2x)(1 + \cos 2x)} dx \\
&= \int \frac{2 \times (2 \cos^2 x - 1) dx}{(\sin x + \cos x)^2 \times 2 \cos^2 x} \\
&= \int \frac{(2 \cos^2 x - 1)}{(\sin x + \cos x)^2} \sec^2 x dx \\
&= \int \frac{(2 - \sec^2 x)}{(1 + \tan x)^2} \sec^2 x dx \\
&= \int \frac{(1 - \tan^2 x)}{(1 + \tan x)^2} \sec^2 x dx \\
&= \int \left( \frac{1 - \tan x}{1 + \tan x} \right) \sec^2 x dx \\
&= \int \left( \frac{1 - t}{1 + t} \right) dt \quad [\because \tan x = t] \\
&= - \int \left( \frac{t - 1}{1 + t} \right) dt = - \int \left( 1 - \frac{2}{1 + t} \right) dt \\
&= -t + 2 \log(1 + t) + C \\
&= -\tan x + 2 \log(1 + \tan x) + C
\end{aligned}$$


---

## Question 96

$$\int \left( \frac{x}{x \cos x - \sin x} \right)^2 dx =$$

### AP EAPCET 2024 - 19th May Evening Shift

Options:

- A.  $\frac{x \operatorname{cosec} x}{x \cos x - \sin x} + \cot x + c$
- B.  $\frac{x \operatorname{cosec} x}{x \cos x - \sin x} - \cot x + c$
- C.  $\frac{x \operatorname{cosec} x}{x \cos + \sin x} + \cot x + c$
- D.  $\frac{x}{x \cos x - \sin x} - \cot x + c$

**Answer: B**

**Solution:**

$$\begin{aligned}
\text{Let } I &= \int \left( \frac{x}{x \cos x - \sin x} \right)^2 dx \\
&= \int \frac{x^2}{(x \cos x - \sin x)^2} dx \\
&= \int (x \operatorname{cosec} x) \cdot \frac{(x \sin x)}{(x \cos x - \sin x)^2} \dots (i)
\end{aligned}$$

$$\text{Let, } I_1 = \int \frac{x \sin x}{(x \cos x - \sin x)^2} dx$$

$$\text{Let, } x \cos x - \sin x = t \Rightarrow x \sin x dx = -dt$$

$$\Rightarrow = - \int \frac{1}{t^2} dt$$

$$= \frac{1}{x \cos x - \sin x}$$

From Eq. (i), Integrate  $I$ , we get

$$I = (x \operatorname{cosec} x) \int \frac{x \sin x dx}{(x \cos x - \sin x)^2}$$

$$- \int \left[ (x \operatorname{cosec} x)^2 \int \frac{x \sin x dx}{(x \cos x - \sin x)^2} \right] dx$$

$$= \frac{x \operatorname{cosec} x}{x \cos x - \sin x}$$

$$- \int \left[ \frac{-x \operatorname{cosec} x \cdot \cot x + \operatorname{cosec} x}{x \cos x - \sin x} \right] dx$$

$$= \frac{x \operatorname{cosec} x}{x \cos x - \sin x} + \int \frac{1}{\sin^2 x} dx + C$$

$$= \frac{x \operatorname{cosec} x}{x \cos x - \sin x} - \cot x + C$$


---

## Question97

$$\int \frac{1}{x^5 \sqrt[5]{x^5 + 1}} dx =$$

### AP EAPCET 2024 - 18th May Morning Shift

**Options:**

A.  $\frac{4}{\sqrt{x^5+1}} + c$

B.  $4x^4(x^5 + 1)^{4/3} + 0$

C.

$-\frac{(x^5 + 1)^{4/5}}{4x^4} + c$

D.  $-\frac{(x^5+1)^{45}}{4x^5} + c$

**Answer: C**

**Solution:**

We have,

$$\text{Let } I = \int \frac{1}{x^3(x^3 + 1)^{1/5}} dx$$

$$= \int \frac{1}{x^2(1 + \frac{1}{x^3})^{1/5}} dx$$

$$\text{Let } 1 + \frac{1}{x^3} = t^3$$

$$-\frac{5}{x^6}dx = 5t^4 dt \Rightarrow \frac{1}{x^6}dx = -t^4 dt$$

$$I = - \int \frac{1}{t} t^4 dt$$

$$= - \int t^3 dt = -\frac{t^4}{4} + C$$

$$= -\frac{1}{4} \left[ 1 + \frac{1}{x^5} \right]^{4/5} + C$$

$$= -\frac{1}{4} \frac{(x^5 + 1)^{4/5}}{x^4} + C$$

## Question98

$$\int \frac{x+1}{\sqrt{x^2+x+1}} dx =$$

### AP EAPCET 2024 - 18th May Morning Shift

Options:

A.  $\frac{1}{2}\sqrt{x^2+x+1} + \frac{1}{2}\cosh^{-1}\left(\frac{x+2}{\sqrt{3}}\right) + c$

B.  $\frac{1}{2}\sqrt{x^2+x+1} + \frac{2}{\sqrt{3}}\tan^{-1}\left(\frac{2x+1}{\sqrt{3}}\right) + c$

C.  $\sqrt{x^2+x+1} + \frac{2}{\sqrt{3}}\log|x^2+x+1| + c$

D.  $\sqrt{x^2+x+1} + \frac{1}{2}\sinh^{-1}\left(\frac{2x+1}{\sqrt{3}}\right) + c$

**Answer: D**

**Solution:**

$$\text{Let } t = \int \frac{x+1}{\sqrt{x^2+x+1}} dx$$

$$= \frac{1}{2} \int \frac{2x+1}{\sqrt{x^2+x+1}} dx + \frac{1}{2} \int \frac{1}{\sqrt{x^2+x+1}} dx$$

$$I = I_1 + I_2 \text{ (Let)} \quad \dots \text{ (i)}$$

$$\text{Let } I_1 = \frac{1}{2} \int \frac{2x+1}{\sqrt{x^2+x+1}} dx$$

$$\text{Let } x^2 + x + 1 = t^2$$

$$(2x+1)dr = 2dt$$

$$t_1 = \frac{1}{2} \int \frac{x dt}{t}$$

$$= \int dt = t = \sqrt{x^2+x+1}$$

$$I_2 = \frac{1}{2} \int \frac{1}{\sqrt{x^2+x+1}} dr$$

$$= \frac{1}{2} \int \frac{1}{\sqrt{\left(x+\frac{1}{2}\right)^2 + \frac{3}{4}}} dx$$

Let  $x + \frac{1}{2} = \cos dx = dx$

$$\begin{aligned} f_2 &= \frac{1}{2} \int \frac{1}{\sqrt{N^2 + \frac{3}{4}}} \\ &= \frac{1}{2} \sinh^{-1} \left( \frac{\pi}{\sqrt{3}} \right) \\ &= \frac{1}{2} \sinh^{-1} \left( \frac{2x+1}{\sqrt{3}} \right) \end{aligned}$$

Hence.

$$I = \sqrt{x^2 + x + 1} + \frac{1}{2} \sinh^{-1} \left( \frac{2x+1}{\sqrt{3}} \right) + C$$


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## Question99

$$\int (\tan^9 x + \tan x) dx = 0$$

### AP EAPCET 2024 - 18th May Morning Shift

Options:

A.  $\frac{\tan^2 x}{12} (2 \tan^3 x - 3 \tan^2 x + 6) + c$

B.  $\frac{\tan^2 x}{6} - \frac{\tan^5 x}{4} + \frac{\tan^2 x}{2} + c$

C.  $\frac{\tan^2 x^2}{6} \tan^4 x + 3 \tan^2 x + 4) + c$

D.  $\frac{\tan x}{12} \tan^4 x - 3 \tan^2 x + 6 + c$

**Answer: A**

**Solution:**

$$\begin{aligned} \text{Let } I &= \int (\tan^3 x + \tan x) dx \\ &= \int \tan x (\tan^6 x + 1) dx \\ &= \int \tan x ((\tan^2 x)^3 + 1^3) dx \\ &= \int \tan x (\tan^2 x + 1) (\tan^4 x + 1 - \tan^2 x) dx \\ &= \int (\tan^3 x - \tan^3 x + \tan x) \sec^2 x dx \end{aligned}$$

Let  $\tan x = t$

$\sec^2 x, dx = dt$

$$\begin{aligned} I &= \int (t^5 - t^3 + t) dt \\ &= \frac{t^6}{6} - \frac{t^4}{4} + \frac{t^2}{2} + C = \frac{t^2 (2t^3 - 3t^2 + 6)}{12} + C \\ &= \frac{\tan^2 x}{12} (2 \tan^3 x - 3 \tan^2 x + 6) + C \end{aligned}$$


---

## Question100

$$\int \frac{\operatorname{cosec} x}{3 \cos x + 4 \sin x} dx =$$

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**Options:**

A.  $\frac{1}{2} \log \left| \frac{\cos x}{3 \sin x + 4 \cos x} \right| + c$

B.  $\frac{1}{3} \log \left| \frac{\sin x}{3 \cos x + 4 \sin x} \right| + c$

C.  $\frac{1}{3} \log \left| \frac{3 \cos x + \sin x}{3 \cos x + 4 \sin x} \right| + c$

D.  $\frac{1}{2} \log \left| \frac{\cos x + 4 \sin x}{3 \cos x + 4 \sin x} \right| + c$

**Answer: B**

**Solution:**

Let

$$\begin{aligned} I &= \int \frac{\operatorname{cosec} x}{3 \cos x + 4 \sin x} dx \\ &= \int \frac{\operatorname{cosec}^2 x}{3 \cot x + 4} dx \end{aligned}$$

$$\begin{aligned} \text{Let } 3 \cot x + 4 &= t \\ -3 \operatorname{cosec}^2 x dx &= dt \end{aligned}$$

$$\operatorname{cosec}^2 x dx = \frac{-1}{3} dt$$

$$\begin{aligned} I &= -\frac{1}{3} \int \frac{1}{t} dt \\ &= \frac{-1}{3} \log |t| + C = \frac{1}{3} \log \left| \frac{1}{t} \right| + C \\ &= \frac{1}{3} \log \left| \frac{1}{3 \cot x + 4} \right| + C \\ &= \frac{1}{3} \log \left| \frac{\sin x}{3 \cos x + 4 \sin x} \right| + C \end{aligned}$$

---

## Question101

$$\int e^{2x+3} \sin 6x dx =$$

**AP EAPCET 2024 - 18th May Morning Shift**

**Options:**

A.  $\frac{e^{2x+3}}{40} (2 \sin(x) + 6 \cos 6x) + c$

B.  $\frac{e^{2v+3}}{40}(2 \cos 6x + 6 \sin 6x) + c$

C.  $\frac{e^{2n+3}}{20}(\sin 6x - 3 \cos 6x) + c$

D.  $\frac{e^{2n3}}{20}(\cos 8x - 3 \sin 6x) + c$

**Answer: C**

**Solution:**

Let  $I = \int e^{2x+3} \sin 6x dx$

$$= \frac{e^{2x+3} \sin 6x}{2} - \int 3e^{2x+3} \cdot \cos 6x dx$$

$$= \frac{e^{2x+3} \sin 6x}{2}$$

$$- \frac{\left[ \frac{3e^{2x+3} \cos 6x}{2} - \int -9e^{2x+3} \sin 6x dx \right]}{2} = \frac{e^{2x+3} \sin 6x}{2} - \frac{3e^{2x+3} \cos 6x}{2} - 9I$$

$$\Rightarrow 10I = \frac{e^{2x+3}(\sin 6x - 3 \cos 6x)}{2}$$

$$\therefore I = \frac{e^{2x+3}(\sin 6x - 3 \cos 6x)}{20} + C$$

## Question102

$$\frac{2x^2+1}{x^3-1} = \frac{A}{x-1} + \frac{Bx+C}{x^2+x+1} \Rightarrow 7A + 2B + C =$$

## AP EAPCET 2022 - 5th July Morning Shift

**Options:**

A. 8

B. 9

C. 10

D. 11

**Answer: B**

**Solution:**

Here,  $\frac{2x^2+1}{x^3-1} = \frac{A}{x-1} + \frac{Bx+C}{x^2+x+1}$

Resolving into partial fractions,

$$\frac{2x^2+1}{x^3-1} = \frac{A(x^2+x+1)+(Bx+C)(x-1)}{(x-1)(x^2+x+1)}$$

$$\Rightarrow 2x^2 + 1 = Ax^2 + Ax + A + Bx^2 - Bx + Cx - C$$

$$\Rightarrow 2x^2 + 1 = x^2(A + B) + x(A - B + C) + A - C$$

Comparing with coefficient, we get

$$A + B = 2, A - B + C = 0 \text{ and } A - C = 1$$

Clearly,  $A = 1, B = 1$  and  $C = 0$

$$\text{So, } 7A + 2B + C = 7(1) + 2(1) + 0 = 9$$

## Question 103

$$\int \frac{3x+4}{x^3-2x+4} dx = \log f(x) + C \Rightarrow f(3) =$$

### AP EAPCET 2022 - 5th July Morning Shift

**Options:**

A.  $\frac{1}{\sqrt{17}}$

B.  $\frac{1}{17}$

C.  $\frac{2}{15}$

D.  $\frac{2}{17}$

**Answer: A**

**Solution:**

$$\int \frac{3x+4}{x^3-2x+4} dx = \log f(x) + C$$

$$I = \int \frac{3x+4}{x^3-2x+4}$$

$$\text{Since, } x^3 - 2x - 4 = (x - 2)(x^2 + 2x + 2)$$

$$\frac{3x+4}{x^3-2x+4} = \frac{A}{(x-2)} + \frac{Bx+C}{(x^2+2x+2)}$$

$$\Rightarrow 3x+4 = A(x^2+2x+2) + (Bx+C)(x-2)$$

On putting  $x = 2$ ,  $A = 1$  and comparing coefficients of  $x^2$ , we get

$$0 = A + B \Rightarrow B = -1$$

On putting  $x = 0$ , we have  $2A - 2C$

$$\Rightarrow C = -1$$

$$\int \frac{3x+4}{x^3-2x+4} dx = \int \frac{dx}{x-2} - \int \frac{x+1}{x^2+2x+2} dx$$

$$= \log |x-2| - \frac{1}{2} \log |(x^2+2x+2)| + C$$

$$= \log f(x) + C = \log \frac{|x-2|}{\sqrt{(x^2+2x+2)}} + C$$

$$f(x) = \frac{|x-2|}{\sqrt{(x^2+2x+2)}}$$

So,  $f(3) = \frac{|3-2|}{\sqrt{((3)^2+2(3)+2)}}$

$$f(3) = \frac{1}{\sqrt{9+6+2}} = \frac{1}{\sqrt{17}}$$


---

## Question104

$$\int \frac{e^{\tan^{-1} x}}{1+x^2} \left[ \left( \sec^{-1} \sqrt{1+x^2} \right)^2 + \cos^{-1} \left( \frac{1-x^2}{1+x^2} \right) \right] dx =$$

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**Options:**

- A.  $e^{\tan^{-1} x} (\tan^{-1} x)^2 + C$
- B.  $e^{\tan^{-1} x} (\sec^{-1} x)^2 + C$
- C.  $e^{\tan^{-1} x} \left( \sec^{-1} \left( \sqrt{1+x^2} \right) \right) + C$
- D.  $e^{\tan^{-1} x} \times \left( \cos^{-1} \left( \frac{1-x^2}{1+x^2} \right) \right) + C$

**Answer: A**

**Solution:**

Let

$$I = \int \frac{e^{\tan^{-1}(x)}}{1+x^2} \left[ \left( \sec^{-1} \sqrt{1+x^2} \right)^2 + \cos^{-1} \left( \frac{1-x^2}{1+x^2} \right) \right] dx$$

On putting  $\tan^{-1} x = t$  and  $\frac{dx}{1+x^2} = dt$ , we get

$$I = \int e^t [t^2 + 2t] dt \Rightarrow I = \int (t^2 e^t + 2te^t) dt$$

By integration by parts, we get

$$I = t^2 e^t - \int 2te^t dt - \int 2te^t dt + C$$

$$I = t^2 e^t + C = e^{\tan^{-1}(x)} (\tan^{-1} x)^2 + C$$


---

## Question105

$$\int \frac{dx}{(x-3)^{\frac{4}{5}} (x+1)^{\frac{6}{5}}} =$$

## AP EAPCET 2022 - 5th July Morning Shift

Options:

A.  $\frac{5}{4} \sqrt[5]{\frac{x-3}{x+1}} + C$

B.  $\frac{5}{4} \left( \frac{x+1}{x-3} \right)^{\frac{1}{5}} + C$

C.  $\frac{1}{5} \left( \frac{x-3}{x+1} \right)^{\frac{1}{5}} + c$

D.  $\frac{5}{4} \left( \frac{x-3}{x+4} \right)^{\frac{4}{5}} + C$

Answer: A

Solution:

$$\text{Let } I = \int \frac{dx}{(x+1)^{\frac{6}{5}}(x-3)^{\frac{4}{5}}}$$
$$I = \int \frac{dx}{(x+1)^2 \left( \frac{x-3}{x+1} \right)^{\frac{4}{5}}}$$

On putting  $\frac{(x-3)}{(x+1)} = t$ , then

$$dt = \frac{4}{(x+1)^2} dx$$

$$I = \int \frac{dt}{4t^{\frac{4}{5}}} = \frac{5}{4} t^{\frac{1}{5}} + C = \frac{5}{4} \left( \frac{x-3}{x+1} \right)^{\frac{1}{5}} + C$$

---

## Question106

If  $I_n = \int (\cos^n x + \sin^n x) dx$  and  $I_n - \frac{n-1}{n} I_{n-2} = \frac{\sin x \cos x}{n} f(x)$ , then  $f(x) =$

## AP EAPCET 2022 - 5th July Morning Shift

Options:

A.  $\cos^{n-2} x + \sin^{n-2} x$

B.  $\cos^{n-2} x - \sin^{n-2} x$

C.  $\frac{\cos^{n-2} x - \sin^{n-2} x}{n}$

D.  $\frac{\cos^{n-2} x + \sin^{n-2} x}{n}$

Answer: B

## Solution:

Here,  $I_n = \int (\cos^n x + \sin^n x) dx$

$$I_n = \int \cos^{n-1} x \cos x dx + \int \sin^{n-1} x \sin x dx$$

Using by parts,

$$I_n = \cos^{n-1} x \int \cos x dx - \int \left( \frac{d}{dx} (\cos^{n-1} x) \int \cos x dx \right) dx \\ + \sin^{n-1} x \int \sin x dx - \int \left( \frac{d}{dx} (\sin^{n-1} x) \int \sin x dx \right) dx$$

$$\Rightarrow I_n = \cos^{n-1} x \sin x + \int (n-1) \cos^{n-2} x \sin^2 x dx$$

$$- \sin^{n-1} x \cos x + \int (n-1) \sin^{n-2} x \cos^2 x dx$$

$$\Rightarrow I_n = \cos^{n-1} x \sin x - \sin^{n-1} x \cos x$$

$$+ \int (n-1) \cos^{n-2} x (1 - \cos^2 x) dx$$

$$+ \int (n-1) \sin^{n-2} x (1 - \sin^2 x) dx$$

$$\Rightarrow I_n = \cos^{n-1} x \sin x - \sin^{n-1} x \cos x$$

$$+ \int (n-1) \cos^{n-2} x dx$$

$$- \int (n-1) \sin^{n-2} x dx + \int (n-1) \sin^{n-2} x dx$$

$$- \int (n-1) \sin^n x dx$$

$$\Rightarrow I_n = \cos^{n-1} x \sin x - \sin^{n-1} x \cos x$$

$$+ \int (n-1) \cos^{n-2} x dx + \int (n-1) \sin^{n-2} x dx$$

$$- [(n-1) \int (\cos^n x + \sin^n x) dx]$$

$$\Rightarrow I_n = \cos^{n-1} x \sin x - \sin^{n-1} x \cos x$$

$$+ (n-1)I_{n-2} - (n-1)I_n$$

$$\Rightarrow I_n(1 + n - 1) - (n-1)I_{n-2}$$

$$= \cos^{n-1} x \sin x - \sin^{n-1} x \cos x$$

$$\Rightarrow n(l_n) - (n-1)I_{n-2} = \cos^{n-1} x \sin x - \sin^{n-1} x \cos x$$

$$\Rightarrow I_n - \left( \frac{n-1}{n} \right) I_{n-2} = \frac{\sin x \cos x}{n} (\cos^{n-2} x - \sin^{n-2} x)$$

$$= \frac{\sin x \cos x}{n} f(x)$$

$$\therefore f(x) = \cos^{n-2} x - \sin^{n-2} x$$

---

## Question107

If  $f(x) = \int x^2 \cos^2 x (2x \tan^2 x - 2x - 6 \tan x) dx$  and  $f(0) = \pi$ , then  $f(x) =$

**Options:**

A.  $x^2 \sin x + \pi$

B.  $\cos x + \pi - 1$

C.  $-x^3 \sin 2x + \pi$

D.  $x^3 \cos 2x + \pi \cos x$

**Answer: C**

**Solution:**

$$\begin{aligned} f(x) &= \int x^2 \cos^2 x (2x \tan^2 x - 2x - 6 \tan x) dx \\ &= \int (2x^3 \sin^2 x - 2x^3 \cos^2 x) dx - 3 \int x^2 (2 \sin x \cdot \cos x) dx \\ &= - \int 2x^3 \cdot \cos 2x dx - 3 \int x^2 \sin 2x dx \\ &= - \left[ 2x^3 \int \cos 2x dx - \int 2 \cdot 3x^2 \frac{\sin 2x}{2} dx \right] - 3 \int x^2 \sin 2x dx \\ &= -x^3 \sin 2x + 3 \int x^2 \sin 2x dx - 3 \int x^2 \sin 2x dx \\ f(x) &= -x^3 \sin 2x + C \\ \therefore f(0) &= \pi \Rightarrow C = \pi \\ \therefore f(x) &= -x^3 \sin 2x + \pi \end{aligned}$$

---

## Question 108

If  $\int \frac{e^{\sqrt{x}}}{\sqrt{x}} (x + \sqrt{x}) dx = e^{\sqrt{x}} [Ax + B\sqrt{x} + C] + K$  then  $A + B + C =$

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**Options:**

A.  $-2$

B.  $2$

C.  $4$

D.  $-4$

**Answer: B**

**Solution:**

$$\int \frac{e^{\sqrt{x}}}{\sqrt{x}} (x + \sqrt{x}) dx$$

$$= \int (\sqrt{x}e^{\sqrt{x}} + e^{\sqrt{x}}) dx = \int \sqrt{x}e^{\sqrt{x}} dx + \int e^{\sqrt{x}} dx$$

$$\text{Now, } \int \sqrt{x} \cdot e^{\sqrt{x}} dx$$

$$\text{Let } x = p^2$$

$$\Rightarrow dx = 2p dp$$

$$= \int \sqrt{x}e^{\sqrt{x}} dx = \int pe^p(2p dp) = 2 \int p^2 e^p dp$$

$$= 2 \left[ p^2 e^p - 2 \int pe^p dp \right]$$

$$= 2 \left[ p^2 e^p - 2 \{pe^p - e^p\} \right] + K$$

$$= 2 \left[ p^2 e^p - 2pe^p + 2e^p \right] + K$$

$$= 2 \left[ xe^{\sqrt{x}} - 2\sqrt{x}e^{\sqrt{x}} + 2e^{\sqrt{x}} \right] + K$$

$$\int e^{\sqrt{x}} dx = 2 \int pe^p dp = 2 [pe^p - e^p] = 2 \left[ \sqrt{x}e^{\sqrt{x}} - e^{\sqrt{x}} \right]$$

$$\therefore \int \frac{e^{\sqrt{x}}}{\sqrt{x}} (x + \sqrt{x}) dx$$

$$= 2xe^{\sqrt{x}} - 4\sqrt{x}e^{\sqrt{x}} + 4e^{\sqrt{x}} + 2\sqrt{x}e^{\sqrt{x}} - 2e^{\sqrt{x}} + K$$

$$= e^{\sqrt{x}}(2x - 2\sqrt{x} + 2) + K$$

$$\therefore A = 2, B = -2, C = 2$$

$$\therefore A + B + C = 2$$

## Question 109

If  $\int \frac{1+\sqrt{\tan x}}{\sin 2x} dx = A \log \tan x + B \tan x + C$ , then  $4A - 2B =$

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**Options:**

A.  $-1$

B.  $2$

C.  $1$

D.  $-2$

**Answer: A**

**Solution:**

$$\int \frac{1+\sqrt{\tan x}}{\sin 2x} dx$$

$$\begin{aligned}
&= \int \operatorname{cosec} 2x dx + \int \frac{\sqrt{\tan x}}{\sin 2x} dx \Rightarrow \int \frac{1 + \sqrt{\tan x}}{\sin x} dx \\
&= \int \operatorname{cosec} 2x dx + \frac{\sqrt{\tan x}}{\sin 2x} dx \\
&= \frac{1}{2} \log |\tan x| + \frac{1}{2} \int \frac{\sqrt{\sin x}}{\sqrt{\cos x} \sin x \cos x} dx \\
&= \frac{1}{2} \log |\tan x| + \frac{1}{2} \int \frac{dx}{\sqrt{\cos x} \sqrt{\sin x}} \cos x \frac{\sqrt{\cos x}}{\sqrt{\cos x}} \\
&= \frac{1}{2} \log |\tan x| + \frac{1}{2} \int \frac{\sec^2 x dx}{\sqrt{\tan x}} \\
&= \frac{1}{2} \log |\tan x| + \frac{1}{2} (2) \cdot \sqrt{\tan x} + C \\
&= \frac{1}{2} \log |\tan x| + \sqrt{\tan x} + C \\
\therefore 4A - 2B &= 4 \times \frac{1}{2} - 2(l) = 2 - 2 = 0
\end{aligned}$$


---

## Question 110

$$\int \frac{1 + \tan x \tan(x+a)}{\tan x \tan(x+a)} dx =$$

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**Options:**

- A.  $\tan a (\log(\sec(x+a)) + \log \sec x) + C$
- B.  $\cot a (\log |\sin x| - \log |\sin(x+a)|) + C$
- C.  $\tan a \left( \log \left( \frac{\cos x}{\sin(x+a)} \right) \right) + C$
- D.  $\cot a \left( \log \frac{\sin(x+a)}{\cos(x+a)} \right) + C$

**Answer: B**

**Solution:**

$$\begin{aligned}
&\int \frac{1 + \tan x \cdot \tan(x+a)}{\tan x \cdot \tan(x+a)} dx \\
&= \int \frac{\cos(x) \cos(x+a) + \sin x \sin(x+a)}{\sin x \sin(x+a)} dx \\
&= \int \frac{\cos(a)}{\sin x \cdot \sin(x+a)} dx = \frac{\cos a}{\sin a} \int \frac{\sin(x+a-x)}{\sin x \cdot \sin(x+a)} dx \\
&= \cot a \int \frac{\sin(x+a) \cos x - \cos(x+a) \sin x}{\sin x \sin(x+a)} dx \\
&= \cot a \left[ \int \cot x dx - \int \cot(x+a) dx \right] + C \\
&= \cot a [\log |\sin x| - \log |\sin(x+a)|] + C
\end{aligned}$$

---

## Question111

**Assertion (A)** If  $I_n = \int \cot^n x dx$ , then  $I_6 + I_4 = \frac{-\cot^5 x}{5}$

**Reason (R)**  $\int \cot^n x dx = \frac{-\cot^{n-1} x}{n} - \int \cot^{n-2} x dx$

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**Options:**

- A. A is false, R is false
- B. A is true, R is true
- C. A is true, R is false
- D. A is false, R is true

**Answer: C**

**Solution:**

$$I_n = \int \cot^n x dx$$

$$I_6 + I_4 = \int \cot^6 x dx + \int \cot^4 x dx$$

$$= \int \cot^4 x (\cot^2 x + 1) dx$$

$$= - \int \cot^4 x \cdot (-\operatorname{cosec}^2 x) dx$$

$$= - \int (\cot x)^4 \cdot (-\operatorname{cosec}^2 x) dx = - \frac{(\cot x)^5}{5}$$

$\therefore$  Assertion is true.

$$\int \cot^n x dx = \int \cot^{n-2} x \cdot \cot^2 x dx$$

$$= \int \cot^{n-2} x \cdot (\operatorname{cosec}^2 x - 1) dx$$

$$= \int \cot^{n-2} x \cdot \operatorname{cosec}^2 x \cdot dx - \int \cot^{n-2} x dx$$

$$= - \int (\cot x)^{n-2} x \cdot (-\operatorname{cosec}^2 x) dx - \int \cot^{n-2} x dx$$

$$= - \frac{(\cot x)^{n-1}}{n-1} - \int \cot^{n-2} x dx$$

$\therefore$  Reason is false.

---

## Question112

If  $I_n = \int \tan^n x dx$ , and  $I_0 + I_1 + 2I_2 + 2I_3 + 2I_4 + I_5 + I_6 = \sum_{k=1}^n \frac{\tan^k x}{k}$ , then  $n =$

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**Options:**

A. 6

B. 5

C. 4

D. 3

**Answer: B**

**Solution:**

$$\begin{aligned} I_n &= \int \tan^n x dx \\ I_0 + I_1 + 2I_2 + 2I_3 + 2I_4 + I_5 + I_6 &= \int \tan^0 x dx + \int \tan^1 x dx + 2 \int \tan^2 x dx + 2 \int \tan^3 x dx \\ &\quad + 2 \int \tan^4 x dx + \int \tan^5 x \cdot dx + \int \tan^6 x dx \\ &= \int [1 + \tan x + 2 \tan^2 x + 2 \tan^3 x + 2 \tan^4 x + \tan^5 x \\ &\quad \quad \quad + \tan^6 x] dx \\ &= \int [1 + \tan^2 x + (\tan x + \tan^3 x) + (\tan^2 x + \tan^4 x) \\ &\quad \quad \quad + (\tan^3 x + \tan^5 x) + (\tan^4 x + \tan^6 x)] dx \\ &= \int [1 + \tan^2 x + \tan x (1 + \tan^2 x) + \tan^2 x (1 + \tan^2 x) \\ &\quad \quad \quad + \tan^3 x (1 + \tan^2 x) + \tan^4 x (1 + \tan^2 x)] dx \\ &= \int [\sec^2 x + \tan x \sec^2 x + \tan^2 x \cdot \sec^2 x \\ &\quad \quad \quad + \tan^3 x \sec^2 x + \tan^4 x \sec^2 x] dx \\ &= \int [1 + \tan x + \tan^2 x + \tan^3 x + \tan^4 x] \sec^2 x dx \\ &= \int \sec^2 x \cdot dx + \int (\tan x + \tan^2 x + \tan^3 x + \tan^4 x) \sec^2 x dx \\ &= \frac{\tan x}{1} + \frac{\tan^2 x}{2} + \frac{\tan^3 x}{3} + \frac{\tan^4 x}{4} + \frac{\tan^5 x}{5} \\ &= \sum_{k=1}^5 \frac{\tan^k x}{k} = \sum_{k=1}^n \frac{\tan^k x}{k} \Rightarrow n = 5 \end{aligned}$$

---

## Question113

$$\int \frac{e^{\cot x}}{\sin^2 x} (2 \log \operatorname{cosec} x + \sin 2x) dx =$$

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Options:

- A.  $-2e^{\cot x} \log(\operatorname{cosec}^2 x) + C$
- B.  $-2e^{\cot x} \log(\operatorname{cosec} x) + C$
- C.  $-2e^{\cot x} \log(\operatorname{cosec} x + \sin x) + C$
- D.  $-2e^{\cot x} \log(\operatorname{cosec} x - \cot x) + C$

**Answer: B**

**Solution:**

$$\begin{aligned}\text{Let } I &= \int \frac{e^{\cos x}}{\sin^2 x} (2 \log \operatorname{cosec} x + \sin 2x) dx \\ I &= \int e^{\cos x} (\log \operatorname{cosec}^2 x + \sin 2x) \cdot \operatorname{cosec}^2 x dx\end{aligned}$$

$$\text{Let } \cot x = t \Rightarrow -\operatorname{cosec}^2 x dx = dt$$

$$\begin{aligned}I &= - \int e^t \left[ \log(1+t^2) + \frac{2t}{1+t^2} \right] dt \\ \left[ \text{Here, } \frac{2t}{1+t^2} &= \frac{2 \cot x}{1+\cot^2 x} = \frac{2 \cot x}{\operatorname{cosec}^2 x} \right. \\ &= \left. \frac{2 \cos x / \sin x}{1/\sin^2 x} = \sin 2x \right]\end{aligned}$$

$$\Rightarrow I = -e^t \cdot \log(1+t^2) + C$$

$$\left[ \int e^x (f(x) + f'(x)) dx = e^x \cdot f(x) + C \right]$$

$$\Rightarrow I = -e^{\cot x} \log \operatorname{cosec}^2 x + C$$

$$\Rightarrow I = -2e^{\cot x} \log \operatorname{cosec} x + C$$

---

## Question 114

The parametric form of a curve is  $x = \frac{t^3}{t^2-1}$ ,  $y = \frac{t}{t^2-1}$ , then  $\int \frac{dx}{x-3y} =$

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Options:

- A.  $\frac{1}{2} \log(t^2 - 1) + C$
- B.  $2 \log(t(t^2 - 1)) + C$

C.  $\frac{1}{4}\log\left(\frac{t}{t^2-3}\right) + C$

D.  $\frac{5}{2}\log\left(t + \frac{1}{t^2}\right) + C$

**Answer: A**

**Solution:**

$$x = \frac{t^3}{t^2-1}, y = \frac{t}{t^2-1} \text{ and let } I = \int \frac{dx}{x-3y}$$

$$x = \frac{t^3}{t^2-1}$$

On differentiating w.r.t.  $t$ , we get

$$\frac{dx}{dt} = \frac{(t^2-1)(3t^2) - t^3(2t)}{(t^2-1)^2}$$

$$\Rightarrow \frac{dx}{dt} = \frac{3t^4 - 3t^2 - 2t^4}{(t^2-1)^2}$$

$$\Rightarrow dx = \frac{t^4 - 3t^2}{(t^2-1)^2} dt$$

$$\Rightarrow dx = \frac{t^2(t^2-3)}{(t^2-1)^2} dt$$

$$\Rightarrow I = \int \frac{1}{\frac{t^3}{t^2-1} - \frac{3t}{t^2-1}} \cdot \frac{t^2(t^2-3)}{(t^2-1)^2} dt$$

$$= \int \frac{(t^2-1)}{t(t^2-3)} \cdot \frac{t^2(t^2-3)}{(t^2-1)^2} dt$$

$$= \int \frac{t}{t^2-1} dt = \frac{1}{2} \int \frac{2t}{t^2-1} dt$$

$$= \frac{1}{2} \log(t^2-1) + C$$

## Question115

**If**

$$\frac{2x^4 - x^3 + 3x^2 - x + 4}{x^2 - 3x + 2} = f(x) + \frac{A}{x-1} + \frac{B}{x-2}, \text{ then}$$

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**Options:**

A.  $f(x) = 2x^2 + 5x + 14, A + B = 39$

B.  $f(x) = 2x^2 - 5x + 14, A + B = 31$

C.  $f(x) = 2x^2 + 5x + 14, A + B = 31$

D.  $f(x) = 2x^2 + 5x + 14, A = 4, B = 35$

**Answer: C**

**Solution:**

$$\frac{2x^4 - x^3 + 3x^2 - x + 4}{x^2 - 3x + 2}$$

∴ Degree of numerator is greater than the degree of denominator.

∴ First we divide,

$$\begin{array}{r}
 2x^2 + 5x + 14 \\
 x^2 - 3x + 2 \overline{) 2x^4 - x^3 + 3x^2 - x + 4} \\
 \underline{2x^4 - 6x^3 + 4x^2} \phantom{+ 4} \\
 5x^3 - x^2 - x + 4 \\
 \underline{5x^3 - 15x^2 + 10x} \phantom{+ 4} \\
 14x^2 - 11x + 4 \\
 \underline{14x^2 - 42x + 28} \\
 31x - 24
 \end{array}$$

$$\begin{aligned}
 \text{or } &= \frac{2x^4 - x^3 + 3x^2 - x + 4}{x^2 - 3x + 2} \\
 &= (2x^2 + 5x + 14) + \frac{31x - 24}{(x - 2)(x - 1)} \quad \dots(i)
 \end{aligned}$$

$$\text{Let } \frac{31x - 24}{(x - 2)(x - 1)} = \frac{A}{x - 1} + \frac{B}{x - 2}$$

$$\Rightarrow 31 - 24 = -A(x - 2) + B(x - 1) \quad \dots (ii)$$

Put  $x = 1$

$$\Rightarrow 31 - 24 = -A \Rightarrow A = -7$$

Put  $x = 2$  in Eq. (ii)

$$31 \times 2 - 24 = 0 + B \Rightarrow B = 38$$

∴ From Eq. (i)  $f(x) = 2x^2 + 5x + 14$

$$\text{and } A + B = -7 + 38 = 31$$

## Question116

If  $f'(x) = x + \frac{1}{x}$ , then  $f(x)$  is equal to

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**Options:**

A.  $x^2 + \log(x) + c$

B.  $\frac{x^2}{2} + \log(x) + c$

C.  $x + \log(x) + c$

D.  $\frac{x}{2} + \log(x) + c$

**Answer: B**

**Solution:**

$f'(x) = x + \frac{1}{x}$ , then integrate it

$$f(x) = \int \left(x + \frac{1}{x}\right) dx + c = \frac{x^2}{2} + \log x + c$$


---

## Question117

If  $f(x) = \frac{1}{(\cos^2 x)\sqrt{1+\tan x}}$ , then its antiderivative  $F(x) = \dots\dots\dots$ , given,  $F(0) = 4$

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**Options:**

A.  $\sqrt{1 + \tan x} + 4$

B.  $\frac{2}{3}(1 + \tan x)^{\frac{3}{2}}$

C.  $2(\sqrt{1 + \tan x} + 1)$

D.  $\sqrt{1 + \tan x} + 2$

**Answer: C**

**Solution:**

$$f(x) = \frac{1}{\cos^2 x \sqrt{1 + \tan x}} = \frac{\sec^2 x}{\sqrt{1 + \tan x}}$$

$$\Rightarrow \int f(x) dx = \int \frac{\sec^2 x}{\sqrt{1 + \tan x}} dx$$

$$\Rightarrow \text{Let } 1 + \tan x = u, \text{ then } \sec^2 x dx = du$$

$$\Rightarrow \int f(x) dx = \int \frac{du}{\sqrt{u}} = 2\sqrt{u} + C$$

$$\Rightarrow F(x) = 2\sqrt{1 + \tan x} + C$$

$$\text{Given, } F(0) = 4 = 2\sqrt{1} + C \Rightarrow C = 2$$

$$\therefore F(x) = 2(\sqrt{1 + \tan x} + 1)$$


---

## Question118

If the primitive of  $\cos(\log x)$  is  $f(x)\{\cos(g(x)) + \sin(h(x))\}$ , then which among the following is true?

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Options:

A.  $h'(x) = \frac{-1}{x}$

B.  $f'(x) = \frac{1}{2}$

C.  $g'(x) = \log(x)$

D.  $h(x) = \frac{x}{2}$

**Answer: B**

**Solution:**

Given,

$$\begin{aligned}\int \cos(\log x) dx &= f(x)\{\cos(g(x)) + \sin(h(x))\} \\ I &= \int \cos(\log x) dx \\ &\text{{ using by-part method of integration }} \\ &= \cos(\log x) \cdot x + \int \frac{\sin(\log x)}{x} \cdot x dx \\ &= \cos(\log x) \cdot x + \int \sin(\log x) dx \\ &= \cos(\log x) \cdot x + \left[ \sin(\log x) \cdot x - \int \frac{\cos(\log x)}{x} \cdot x dx \right] \\ I &= x \cdot \cos(\log x) + x \cdot \sin(\log x) - I \\ 2I &= x(\cos(\log x) + \sin(\log x)) \\ \Rightarrow I &= \frac{x}{2}(\cos(\log x) + \sin(\log x)) \\ \therefore f(x) &= \frac{x}{2} \Rightarrow f'(x) = \frac{1}{2}\end{aligned}$$

---

## Question119

$\int \frac{\sec x}{\sqrt{\sin(2x+\theta)+\sin \theta}} dx$  is equal to

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**Options:**

A.  $\sqrt{(\tan x + \tan \theta) \sec \theta} + c$

B.  $\sqrt{2(\tan x + \tan \theta) \sec \theta} + c$

C.  $\sqrt{2(\sin x + \tan \theta) \sec \theta} + c$

D.  $\sqrt{2(\cos x + \tan \theta) \sec \theta} + c$

**Answer: B**

**Solution:**

$$\begin{aligned} \int \frac{\sec x}{\sqrt{\sin(2x + \theta) + \sin \theta}} dx &= I \text{ (say)} \\ I &= \int \frac{\sec x}{\sqrt{\sin 2x \cos \theta + \cos 2x \sin \theta + \sin \theta}} dx \\ I &= \int \frac{\sec x}{\sqrt{\sin 2x \cos \theta + (\cos 2x + 1) \sin \theta}} dx \\ &= \int \frac{\sec x}{\sqrt{2 \cos x (\sin x \cos \theta + \cos x \sin \theta)}} dx \quad [\because \cos 2x + 1 = 2 \cos^2 x] \\ &= \int \frac{\sec x}{\sqrt{2 \cos^2 x \cos 2x \theta (\tan x + \tan \theta)}} dx \\ &= \frac{1}{\sqrt{2 \cos \theta}} \int \frac{\sec^2 x}{\sqrt{\tan x + \tan \theta}} dx \\ \text{Let } \tan x + \tan \theta &= u \Rightarrow \sec^2 x dx = du \\ &= \frac{1}{\sqrt{2 \cos \theta}} \int \frac{du}{\sqrt{u}} = \frac{1}{\sqrt{2 \cos \theta}} \cdot 2 \cdot \sqrt{u} + c \\ &= \frac{\sqrt{2}}{\sqrt{\cos \theta}} \cdot \sqrt{\tan x + \tan \theta} + c = \sqrt{\frac{2(\tan x + \tan \theta)}{\cos \theta}} + c \\ &= \sqrt{2(\tan x + \tan \theta) \sec \theta} + c \end{aligned}$$

---

## Question120

**Given,**  $\frac{3x-2}{(x+1)^2(x+3)} = \frac{A}{x+1} + \frac{B}{(x+1)^2} + \frac{C}{x+3}$ , **then**  $4A + 2B + 4C$  **is equal to**

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**Options:**

A. 5

B. -5

C. -3

D. 3

**Answer: B**

**Solution:**

$$\frac{3x-2}{(x+1)^2(x+3)} = \frac{A}{x+1} + \frac{B}{(x+1)^2} + \frac{C}{(x+3)}$$
$$\Rightarrow 3x-2 = A(x+1)(x+3) + B(x+3) + C(x+1)^2$$

Putting  $x = -1$

$$-5 = 2B$$

$$\Rightarrow B = \frac{-5}{2}$$

Putting  $x = -3$

$$-11 = 4C \Rightarrow C = \frac{-11}{4}$$

Putting  $x = -3$

$$-11 = 4C \Rightarrow C = \frac{-11}{4}$$

Putting  $x = 0$

$$-2 = 3A + 3B + C \Rightarrow -2 = 3A - \frac{15}{2} - \frac{11}{4}$$

$$\Rightarrow \frac{33}{4} = 3A \Rightarrow A = \frac{11}{4}$$

$$\Rightarrow 4A + 2B + 4C = 11 - 5 - 11 = -5$$

---

## Question 121

$\int \frac{\sin \alpha}{\sqrt{1+\cos \alpha}} d\alpha$  is equal to

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**Options:**

A.  $-2\sqrt{2} \cos \left( \frac{\alpha}{2} \right) + C$

B.  $2\sqrt{2} \cos \left( \frac{\alpha}{2} \right) + C$

C.  $\sqrt{2} \cos \left( \frac{\alpha}{2} \right) + C$

D.  $-\sqrt{2} \cos \left( \frac{\alpha}{2} \right) + C$

**Answer: A**

**Solution:**

$$\begin{aligned}
 \int \frac{\sin \alpha d\alpha}{\sqrt{1+\cos \alpha}} &= \int \frac{\sin \alpha \sqrt{1-\cos \alpha}}{\sqrt{1-\cos^2 \alpha}} d\alpha \\
 &= \int \sqrt{1-\cos \alpha} d\alpha \\
 &= \sqrt{2} \int \sin \frac{\alpha}{2} d\alpha \\
 &= -2\sqrt{2} \cos \frac{\alpha}{2}
 \end{aligned}$$


---

## Question 122

If  $\int \frac{\cos 4x + 1}{\cot x - \tan x} = k \cos 4x + C$ , then  $k$  is equal to

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Options:

A.  $\frac{-1}{2}$

B.  $\frac{-1}{8}$

C.  $\frac{-1}{3}$

D.  $\frac{-1}{5}$

**Answer: B**

**Solution:**

$$\begin{aligned}
 \int \frac{1+\cos 4x}{\cot x - \tan x} dx &= \int \frac{2 \cos^2 2x}{\frac{\cos 2x}{\cos x \sin x}} dx \\
 &= \int (2 \cos 2x \cos x \sin x) = \frac{1}{2} \int 2 \cos 2x \sin 2x dx \\
 &= \frac{1}{2} \int \sin 4x dx = \frac{-1}{8} \cos 4x + C \Rightarrow k = -\frac{1}{8}
 \end{aligned}$$


---

## Question 123

If  $\int [\cos(x) \cdot \frac{d}{dx}(\operatorname{cosec}(x))] dx = f(x) + g(x) + c$  then  $f(x) \cdot g(x)$  is equal to

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Options:

A.  $x \cot(x)$

B.  $x \tan(x)$

C.  $x \cos(x)$

D. 1

**Answer: A**

**Solution:**

$$\begin{aligned} & \int -\cos x \operatorname{cosec} x \cot x dx \\ &= -\int \cot^2 x dx \\ &= \int (1 - \operatorname{cosec}^2 x) dx = x + \cot x + C \\ &\therefore f(x) = x \text{ and } g(x) = \cot x \\ &f(x) \cdot g(x) = x \cot x \end{aligned}$$

---

## Question124

If  $\int \frac{(2x+1)^6}{(3x+2)^8} dx = P\left(\frac{2x+1}{3x+2}\right)^Q + R$ , then  $\frac{P}{Q}$  is equal to

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**Options:**

A.  $\frac{1}{7^2}$

B.  $\frac{1}{7}$

C.  $7^2$

D. 7

**Answer: A**

**Solution:**

$$\int \frac{(2x+1)^6}{(3x+2)^8} dx$$

$$\int \left( \frac{2x+1}{3x+2} \right)^6 \left( \frac{1}{3x+2} \right)^2 dx$$

Let  $\frac{2x+1}{3x+2} = t$

$$\Rightarrow \frac{(3x+2)2 - (2x+1)3}{(3x+2)^2} = \frac{dt}{dx} \Rightarrow \left( \frac{1}{3x+2} \right)^2 dx = dt$$

$$\Rightarrow \int t^6 dt = \frac{t^7}{7} + C \Rightarrow P = \frac{1}{7}, Q = 7$$

$$P/Q = 1/49 = 1/72$$


---

## Question 125

Which of the following is partial fraction of  $\frac{-x^2+6x+13}{(3x+5)(x^2+4x+4)}$  is equal to

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**Options:**

A.  $\frac{3}{3x+5} + \frac{-1}{x+2} + \frac{2}{(x+2)^2}$

B.  $\frac{2}{3x+5} + \frac{-1}{x+2} + \frac{3}{(x+2)^2}$

C.  $\frac{-1}{3x+5} + \frac{2}{x+2} + \frac{3}{(x+2)^2}$

D.  $\frac{3}{3x+5} + \frac{2}{x+2} + \frac{-1}{(x+2)^2}$

**Answer: B**

**Solution:**

$$\frac{-x^2+6x+13}{(3x+5)(x^2+4x+4)} = \frac{-x^2+6x+13}{(3x+5)(x+2)^2}$$

$$\Rightarrow \frac{-x^2+6x+13}{(3x+5)(x+2)^2} = \frac{A}{3x+5} + \frac{B}{x+2} + \frac{C}{(x+2)^2}$$

$$\Rightarrow -x^2+6x+13 = A(x+2)^2 + B(3x+5)(x+2) + C(3x+5) \dots (i)$$

It is an identity

$\therefore$  True for every value of  $x$ .

Put  $x = -2$  in Eq. (i),

$$-(-2)^2 + 6(-2) + 13 = A \cdot 0 + B \cdot 0 + C(3(-2) + 5)$$

$$\Rightarrow -4 - 12 + 13 = -C \Rightarrow C = 3$$

Put  $x = \frac{-5}{3}$  in Eq. (i), we get

$$\begin{aligned}
& -\left(\frac{-5}{3}\right)^2 + 6\left(\frac{-5}{3}\right) + 13 \\
= & A\left(\frac{-5}{3} + 2\right)^2 + B \cdot 0 + C \cdot 0 \\
\Rightarrow & \frac{-25}{9} - 10 + 13 = A \cdot \frac{1}{9} \\
\Rightarrow & \frac{-25}{9} + 3 = \frac{A}{9} \Rightarrow A = 2
\end{aligned}$$

Put  $x = 0$  in Eq. (i), we get

$$\begin{aligned}
13 &= 4A + 10B + 5C \\
\Rightarrow 13 &= 8 + 10B + 15 \\
\Rightarrow 10B &= 13 - 23 \text{ or } B = -1 \\
&\therefore \frac{-x^2+6x+13}{(3x+5)(x^2+4x+4)} \\
&= \frac{2}{3x+5} + \frac{-1}{x+2} + \frac{3}{(x+2)^2}
\end{aligned}$$


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## Question 126

$\int \frac{1+x+\sqrt{x+x^2}}{\sqrt{x}+\sqrt{1+x}} dx$  is equal to

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**Options:**

- A.  $\frac{1}{2}(\sqrt{1+x}) + c$
- B.  $\sqrt{1+x} + c$
- C.  $2(1+x)^{3/2} + c$
- D.  $\frac{2}{3}(1+x)^{3/2} + c$

**Answer: D**

**Solution:**

$$\begin{aligned}
\text{Let } I &= \int \frac{(1+x) + \sqrt{x+x^2}}{\sqrt{x} + \sqrt{1+x}} dx \\
&= \int \frac{(1+x) + \sqrt{x}\sqrt{1+x}}{\sqrt{x} + \sqrt{1+x}} dx \\
&= \int \frac{\sqrt{1+x}(\sqrt{1+x} + \sqrt{x})}{(\sqrt{1+x} + \sqrt{x})} dx \\
&= \int \sqrt{1+x} dx
\end{aligned}$$

$$\text{Let } 1+x = t^2 \Rightarrow dx = 2t dt$$

$$\therefore I = 2 \int t^2 dt = 2 \frac{t^3}{3} + C = \frac{2}{3}(1+x)^{3/2} + c$$

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## Question127

$\int (\cos x) \log \cot \left( \frac{x}{2} \right) dx$  is equal to

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**Options:**

- A.  $(\sin x) \log \cot \left( \frac{x}{2} \right) + c$
- B.  $(\cos x) \log \cot \left( \frac{x}{2} \right) + c$
- C.  $(\sin x) \log \cot \left( \frac{x}{2} \right) + x + c$
- D.  $(\sin x) \log \cot \left( \frac{x}{2} \right) - x + c$

**Answer: C**

**Solution:**

$$\begin{aligned} \text{Let } I &= \int \frac{(\cos x) \log \cot(x/2)}{\text{II}} \frac{1}{\text{I}} dx \\ &= \log \cot \frac{x}{2} \cdot \int \cos x dx \\ &\quad - \int \left[ \frac{d}{dx} \left( \log \cot \frac{x}{2} \right) \int \cos x dx \right] dx \quad [\text{integrating by parts}] \\ &= \log \left( \cot \frac{x}{2} \right) \cdot \sin x - \int \frac{1}{\cot \frac{x}{2}} \\ &\quad \cdot -\operatorname{cosec} \frac{x}{2} \cdot \frac{1}{2} \cdot \sin x dx \\ &= (\sin x) \log \cot \frac{x}{2} + \int 1 dx \\ &= (\sin x) \log \cot \frac{x}{2} + x + c \end{aligned}$$

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## Question128

$\int \sqrt{e^{4x} + e^{2x}} dx$  is equal to

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**Options:**

- A.  $\frac{1}{2} e^x \left( \sqrt{e^{2x} + 1} \right) + \frac{1}{2} \sinh^{-1} (e^x) + c$

B.  $\frac{1}{2}e^x \left( \sqrt{e^{2x} + 1} \right) + \sinh^{-1}(e^x) + c$

C.  $\frac{1}{2} \left( \sqrt{e^{2x} + 1} \right) + \frac{1}{2} \sinh^{-1}(e^x) + c$

D.  $\sqrt{e^{4x} + e^{2x}} + \sqrt{e^{2x} + 1} + c$

**Answer: A**

**Solution:**

$$I = \int \sqrt{e^{4x} + e^{2x}} dx = \int e^x \sqrt{e^{2x} + 1} dx$$

$$\text{Put } e^x = t \Rightarrow e^x dx = dt$$

$$\therefore I = \int \sqrt{t^2 + 1} dt$$

$$= \frac{1}{2} t \sqrt{t^2 + 1} + \frac{1}{2.1} \sinh^{-1}(t) + c$$

$$= \frac{1}{2} e^x \sqrt{e^{2x} + 1} + \frac{1}{2} \sinh^{-1}(e^x) + c$$

## Question129

If  $\int \frac{1}{1+\sin x} dx = \tan(f(x)) + c$ , then  $f'(0)$  is equal to

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**Options:**

A. 0

B. 1

C.  $\frac{1}{2}$

D.  $\frac{-1}{2}$

**Answer: C**

**Solution:**

$$\int \frac{1}{1 + \sin x} dx = \tan(f(x)) + c$$

$$\Rightarrow \int \frac{1 - \sin x}{\cos^2 x} dx = \tan(f(x)) + c$$

$$\begin{aligned} \Rightarrow \int (\sec^2 x - \sec x \tan x) dx &= \tan(f(x)) + c \\ \Rightarrow \tan x - \sec x + C &= \tan(f(x)) + c \\ \Rightarrow \tan\left(\frac{\pi}{4} + \frac{x}{2}\right) &= \tan(f(x)) \\ \Rightarrow -\tan\left(\frac{-\pi}{4} - \frac{x}{2}\right) &= \tan(f(x)) \\ \Rightarrow f(x) = -\frac{\pi}{4} + \frac{x}{2} &\Rightarrow f'(x) = \frac{1}{2} \\ \text{At } x = 0, f'(0) &= \frac{1}{2} \end{aligned}$$


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## Question130

$\int \frac{e^x(x+3)}{(x+5)^3} dx$  is equal to

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**Options:**

- A.  $\frac{e^x}{(x+5)^2} + C$
- B.  $e^x(x+5)^2 + C$
- C.  $e^x(x+3)^2 + C$
- D.  $\frac{e^x}{(x+3)^2} + C$

**Answer: A**

**Solution:**

$$\begin{aligned} \text{Let } I &= \int \frac{e^x(x+3)}{(x+5)^3} dx = \int e^x \left[ \frac{1}{(x+5)^2} - \frac{2}{(x+5)^3} \right] dx \\ \int [f(x) + f'(x)] e^x dx &= e^x f(x) + C \\ \Rightarrow I &= \frac{e^x}{(x+5)^2} + C \end{aligned}$$


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## Question131

If  $\int \frac{(x-1)^2}{(x^2+1)^2} dx = \tan^{-1}(x) + g(x) + k$ , then  $g(x)$  is equal to

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**Options:**

A.  $\tan^{-1} \left( \frac{x}{2} \right)$

B.  $\frac{1}{x^2+1}$

C.  $\frac{1}{2(x^2+1)}$

D.  $\frac{2}{x^2+1}$

**Answer: B**

**Solution:**

$$\begin{aligned} \int \frac{(x-1)^2}{(x^2+1)^2} dx &= \int \left( \frac{1}{x^2+1} - \frac{2x}{(x^2+1)^2} \right) dx \\ &= \tan^{-1}(x) - \int \frac{2x}{(x^2+1)^2} dx \end{aligned}$$

Let  $x^2 + 1 = t \Rightarrow 2x dx = dt$

$$\begin{aligned} &= \tan^{-1}(x) - \int t^{-2} dt = \tan^{-1}(x) + \frac{1}{t} + k \\ \Rightarrow \int \frac{(x-1)^2}{(x+1)^2} dx &= \tan^{-1}(x) + \frac{1}{x^2+1} + k \end{aligned}$$

Comparing with  $\int \frac{(x-1)^2}{(x^2+1)^2} dx = \tan^{-1}(x) + g(x) + k$  we have  $g(x) = \frac{1}{x^2+1}$

## Question132

If  $\int \frac{1-(\cot x)^{2021}}{\tan x + (\cot x)^{2022}} dx = \frac{1}{A} \log |(\sin x)^{2023} + (\cos x)^{2023}| + c$ , then  $A$  is equal to

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**Options:**

A. 2020

B. 2021

C. 2022

D. 2023

**Answer: D**

**Solution:**

$$\begin{aligned}
& \frac{1}{A} \log |(\sin x)^{2023} + (\cos x)^{2023}| + C \\
&= \int \frac{1 - (\cot x)^{2021}}{\tan x + (\cot x)^{2022}} dx = \int \frac{1 - \frac{(\cos x)^{2021}}{(\sin x)^{2021}}}{\frac{\sin x}{\cos x} + \frac{(\cos x)^{2022}}{(\sin x)^{2022}}} dx \\
&= \int \frac{(\sin x)^{2021} - (\cos x)^{2021}}{(\sin x)^{2023} + (\cos x)^{2023}} \cdot \cos x \sin x dx \\
&= \int \frac{\cos x (\sin x)^{2022} - \sin x (\cos x)^{2022}}{(\sin x)^{2023} + (\cos x)^{2023}} dx \\
&\because \int \frac{f'(x)}{f(x)} dx = \log |f(x)| + C \\
&\Rightarrow \frac{1}{A} \log |(\sin x)^{2023} + (\cos x)^{2023}| + C \\
&= \frac{1}{2023} \log |(\sin x)^{2023} + (\cos x)^{2023}| + C \\
&\Rightarrow A = 2023
\end{aligned}$$


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